CAREERS 360

PREPARATION Series

NEET UG 2027

Human Physiology NEET PYQs

100+ PYQs with Detailed Solutions & Concept Explanations

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Human Physiology Previous Year Questions

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About this eBook

Dear NEET Aspirants,

Welcome to your ultimate Year-wise NEET PYQs Guide for Human Physiology! This eBook is designed to help you master one of the most important sections of Biology, which carries a significant weight in the NEET exam. By focusing on previous years'

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questions (PYQs) organized year-wise, this eBook ensures that you understand the exam pattern, frequently tested topics, and question trends over the years.

What does this eBook contain?

Year-wise NEET PYQs Compilation – We have systematically arranged previous years' questions year-wise to help you analyze the trends and understand how concepts are tested over different NEET exams.

Detailed Solutions – Every question comes with a step-by-step explanation to ensure conceptual clarity, reinforcing the fundamentals of Human Physiology.

Topic-Wise Yearly Breakdown – The eBook covers the following topics:

- **Digestive System** – Year-wise PYQs on enzymes, digestion & absorption, and digestive disorders.
- **Respiratory System** – NEET questions on the mechanism of breathing, gas exchange, and transport of gases.
- **Circulatory System** – Questions from past years on heart anatomy, blood circulation, ECG, and blood pressure.
- **Excretory System** – Yearly NEET PYQs on urine formation, kidney function, and osmoregulation.
- **Nervous System** – Questions covering neurons, impulse conduction, brain functions, and reflex actions.
- **Endocrine System** – PYQs on hormones, feedback mechanisms, and hormonal disorders.
- **Muscular System** – Year-wise questions on muscle contraction, types of muscles, and neuromuscular coordination.

How Will This eBook Benefit You?

- **NEET-Oriented Preparation** – By focusing on **PYQs**, you'll develop a deep understanding of how questions are framed and what to expect in the exam.

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- **Higher Accuracy & Speed** – Practicing past questions improves your **problem-solving speed** and accuracy, giving you an edge over others.
- **Boosts Confidence** – Since these questions have appeared in **previous NEET exams**, solving them will strengthen your confidence and enhance your exam strategy.
- **Smart & Efficient Learning** – Instead of studying randomly, this eBook ensures **structured, high-yield learning**, helping you focus on **scoring** topics without unnecessary distractions.

Mastering **Human Physiology PYQs** will give you the **competitive advantage** needed to excel in NEET. Study smart, stay consistent, and take one step closer to your dream of becoming a medical professional!

RE NEET 2026

Q 1. The number of vertebrae in a human is _____. (Locomotion and Movement)

(1) 26
(2) 12
(3) 206
(4) 7

Correct Answer: (1) 26

Solution:

The human vertebral column develops with 33 vertebrae, but the 5 sacral vertebrae fuse into the sacrum and the 4 coccygeal vertebrae fuse into the coccyx. So the adult vertebral column is considered to have 26 vertebrae: 7 cervical + 12 thoracic + 5 lumbar + 1 sacrum + 1 coccyx = 26.

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Q 2. Which of the following is not a part of human central neural system?(Neural Control and Coordination)

(1) Arachnoid
(2) Dura mater
(3) Pia mater
(4) Pericardium

Correct Answer: (4) Pericardium

Solution:

Dura mater, arachnoid and pia mater are the three meninges — protective and nutritive membranes covering the brain and spinal cord (the CNS). The pericardium is the membrane covering the heart and has no connection with the neural system.

Q 3. Which of the following statements related to pituitary gland are correct?(Chemical Coordination and Integration)

(a) It is divided anatomically into adenohypophysis and neurohypophysis

(b) It secretes follicle stimulating hormone

(c) It secretes melanocyte stimulating hormone

(d) It does not secrete prolactin

(1) (a) and (b) only
(2) (a), (b) and (c) only
(3) (b) and (c) only
(4) (a), (b) and (d) only

Correct Answer: (2) (a), (b) and (c) only

Solution:

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The pituitary gland is anatomically divided into adenohypophysis (anterior pituitary) and neurohypophysis (posterior pituitary) — (a) is correct. The adenohypophysis secretes FSH — (b) is correct — and its pars intermedia secretes MSH — (c) is correct. The pituitary DOES secrete prolactin (from the pars distalis), so statement (d) is false. Hence (a), (b) and (c) only are correct.

Q 4. Which of the following represents the correct sequence of arrangement of bones in the lower limb of humans? (Locomotion and Movement)

(1) Femur-tibia-patella-tarsal
(2) Patella-femur-tibia-tarsal
(3) Femur-patella-tibia-tarsal
(4) Femur-tarsal-patella-tibia

Correct Answer: (3) Femur-patella-tibia-tarsal

Solution:

From proximal (hip) to distal (ankle), the lower limb bones are arranged as: Femur (thigh bone) → Patella (knee cap) → Tibia/Fibula (leg bones) → Tarsals (ankle bones).

Q 5. The number of action potentials generated by sino-atrial node (SAN) in a healthy human is _____ per minute. (Body Fluids and Circulation)

(1) 28 - 30
(2) 70 - 75
(3) 100 - 110
(4) 120 - 140

Correct Answer: (2) 70 - 75

Solution:

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The SAN (sino-atrial node) is the natural pacemaker of the heart. It generates the maximum number of action potentials, about 70-75 per minute, without any external nervous stimulation, and sets the pace of the heart's activity — the normal sinus rhythm.

Q 6. Which of the following statements regarding blood coagulation in response to an injury is incorrect? (Body Fluids and Circulation)

(1) Blood coagulates in response to an injury
(2) Blood clot consists of fibrins
(3) Fibrin is produced from fibrinogen
(4) Fibrinogen is produced from fibrin

Correct Answer: (4) Fibrinogen is produced from fibrin

Solution:

During blood clotting, the enzyme thrombin converts the soluble plasma protein fibrinogen into insoluble fibrin threads, which form the clot — not the reverse. Hence, "Fibrinogen is produced from fibrin" is the incorrect statement.

Q 7. Which of the following is used as a clot buster? (Body Fluids and Circulation)

(1) Streptokinase
(2) Penicillin
(3) Cyclosporin A
(4) Statins

Correct Answer: (1) Streptokinase

Solution:

Streptokinase, produced by the bacterium Streptococcus (and modified using genetic engineering), is used as a clot-buster drug to dissolve clots in patients who have undergone myocardial infarction. Penicillin is an antibiotic, Cyclosporin A is an

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immunosuppressant used in organ transplants, and statins are cholesterol-lowering drugs.

Q 8. Match List-I with List-II.

List-I:
A. Excess growth hormone
B. Luteinizing hormone
C. Vasopressin
D. Oxytocin

List-II:
I. Reabsorption of water and electrolytes in kidney
II. Contraction of uterus during child birth
III. Acromegaly
IV. Ovulation

Choose the correct answer from the options given below: (Chemical Coordination and Integration)

(1) A-III, B-IV, C-II, D-I
(2) A-III, B-IV, C-I, D-II
(3) A-II, B-IV, C-I, D-III
(4) A-IV, B-III, C-I, D-II

Correct Answer: (2) A-III, B-IV, C-I, D-II

Solution:

Excess growth hormone secretion in adults causes Acromegaly (A-III). The LH surge triggers ovulation (B-IV). Vasopressin (ADH) promotes reabsorption of water and electrolytes in the kidney (C-I). Oxytocin causes contraction of the uterus during childbirth (D-II). So the correct match is A-III, B-IV, C-I, D-II.

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Q 9. The opening between the right atrium and the right ventricle is guarded by _____. (Body Fluids and Circulation)

(1) bicuspid valve
(2) tricuspid valve
(3) semilunar valve
(4) sino-atrial node

Correct Answer: (2) tricuspid valve

Solution:

The right atrioventricular opening is guarded by the tricuspid valve (three cusps), while the left atrioventricular opening is guarded by the bicuspid (mitral) valve.

Q10. Muscle contraction is initiated by a signal sent by the central nervous system by the release of _____. (Neural Control and Coordination)

(1) acetyl choline
(2) acetyl coenzyme A
(3) cyclic guanine monophosphate
(4) cyclic adenine monophosphate

Correct Answer: (1) acetyl choline

Solution:

A signal from the CNS travels through a motor neuron to the neuromuscular junction, where the neurotransmitter acetylcholine is released. This generates an action potential in the muscle fibre (sarcolemma), which ultimately initiates muscle contraction.

Q 11. Which of the following statements about the reabsorption process in Henle's loop are correct?

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(a) The descending limb of Henle's loop is permeable to water but almost impermeable to electrolytes.

(b) Urine gets concentrated in Henle's loop.

(c) Reabsorption of Na+ and water takes place in Henle's loop.

(d) Active or passive transport of electrolytes occurs in the ascending limb of Henle's loop. (Excretory Products and their Elimination)

Choose the correct answer from the options given below:

(1) (a) and (b) only
(2) (b), (c) and (d) only
(3) (a), (b) and (c) only
(4) (a), (b) and (d) only

Correct Answer: (4) (a), (b) and (d) only

Solution:

The descending limb of Henle's loop is permeable to water but almost impermeable to electrolytes, so water moves out as the filtrate passes down — statement (a) is correct, and this progressively concentrates the filtrate/urine — statement (b) is correct. The ascending limb is impermeable to water but actively/passively transports electrolytes out of the filtrate — statement (d) is correct. Statement (c) is not precisely correct because Na+ and water are not reabsorbed together in the same segment — water leaves via the descending limb while electrolytes leave via the ascending limb. Hence (a), (b) and (d) only are correct.

Q 12. Which of the following is the correct order of arrangement of the vertebral column from the head to toe? (Locomotion and Movement)

(1) Cervical vertebra, thoracic vertebra, sacrum, lumbar vertebra
(2) Sacrum, lumbar vertebra, thoracic vertebra, cervical vertebra

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(3) Cervical vertebra, lumbar vertebra, thoracic vertebra, sacrum
(4) Cervical vertebra, thoracic vertebra, lumbar vertebra, sacrum

Correct Answer: (4) Cervical vertebra, thoracic vertebra, lumbar vertebra, sacrum

Solution:

From head to toe, the regions of the vertebral column are arranged as: Cervical (7 vertebrae) → Thoracic (12 vertebrae) → Lumbar (5 vertebrae) → Sacral (fused, forming the sacrum) → Coccygeal (fused, forming the coccyx).

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Q1. The following are the steps involved in the Renin-Angiotensin mechanism. Arrange them in the correct order :

A. Increase in blood pressure and Glomerular filtration rate.

B. Reabsorption of Na^+ and water from distal parts of tubule due to Aldosterone.

C. Fall in Glomerular filtration rate.

D. Vasoconstriction by Angiotensin II and release of Aldosterone.

E. Renin converts Angiotensinogen into Angiotensin I, followed by Angiotensin II.

Choose the correct answer from the options given below :

(1) C, A, B, D, E

(2) A, D, B, E, C

(3) A, C, E, B, D

(4) C, E, D, B, A

Correct Answer: (4) C, E, D, B, A

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Solution:

A fall in Glomerular Filtration Rate (GFR) (C) is sensed by the juxtaglomerular cells, which release Renin. Renin converts Angiotensinogen (a plasma protein) into Angiotensin I, which is then converted to Angiotensin II (E). Angiotensin II is a powerful vasoconstrictor that raises blood pressure and also stimulates the adrenal cortex to release Aldosterone (D). Aldosterone causes reabsorption of Na^+ and water from the distal parts of the tubule (B), which ultimately causes an increase in blood pressure and GFR (A), restoring the fluid balance.

Q2. Match List I with List II :

	List I (Respiratory Volume)		**List II (Capacity in mL)**
A.	ERV (Expiratory Reserve Volume)	I.	2500 – 3000 mL
B.	RV (Residual Volume)	II.	500 mL
C.	IRV (Inspiratory Reserve Volume)	III.	1000 – 1100 mL
D.	TV (Tidal Volume)	IV.	1100 – 1200 mL

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Choose the correct answer from the options given below :

(1) A-III, B-I, C-IV, D-II

(2) A-I, B-III, C-II, D-IV

(3) A-III, B-IV, C-I, D-II

(4) A-I, B-II, C-III, D-IV

Correct Answer: (3) A-III, B-IV, C-I, D-II

Solution:

Standard human respiratory volumes: Tidal Volume (TV) ≈ 500 mL; Inspiratory Reserve Volume (IRV) ≈ 2500–3000 mL; Expiratory Reserve Volume (ERV) ≈ 1000–1100 mL; Residual Volume (RV) ≈ 1100–1200 mL. Hence, ERV-III, RV-IV, IRV-I, TV-II.

Q3. The WBC count of a person's blood sample is 8000/cu.mm. How many eosinophils and lymphocytes would be in the same blood sample, approximately ?

(1) 160 – 240/cu.mm and 1600 – 2000/cu.mm, respectively

(2) 100 – 120/cu.mm and 160 – 200/cu.mm, respectively

(3) 300 – 500/cu.mm and 500 – 700/cu.mm, respectively

(4) 300 – 500/cu.mm and 1200 – 1500/cu.mm, respectively

Correct Answer: (1) 160 – 240/cu.mm and 1600 – 2000/cu.mm, respectively

Solution:

Eosinophils normally constitute about 2–3% of total WBCs, and lymphocytes about 20–25% of total WBCs. For a total WBC count of 8000/cu.mm: Eosinophils ≈ 2–3% of 8000 = 160–240/cu.mm; Lymphocytes ≈ 20–25% of 8000 = 1600–2000/cu.mm.

Q4. Match List I with List II related to muscular/skeletal system :

	List I		**List II**
A.	Tetany	I.	Inflammation of joints
B.	Arthritis	II.	Autoimmune disorder affecting neuromuscular junction
C.	Myasthenia gravis	III.	Wild contraction in muscle due to low Ca^{2+} in body fluid
D.	Muscular dystrophy	IV.	Progressive degeneration of skeletal muscle

Choose the correct answer from the options given below :

(1) A-III, B-I, C-II, D-IV

(2) A-IV, B-III, C-II, D-I

(3) A-I, B-II, C-III, D-IV

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(4) A-III, B-II, C-I, D-IV

Correct Answer: (1) A-III, B-I, C-II, D-IV

Solution:

Tetany results from low Ca^{2+} in body fluids, causing wild/rapid muscle contractions. Arthritis is inflammation of joints. Myasthenia gravis is an autoimmune disorder that affects the neuromuscular junction, causing muscle weakness. Muscular dystrophy is a progressive degeneration of skeletal muscle, mostly genetic in origin.

Q5. In humans, respiration occurs in the following steps. Arrange these steps in the correct order.

A. Diffusion of O_2 and CO_2 between blood and tissues

B. Diffusion of O_2 and CO_2 across alveolar membrane

C. Pulmonary ventilation by which atmospheric air is drawn in and CO_2 rich alveolar air is released out

D. Cellular respiration

E. Transport of gases by the blood

Choose the correct answer from the options given below :

(1) A, B, C, D, E

(2) E, A, C, D, B

(3) C, A, B, E, D

(4) C, B, E, A, D

Correct Answer: (4) C, B, E, A, D

Solution:

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Breathing/respiration in humans proceeds as: Pulmonary ventilation (C) → Diffusion of O_2 and CO_2 across the alveolar membrane (B) → Transport of gases by the blood (E) → Diffusion of O_2 and CO_2 between blood and tissues (A) → Utilisation of O_2 by cells for catabolic reactions, i.e. Cellular respiration (D).

Q6. Select the incorrect statements with reference to Rh grouping.

A. Erythroblastosis foetalis is a condition observed having foetus with Rh^{-ve} blood and mother with Rh^{+ve} blood.

B. Rh antigen is observed on RBCs in the majority of human beings.

C. Before blood transfusion, Rh group should also be matched.

D. Rh incompatibility is observed when a pregnant mother is Rh^{-ve} and the foetus is Rh^{+ve}.

E. Erythroblastosis foetalis can be avoided by administering anti-Rh antibodies to the mother immediately after the delivery of the second child.

Choose the answer from the options given below :

(1) A and E only

(2) A and B only

(3) B and C only

(4) C and D only

Correct Answer: (1) A and E only

Solution:

Statement A is incorrect: erythroblastosis foetalis actually occurs when the mother is Rh^{-ve} and the foetus is Rh^{+ve} (not the reverse, as stated). Statement E is incorrect: anti-Rh antibodies must be administered to the Rh^{-ve} mother immediately after the delivery of the FIRST child, not the second, to prevent the condition in a subsequent

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pregnancy. Statements B, C and D correctly describe Rh antigen prevalence, the need to match Rh group before transfusion, and the cause of Rh incompatibility.

Q7. Which of the following statements are correct with reference to human endoskeleton ?

A. Human skull is monocondylic.

B. The joint between any two adjoining vertebrae is a cartilaginous joint.

C. In human beings, the number of cervical vertebrae is seven.

D. All ribs except the last 2 pairs are bicephalic.

E. The occipital bone of skull is articulated with atlas vertebra.

Choose the correct answer from the options given below :

(1) A, B and D only

(2) B and E only

(3) B, C and E only

(4) C, D and E only

Correct Answer: (3) B, C and E only

Solution:

The human skull is dicondylic (has two occipital condyles), not monocondylic — A is false. The joint between adjoining vertebral bodies is a cartilaginous joint — B is correct. Humans have seven cervical vertebrae — C is correct. Of the 12 pairs of ribs, only ribs 2–10 are bicephalic (have two articulating heads); ribs 1, 11 and 12 are monocephalic — so D, as stated ('all ribs except the last 2 pairs'), is false. The occipital condyles articulate with the atlas vertebra — E is correct.

Q8. Match List I with List II :

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	List I (Bioactive molecule)		List II (Function)
A.	Cortisol	I.	Stimulates the formation of alveoli in mammary glands
B.	Aldosterone	II.	Produces anti-inflammatory reactions
C.	Cholecystokinin	III.	Stimulates reabsorption of Na^+ and water from renal tubule
D.	Progesterone	IV.	Stimulates secretion of pancreatic enzymes and bile juice

Choose the correct answer from the options given below :

(1) A-III, B-II, C-IV, D-I

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(2) A-II, B-III, C-IV, D-I

(3) A-IV, B-II, C-I, D-III

(4) A-II, B-III, C-I, D-IV

Correct Answer: (2) A-II, B-III, C-IV, D-I

Solution:

Cortisol produces anti-inflammatory reactions (A-II). Aldosterone stimulates reabsorption of Na^+ and water from the renal tubule (B-III). Cholecystokinin (CCK) stimulates secretion of pancreatic enzymes and bile juice (C-IV). Progesterone stimulates the formation/growth of alveoli in mammary glands during pregnancy (D-I).

Q9. The specific receptors for neurotransmitters in a synapse are present on _____.

(1) Post-synaptic membrane

(2) Pre-synaptic membrane

(3) Myelin sheath

(4) Schwann cell

Correct Answer: (1) Post-synaptic membrane

Solution:

Neurotransmitters released from synaptic vesicles at the pre-synaptic membrane diffuse across the synaptic cleft and bind to specific receptor proteins located on the post-synaptic membrane, generating a new potential in the post-synaptic neuron.

Q10. Choose the correct statements regarding muscle contraction.

A. A motor neuron carries a signal sent by the Central Nervous System (CNS) to the sarcolemma of the muscle fibre.

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B. The neural signal generates an action potential which causes the release of Ca^{2+} into sarcoplasm.

C. Increase in Ca^{2+} inactivates the actin for breaking cross bridges.

D. Actin binds to the myosin head to form a cross bridge.

E. Shortening of sarcomere takes place, by pulling actin filaments towards the centre of 'A' band.

Choose the correct answer from the options given below :

(1) A, B, D and E only

(2) C and D only

(3) C and E only

(4) A and B only

Correct Answer: (1) A, B, D and E only

Solution:

Statement C is incorrect: an increase in Ca^{2+} does not inactivate actin — instead, Ca^{2+} binds to troponin, causing tropomyosin to shift and EXPOSE the actin binding sites for myosin, which activates cross-bridge formation. All other statements correctly describe the sequence of events in the sliding filament mechanism of muscle contraction: signal transmission from CNS via motor neuron to sarcolemma (A), Ca^{2+} release into sarcoplasm (B), actin-myosin cross-bridge formation (D), and sarcomere shortening as actin filaments slide towards the centre of the A band (E).

Q11. The JGA (Juxta Glomerular Apparatus) is a special sensitive region formed by cellular modifications in _____ related to the same nephron.

(1) Distal convoluted tubule and efferent renal arteriole

(2) Proximal convoluted tubule and afferent renal arteriole

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(3) Distal convoluted tubule and afferent renal arteriole

(4) Proximal convoluted tubule and efferent renal arteriole

Correct Answer: (3) Distal convoluted tubule and afferent renal arteriole

Solution:

The Juxtaglomerular Apparatus (JGA) is formed at the point of contact between the afferent arteriole (at the vascular pole of the glomerulus) and the distal convoluted tubule (DCT) of the same nephron. Cellular modifications at this site (macula densa in the DCT and juxtaglomerular cells in the afferent arteriole) help regulate GFR and blood pressure via the renin-angiotensin mechanism.

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Q1. What is the name of the blood vessel that carries deoxygenated blood from the body to the heart in a frog?

Option 1: Vena Cava
Option 2: Aorta
Option 3: Pulmonary artery
Option 4: Pulmonary vein

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Chapter Name - Body Fluids and Circulation
Concept - The four chambers of the heart are connected to major blood vessels as follows:

1. **Systemic aorta (systemic arch):** Arises from the left ventricle and carries oxygenated blood to the body (except lungs).
2. **Pulmonary trunk (pulmonary arch):** Originates from the right ventricle and splits into two pulmonary arteries that carry deoxygenated blood to the lungs.
3. **Superior vena cava (precaval):** Brings deoxygenated blood from the head and upper body to the right atrium.
4. **Inferior vena cava (postcaval):** Brings deoxygenated blood from the lower body to the right atrium.
5. **Pulmonary veins:** Four in number, these carry oxygenated blood from the lungs to the left atrium.

Solution - Name of the blood vessel: Precaval and postcaval veins (vena cavae)

1. Function: Carry deoxygenated blood from the body to the right atrium of the heart in a frog.

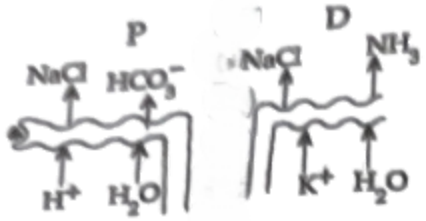
Frogs have a three-chambered heart with two atria and one ventricle. Oxygenated blood, after oxygenation in the skin and lungs, is circulated to the left atrium of the heart. At the same time, deoxygenated blood returning from the body flows into the right atrium. Both types of blood partially mix in the ventricle and are then pumped to different parts of the body.

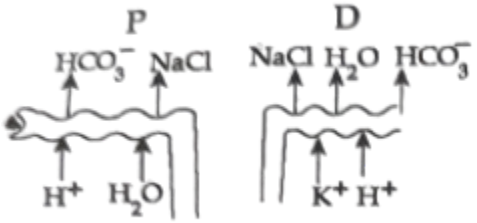
Hence, the correct answer is option 1) Vena Cava.

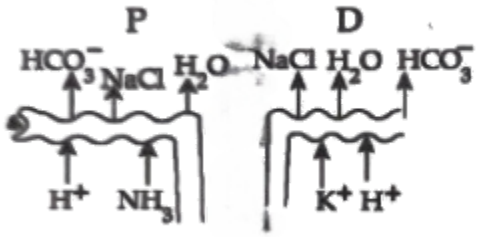
Ques:2 Which of the following diagrams is correct with regard to the proximal (P) and Distal (D) Tubule in the Nephron?

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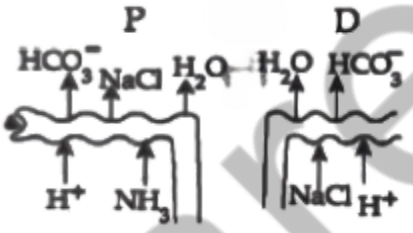
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Chapter Name: Excretory Products and Elimination

Difficulty: Medium

Concept: Proximal Convoluted Tubule (PCT): The tubule continues further to form a highly coiled network – proximal convoluted tubule. It is called convoluted due to its tortuous path.

Loop of Henle: It starts at the end of the Loop of Henle. Its major portion lies in the medulla of the kidney. It consists of a descending limb and an ascending limb.

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Distal Convoluted Tubule (DCT): It lies in the cortex of the kidney. It is lined by cuboidal epithelium that has few, small and irregular microvilli.

Collecting Duct: It is also lined by cuboidal or columnar epithelium. Many distal convoluted tubules combine to form the collecting duct.

Solution: Proximal Tubule (P):
Reabsorption: It reabsorbs Sodium chloride (NaCl), bicarbonate ions (HCO_3^-), and water (H_2O) into the blood from the filtrate.
Secretion: Hydrogen ions (H^+) and ammonia (NH_3) are secreted into the tubular fluid, helping to keep pH stable and to eliminate the waste..

Distal Tubule (D):
Reabsorption: It reabsorbs NaCl, HCO_3^-, and water to maintain the body's electrolyte and water balance.
Secretion: Distal Tubule excretes H^+ and potassium ions (K^+) into the filtrate, which helps in acid-base balance and potassium control.

Hence, the correct answer is option (3).

Ques: 3 Why can't insulin be given orally to diabetic patients?
Option 1: Its bioavailability will be increased.
Option 2: The human body will elicit a strong immune response.
Option 3: It will be digested in the Gastrointestinal (GI) tract.
Option 4: Because of structural variation.
Chapter Name: Chemical Cordination and Integration
Difficulty: Medium
Concept: Mechanism of Hormone Action -

- The hormones can be of two types:
    - Water-soluble, such as amino acid derivatives, peptide hormones (insulin, oxytocin), and protein hormones (GH)
    - Lipid soluble, such as steroid hormones

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- The water-soluble hormones need an extracellular receptor and a secondary messenger for carrying out their activity.
- Lipid-soluble hormones can act directly on the DNA.

Solution: Insulin is a protein hormone. When taken by mouth, it is digested by enzymes in the intestine and stomach like any other protein, and thus it will never enter the bloodstream and show its effect on blood sugar. Insulin is given by injection. Its mechanism of action requires an extracellular receptor and a secondary messenger to show its activity. In response to binding with its cell surface receptor, it initiates a mixture of intracellular signals that increase glucose uptake, particularly in muscle and adipose cells.

Hence, the correct answer is option 3) It will be digested in the Gastrointestinal (GI) tract.

Ques: 4 After maturation, in primary lymphoid organs, the lymphocytes migrate to interact with antigens in secondary lymphoid organ (s) / tissue(s) like:

A. Thymus

B. Bone marrow

C. Spleen

D. Lymph nodes

E. Peyer's patches

Choose the correct answer from the options given below

Chapter Name: Body Fluids and Circulation

Difficulty Level: Medium

Option 1: C, D, E only

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Option 2: B, C, D only

Option 3: A, B, C only

Option 4: E, A, B only

Concept: Lymph nodes and Lymph organs are Lymphatic tissues

- Lymph nodes are oval or bean-shaped structures located along the length of lymphatic vessels.
- Lymph filters through the lymph nodes
- Lymph nodes also contain phagocytic WBCs and macrophages, which consume microorganisms and foreign particles from the lymph
- The thymus, Spleen, and Tonsils are Lymphatic organs, and these also help to filter out pathogens

Solution: After maturing in the primary lymphoid organs -bone marrow and thymus, the lymphocytes move to secondary lymphoid organs like the spleen, lymph nodes, and Peyer's patches. These are the sites where they encounter antigens and start immune responses.

1. Thymus (A) and Bone marrow (B) are primary lymphoid organs, where lymphocytes mature. They are not sites where lymphocytes interact with antigens.
2. Spleen (C), Lymph nodes (D), and Peyer's patches (E) are secondary lymphoid organs. These are the sites where mature lymphocytes interact with antigens to initiate immune responses.

Hence, the correct answer is option 1) C, D, E only.

Ques: 5 Which of the following hormones released from the pituitary is actually synthesized in the hypothalamus?

Chapter: Chemical Coordination and Integration

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Difficulty: Medium

Option 1: Adrenocorticotrophic hormone (ACTH)

Option 2: Luteinizing hormone (LH)

Option 3: Anti-diuretic hormone (ADH)

Option 4: Follicle-stimulating hormone (FSH)

Concept: The Hypothalamus -

- The hypothalamus is the basal part of the diencephalon of the forebrain.
- It develops from the ectoderm.
- It regulates a wide spectrum of body functions.
- It contains several groups of neurosecretory cells called nuclei, which produce hormones.
- These hormones regulate the synthesis and secretion of pituitary hormones.
- The hormones produced by the hypothalamus are of two types:
    - The releasing hormones (which stimulate the secretion of pituitary hormones)
    - The inhibiting hormones (which inhibit secretions of pituitary hormones).

Solution: Anti-diuretic hormone (ADH) is synthesised in the hypothalamus but secreted and released from the posterior pituitary. The remaining hormones (ACTH, LH, FSH) are synthesised and secreted by the anterior pituitary. ADH is unique because it is synthesised by the hypothalamus but only secreted by the pituitary. ADH plays an essential role in water balance in the body by promoting the reabsorption of water in the kidneys. Its secretion happens when the body requires water preservation, like in a situation of dehydration.

Hence, the correct answer is option (3) Anti-diuretic hormone (ADH).

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Ques: 6 Match the following

List - I	List - II
A. Emphysema	**I Rapid spasms in the muscle due to low Ca^{2+} in the body fluid**
B. Angina Pectoris	**II. Damaged alveolar walls and decreased respiratory surface**
C.Glomerulonephritis	**III. Acute chest pain occurs when not enough oxygen is reaching the heart muscle**
D. Tetany	**IV. Inflammation of the glomeruli of the kidney**

Chapter: Locomotion and Movement

Difficulty: Medium

Option 1: A-II, B-III, C-IV, D-I.

Option 2: A-III, B-I, C-IV, D-II.

Option 3: A-II, B-I, C-II, D-IV.

Option 4: A-II, B-IV, C-III, D-I.

Concept: Some disorders of the Muscular and Skeletal System-

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- Myasthenia gravis: Autoimmune disorder affecting the neuromuscular junction, leading to fatigue, weakening, and paralysis of skeletal muscle.
- Muscular dystrophy: Progressive degeneration of skeletal muscle, especially due to a genetic disorder.
- Tetany: Rapid spasms (wild contractions) in muscle due to low Ca++ in body fluid.

Solution: A. Emphysema → II - It affects alveolar walls and reduces surface area for gas exchange

B. Angina Pectoris → III - Chest pain due to reduced oxygen supply to the heart muscle

C. Glomerulonephritis → IV - Inflammation of glomeruli in kidneys

D. Tetany → I - Muscle spasms due to low calcium levels

Hence, the correct answer is option 1) A-II, B-III, C-IV, D-I.

Ques: 7 Match the following:

List – I	List – II
A. Progesterone	**I. Pars intermedia**
B. Relaxin	**II. Ovary**
C. Melanocyte-stimulating hormone (MSH)	**III. Adrenal Medulla**

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D. Catecholamines	**IV. Corpus luteum**

Chapter: Chemical Control and Integration

Difficulty: Medium

Option 1: A–III, B–II, C–IV, D–I

Option 2: A–IV, B–II, C–I, D–III

Option 3: A–IV, B–II, C–III, D–I

Option 4: A–II, B–IV, C–I, D–III

Concept: Properties of Hormones:

- They are synthesised and secreted by endocrine cells
- Hormones are non-nutrient chemicals.
- Hormones are transported by the bloodstream from the endocrine cells to serve as 'chemical messengers' which act on target cells or organs.
- They may accelerate or inhibit a specific physiological process.
- Hormones are often used up in their regulatory activities.

Mode of Action:

- The three most common types of interaction are as follows:
- The permissive effect, in which the presence of one hormone enables another hormone to act.
- The synergistic effect in when two hormones with similar effects produce an amplified response.
- The antagonistic effect includes two hormones having opposing effects.

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Solution: Correct Matching:

A. Progesterone → IV. Corpus luteum
B. Relaxin → II. Ovary
C. Melanocyte Stimulating Hormone (MSH) → I. Pars intermedia
D. Catecholamines → III. Adrenal medulla

Explanation:

A. Progesterone → Corpus luteum: Progesterone is a steroid hormone produced by the corpus luteum of the ovary following ovulation. It supports the maintenance of the uterine lining for implantation and early pregnancy.

B. Relaxin → Ovary: Relaxin is an ovarian hormone, mainly secreted during pregnancy. Relaxin causes relaxation of the ligaments of the pelvis and softening of the cervix to ease labor.

C. Melanocyte Stimulating Hormone (MSH) → Pars intermedia: MSH is synthesised by the pars intermedia of the pituitary. MSH acts on melanocytes to synthesise melanin that influences pigmentation of the skin.

D. Adrenal medulla → Catecholamines: The adrenal medulla secretes catecholamines like adrenaline and noradrenaline when the body is under stress. They condition the body for the 'fight or flight' response

Hence, the correct answer is option 2) A–IV, B–II, C–I, D–III

Ques: 8 Cardiac activities of the heart are regulated by:

A. Nodal tissue
B. A special neural centre in the medulla oblongata
C. Adrenal medullary hormones
D. Adrenal cortical hormones

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Choose the correct answer from the options given below:

Chapter: Body Fluids and Circulation

Difficulty: Medium

Concept: Regulation of cardiac activity is done at the neural and chemical levels

Nervous regulation

- The sympathetic and parasympathetic nervous system regulates the cardiac activity.
- The changes in arterial pressure can be measured through baroreceptors

Baro-receptors

- These are the endings of the nerves that lie in the walls of the arteries.
- A rise in arterial pressure causes them to transmit signals into the CNS.
- In response to these signals, feedback signals are then sent back via the autonomic nervous system to the circulation to reduce the arterial pressure to the normal level.
- The signals are transmitted from each carotid sinus to the glossopharyngeal nerve and then to the tractus solitarius in the medulla oblongata of the brain.
- Signals from the aortic arch transmit through the vagus nerve to the tractus solitarius in the medulla oblongata of the brain.

Solution: A. Nodal tissue: It generates and conducts electrical impulses that control the heartbeat (e.g., SA node, AV node).

B. Medulla oblongata: It has the cardiovascular center, which regulates heart rate through autonomic nerves.

C. Adrenal medullary hormones: The Adrenal medulla secretes adrenaline and noradrenaline, which increase the heart rate and cardiac output.

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D. Adrenal cortical hormones: These hormones are primarily responsible for metabolism and for the maintenance of electrolytes in the human body.

Hence, the correct answer is option 2) A, B, and C only.

Ques: 9 Match the following:

List – I	List – II
A. Heart	**I. Erythropoietin**
B. Kidney	**II. Aldosterone**
C. Gastro-intestinal tract	**III. Atrial natriuretic factor**
D. Adrenal Cortex	**IV. Secretin**

Option 1: A–III, B–I, C–IV, D–II

Option 2: A–II, B–I, C–III, D–IV

Option 3: A–IV, B–III, C–II, D–I

Option 4: A–I, B–III, C–IV, D–II

Chapter: Chemical Control and Cordination

Difficulty: Medium

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Concept: Heart:

- Cardiocytes release a peptide hormone called atrial natriuretic factor (ANF).
- ANF inhibits the release of renin from the juxtaglomerular cells of the kidneys.
- It lowers the blood pressure.

Gastrointestinal Tract:

- The cells of the mucosa in the stomach and intestine release some important hormones as follows:
    - Gastrin
    - Enterogastrone or Gastric Inhibitory Peptide
    - Secretin
    - Cholecystokinin

Solution: A. Heart → III. Atrial natriuretic factor (ANF):

The heart releases ANF from the wall of the atrium, which helps in the regulation of blood pressure by initiating the excretion of sodium.

B. Kidney → I. Erythropoietin:

The kidney secretes erythropoietin, a hormone that causes red blood cell formation in the bone marrow.

C. Gastro-intestinal tract → IV. Secretin:

The GI tract releases secretin, which controls the pH of the small intestine by the release of bicarbonate from the pancreas.

D. Adrenal Cortex → II. Aldosterone:

Aldosterone, the mineralocorticoid hormone that regulates sodium and potassium levels in the body, is produced by the adrenal cortex.

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Hence, the correct answer is option 1)A–III, B–I, C–IV, D–II.

Ques: 10 Consider the following statements regarding the function of adrenal medullary hormones:

A. It causes pupillary constriction

B. It is a hyperglycemic hormone

C. It causes piloerection

D. It increases the strength of heart contraction

Choose the correct answer from the options given below:

Chapter: Chemical Control and Integration

Difficulty: Medium

Concept: Adrenal Glands or Suprarenal Glands: Adrenal Medulla -

Norepinephrine or Noradrenaline:

- It regulates blood pressure under normal conditions.
- It causes constriction of essentially all the blood vessels of the body.
- It causes increased activity of the heart, inhibition of the gastrointestinal tract, dilation of the pupils of the eyes and so forth.

Epinephrine or Adrenaline:

- It is secreted at the time of an emergency. Hence, it is also called an emergency hormone.
- It has a greater effect on cardiac activity than norepinephrine.
- Both adrenaline and noradrenaline act on the cells of skeletal, cardiac and smooth muscles and blood vessels and fat cells.

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- Because of the role of their hormones, the adrenal glands are also called ‘glands of emergency’.

Solution: The adrenal medullary hormones, e.g., adrenaline (epinephrine), cause the following effects:

B. Hyperglycemic effect: They increase blood glucose levels by causing the breakdown of glycogen and suppressing insulin release.

C. Piloerection: They cause contraction of small muscles at the base of hair follicles, resulting in hair standing on end (goosebumps), usually during stress or cold.

D. Increased cardiac function: They raise the heart rate and the strength of heart contractions, which increases the blood flow in cases of emergencies.

A. Constriction of pupils : Incorrect: Adrenaline induces pupillary dilation (mydriasis), permitting a greater light entry into the eyes in response to the "fight or flight" stimulus.

Hence, the correct answer is option 3) B, C, and D only.

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Ques:1 As per the ABO blood grouping system, the blood group of the father is B^+, the mother is A^+ and the child is O^+ Their respective genotype can be

A . $I^Bi/I^Ai/ii$

B. $I^BI^B/I^AI^A/ii$

C. $I^AI^B/iI^A/I^Bi$

D. $I^Ai/I^Bi/I^Ai$

E. $iI^B/iI^A/I^AI^B$

Choose the most appropriate answer from the options given below :

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Ques:1 As per the ABO blood grouping system, the blood group of the father is B^+, the mother is A^+ and the child is O^+ Their respective genotype can be

A . $I^Bi/I^Ai/ii$

B. $I^BI^B/I^AI^A/ii$

C. $I^AI^B/iI^A/I^Bi$

D. $I^Ai/I^Bi/I^Ai$

E. $iI^B/iI^A/I^AI^B$

Choose the most appropriate answer from the options given below :

1) A only
2) B only
3) C & B only
4) D & E only

Chapter Name - Body fluids and circulation: Composition of blood, blood groups, coagulation of blood

Difficulty - Medium

Concept - Blood groups: ABO Blood Groups

- There are more than 30 antigens present on the surfaces of blood cells. These antigens give rise to blood groups.

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- Antigens can induce an immune response by secreting a class of proteins called antibodies that bind in a specific fashion with antigens.
- Various types of blood grouping have been done depending on the nature of antigens present on the membrane of RBCs. Two such blood group systems are ABO (antigen A, B) and Rh (antigen Rh) blood grouping.

ABO blood group:

- ABO blood groups in human beings were reported for the first time by Karl Landsteiner. According to this system, humans can have 4 types of blood groups, A, B, AB, and .O.
- Landsteine discovered A, B, and O blood groups, while de Castello and Steini discovered AB blood groups.
- ABO blood groups are determined by the gene Ig which has three alleles Ia, Ib, and Io of this gene
- Allele Ia and Ib produce antigen A and antigen B respectively.
- RBCs of each blood group contain some antigens and they possess antibodies against other antigens in their plasma.

RELATIONSHIPS BETWEEN BLOOD TYPES AND ANTIBODIES

Blood Type	Antigens on Red Blood Cell	Can Donate Blood To	Antibodies in Cerum	Can Recieve Blood From
A	A	A, AB	Anti-B	A, O
B	B	B, AB	Anti-A	B, O
AB	A and B	AB	None	AB, O
O	None	A, B, AB, O	Anti-A and Anti-B	O

- During blood transfusion, it is very important to know the blood groups to avoid a clumping reaction.

- Clumping is also known as Agglutination. It occurs when Antibodies stick to the wrong Antigens and this can result in clumps which can be fatal. Hence blood group recognition is very important before transfusion (Antigens are Agglutinogen and Antibodies are Agglutinins)
- Since individuals with blood group AB do not have antibodies against any of the antigens, they can receive blood from individuals with A, B, and O blood groups. This is why blood group AB is called the universal recipient.
- Individuals with blood group O do not have any kind of Antigen and thus can donate blood to individuals with A, B, and AB blood groups. Hence, blood group O is called the universal donor.

Solution:

In the ABO blood grouping system, the blood group of a person is determined by the alleles inherited from both parents. The possible alleles are:

- $\mathbf{I^A}$: A allele
- $\mathbf{I^B}$: B allele
- **i**: O allele (recessive)

The genotype of the father with blood group $B^+ = I^Bi$
The genotype of the mother with blood group $A^+ = I^Ai$
The genotype of the child with blood group $O^+ = ii$

Hence, the correct answer is Option (1)A only.

Ques:2 Choose the correct statement given below regarding the juxtamedullary nephron.

1) Juxta medullary nephrons are located in the columns of Bertini.

2) The renal corpuscle of the juxta medullary nephron lies in the outer portion of the renal medulla.

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3) The loop of Henle of Juxta medullary nephron runs deep into the medulla.

4) Juxta medullary nephrons outnumber the cortical nephrons.

Chapter Name - Excretory Products & Their Elimination
Difficulty - Difficult
Concept - Juxtamedullary Nephrons:

- They form about 15% of the total nephrons.
- Their glomeruli are found at the junction of the cortex and medulla.
- These are larger in size and the loop of Henle is present deeper into the medulla.
- The loop of Henle is associated with the vasa recta.
- They are responsible for controlling the volume of plasma when the water supply is short.

Cortical Nephrons:

- They form about 85% of the total nephrons.
- They lie in the renal cortex. The glomeruli are present in the outer cortex.
- The loop of Henle is short and extends a short distance into the medulla.
- The vasa recta are absent.

Solution:
Longer than the length of the loop of Henle of the cortical nephron and runs deep into the medulla.
Option (1) is incorrect as juxta medullary nephrons are not present in the columns of Bertini.
Option (2) is incorrect because the renal corpuscle of the juxta medullary nephron lies in the inner cortical region.
Option (4) is incorrect as juxta medullary nephrons are lesser in number than cortical nephrons.

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Hence, the correct answer is option 3) Loop of Henle of Juxta medullary nephron runs deep into the medulla.

Ques:3 Match List I with List II :

	List I		List II
A	**P wave**	**I**	**Heart muscles are electrically silent**
B	**QRS complex**	**II**	**Depolarisation of ventricles.**
C	**T wave**	**III**	**Depolarisation of atria.**
D	**T-P gap**	**IV**	**Repolarisation of ventricles.**

Choose the correct answer from the options given below :

Option 1) A-I, B-III, C-IV, D-II

Option 2) A-III, B-II, C-IV, D-I

Option 3) A-II, B-III, C-I, D-IV

Option 4) A-IV, B-II, C-I, D-III

Chapter Name - Human Physiology
Difficulty - Medium
Concept - ECG

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- The representation of the electric current produced by the excitation of the cardiac muscles in the form of a graph is called an electrocardiogram (ECG)
- The machine by which the electrocardiogram is recorded is called an electrocardiograph.
- The electrocardiograph was invented by Einthoven and he is known as the father of electrocardiography.
- An electrocardiogram shows three consecutive waves namely the P wave, the QRS wave, and the T wave.

1. P wave-

- It is a small upward wave that indicates the depolarisation of the atria leading to contraction of both atria. It is caused by the activation of the SA node.

2. QRS wave

- It begins after a fraction of a second of the P wave
- It begins as a small downward deflection (Q) and continues as a large upright (R) and triangular wave, ending as a downward wave (S) at its base
- QRS wave indicates depolarization of ventricles
- It is caused by the impulses of the contraction from the AV node through the Bundle of His and Purkinje fibres and the contraction of ventricular muscles
- In this wave, there is a spread of electrical impulses through ventricles
- The normal duration of the QRS complex is <0.10 seconds

3. T wave

- The T wave is dome-shaped and it indicates ventricular repolarization (ventricular relaxation)
- The end of T waves marks systole

P-R interval

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- It is the time required for an impulse to travel through the atria and AV node to the remaining conductive tissue.
- The normal duration of the P-R interval is <0.12 to 0.2 seconds
- The P-R interval is lengthened during conditions like arteriosclerotic heart disease and rheumatic fever due to the inflammation of atria and AV node

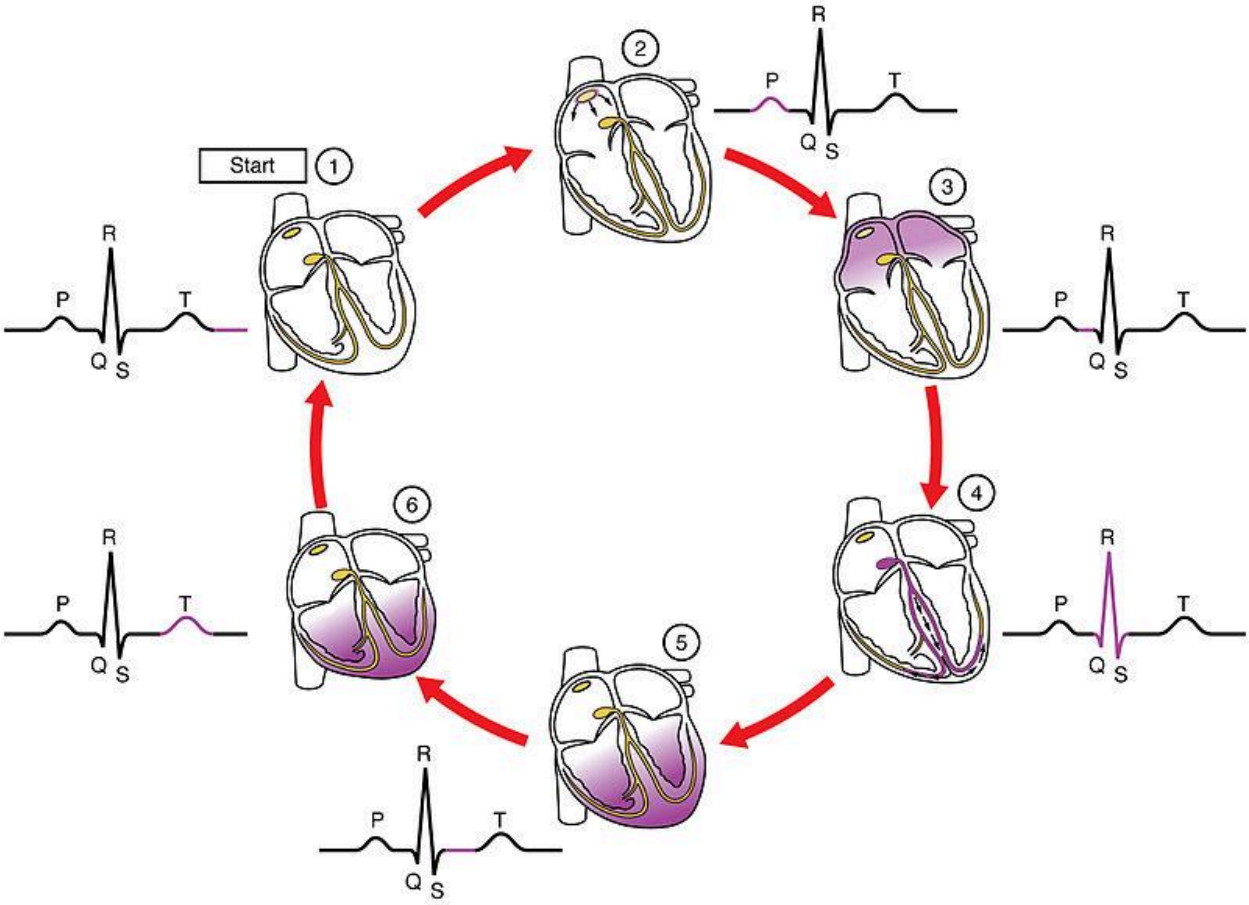
S-T interval

- It is the time between the end of the spread of impulse through ventricles and its repolarization.
- It starts at the end of the S wave and terminates at the beginning of the T wave.
- During acute myocardial infarction, the S-T segment is elevated while it is depressed during the insufficient supply of oxygen to the muscles.

Q-T interval

- The QT interval represents the time of ventricular activity including both depolarization and repolarization.
- It is measured from the beginning of the QRS complex to the end of the T wave.
- The QT interval will vary with the patient's gender, age, and heart rate.
- The normal duration of the Q-T interval is <0.42 seconds.

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Solution:

A. P wave - III. Depolarisation of atria.
B. QRS complex - II. Depolarisation of ventricles.
C. T wave - IV. Repolarisation of ventricles.
D. T-P gap - I. Heart muscles are electrically silent.

Hence, the correct answer is option 2) A-III, B-II, C-IV, D-I

Ques:4 Given below are two statements:

Statement I: The cerebral hemispheres are connected by a nerve tract known as the corpus callosum.

Statement II: The brain stem consists of the medulla oblongata, pons and cerebrum.

In the light of the above statements, choose the most appropriate answer from the options given below:

Option 1) Both Statement I and Statement II are correct.

Option 2) Both Statement I and Statement II are incorrect.

Option 3) Statement I is correct but Statement II is incorrect.

Option 4) Statement I is incorrect but Statement II is correct.

Chapter Name - Neural Control & Coordination
Difficulty - Medium
Concept - Human Brain: Location and Coverings

- The brain is located in the cranial cavity (cranium) of the skull.
- It is enclosed in a bony case called cranium which protects the brain against external injury.
- The brain is surrounded by three membranes called meninges (sing-meninx).
- Outer dura mater is a tough white membrane present below the cranium, closely adhering to the cranium.
- The middle arachnoid layer has a network of fibres.
- Inner diameter is a thin, transparent, and vascular membrane close to the surface of the brain and spinal cord.
- The space between the dura mater and arachnoid is the subdural space.
- The space between the arachnoid and pia mater is known as subarachnoid space.
- These spaces surrounding the brain as well as the cavities within the brain are filled with a fluid.
- Subarachnoid space is filled with fluid called cerebrospinal fluid (CSF).
- The cerebrospinal fluid is also present within the spinal cord.
- The functions of CSF are :

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1. Mechanical cushion to brain
2. Source of nutrition to brain
3. Excretion of metabolic waste products
4. Intra-cerebral transport medium
5. Control of the chemical environment
6. Auto-regulation of intracranial pressure

- The brain structure is composed of three main parts: the forebrain, midbrain, and hindbrain, each with multiple parts.

Solution:

In the human brain, a deep cleft divides the cerebrum longitudinally into two halves, which are termed.
The left and right cerebral hemispheres. The cerebral hemispheres are connected by a tract of nerve.
Fibres are called the corpus callosum.
Three major regions make up the brain stem i.e. midbrain, pons, and medulla oblongata.
The cerebrum is a part of the forebrain that does not form a brain stem.
Options (1), (2) and (4) are incorrect.

Hence, the correct answer is option (3) as statement I is correct but statement II is incorrect.

Ques:5 Match List I with List II :

	List I		**List II**
A	**Exophthalmic goiter**	**I**	**Excess secretion of cortisol, moon face & hypergylcemia.**
B	**Acromegaly**	**II**	**Hypo-secretion of thyroid hormone and stunted growth.**

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C	**Cushing's syndrome**	**II I**	**Hypersecretion of thyroid hormone & protruding eye balls.**
D	**Cretinism**	**I**	**Excessive secretion of growth hormone.**

Choose the correct answer from the options given below :

Option 1) A-I, B-III, C-II, D-IV
Option 2) A-IV, B-II, C-I, D-III
Option 3) A-III, B-IV, C-II, D-I
Option 4) A-III, B-IV, C-I, D-II
Chapter Name - Neural Control & Coordination
Difficulty - Medium
Concept - Hyperthyroidism:

- Hypersecretion of thyroid hormone is also called Exophthalmic goitre Graves' disease or Basedow's disease or Parry's disease.
- It is a thyroid enlargement (goitre) in which the thyroid secretes excessive amounts of thyroid hormone.
- It is characterised by exophthalmia (protrusion of eyeballs because of fluid accumulation behind them), loss of weight, a slight rise in the body temperature, excitability, rapid heartbeat, nervousness and restlessness.

Hypothyroidism:

- Cretinism:
    - This disorder is caused by a deficiency of thyroid hormone in infants.
    - Affected infants have slow body growth and mental development, and reduced metabolic rate
- Myxoedema or Gull's disease:
    - It is caused by a deficiency of thyroid hormone in adults.

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    - This disease is characterized by a puffy appearance due to the accumulation of fat in the subcutaneous tissue because of a low metabolic rate.
    - The patient lacks alertness, intelligence and initiative.
    - He also suffers from slow heartbeat, low body temperature and retarded sexual development.
- Simple Goitre:
    - It is caused by a deficiency of iodine in the diet because iodine is needed for the synthesis of thyroid hormone.
    - It causes thyroid enlargement.
    - It may lead to cretinism or myxoedema.
    - This disease is common in hilly areas.
- Hashimoto's disease:
    - In this disease, all the aspects of thyroid function are impaired.
    - It is an autoimmune disease in which the thyroid gland is destroyed by autoimmunity.

Solution:

A	Exophthalmic goiter	I	Hypersecretion of thyroid hormone and characterized by protruding eyeballs
B	Acromegaly	II	Excessive secretion of growth hormone
C	Cushing's syndrome	III	Excess secretion of cortisol,

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			moon face and hyperglycaemia
D	Cretinism	IV	Hypo-secretion of thyroid hormone and characterized by stunted growth

Hence, the correct answer is 4) A-III, B-IV, C-I, D-II

Ques:6 Match List I with List II :

	List I		List II
A	**Fibrous joints**	**I**	**Adjacent vertebrae, limited movement**
B	**Cartilaginous joints**	**II**	**The humerus and Pectoral girdle, rotational movement**
C	**Hinge joints**	**III**	**Skull, don't allow any movement**
D	**Ball and socket joints**	**IV**	**The knee helps in locomotion**

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Choose the correct answer from the options given below :

Option 1) A-IV, B-II, C-III, D-I

Option 2) A-I, B-III, C-II, D-IV

Option 3) A-II, B-III, C-I, D-IV

Option 4) A-III, B-I, C-IV, D-II

Chapter Name - Locomotion & Movement
Difficulty - Medium
Concept - Gliding Joint: Flat or slightly curved surfaces of bones articulate. Sliding or twisting in various planes is possible. Examples: the joints between the bones of the wrist and between the bones of the ankle.

- Hinge Joint: The convex surface of one bone articulates with the concave surface of another. Up-and-down motion in one plane is possible. Examples: the elbow and knee joints.
- Ball-and-socket joint: The ball-shaped head of one bone fits into the cup-shaped socket of another. Movement in all planes, as well as rotation, is possible. Examples: the shoulder and hip joints.
- Saddle joint: Each bone is saddle-shaped and fits into the complementary regions of the other. A variety of movements are possible. Example: the joint between the carpal and metacarpal bones of the thumb.
- Pivot joint: A small, cylindrical projection of one bone pivots within the ring formed of bone and ligament of another bone. Only rotation is possible. Examples: the joint between the proximal ends of the radius and ulna, and the joint between the atlas and axis.
- Condyloid joint: The oval-shaped condyle of one bone fits into the elliptical cavity of another. Movement in different planes is possible, but rotation is not. Examples: the joints between the metacarpals and phalanges.

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Solution: Fibrous joints do not allow any movement. This type of joint is shown by the flat skull bones which fuse end-to-end with the help of dense fibrous connective tissues in the form of sutures.

- The cartilaginous joint is present between the adjacent vertebrae in the vertebral column and this permits limited movements.
- The hinge joint is a type of synovial joint present in the knee, that helps in locomotion.
- The ball and socket joint is also a type of synovial joint present between the humerus and pectoral girdle and allows rotational movement.

Hence, the correct answer is option 4) A-III, B-I, C-IV, D-II.

Ques:7 Following are the stages of the pathway for the conduction of an action potential through the heart

A. AV bundle

B. Purkinje fibres

C. AV node

D. Bundle branches

E. SA node

Choose the correct sequence of pathways from the options given below

Option 1) E-C-A-D-B

Option 2) A-E-C-B-D

Option 3) B-D-E-C-A

Option 4) E-A-D-B-C

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Chapter Name - Human circulatory system, structure of human heart and blood vessels
Difficulty - Easy
Concept - Conducting system of the human heart

- In the human heart, small groups of specialised neuromuscular cells are present within the myocardium.
- These neuromuscular cells initiate and conduct nerve impulses due to which a coordinated and synchronised contraction of the heart muscles is seen
- The components of the conducting system of the human heart include the sinoatrial node or SA node, the Atrioventricular node or AV node, a Bundle of His and Purkinje fibres

1. Sinoatrial node or SA node-

- It includes a mass of specialised self-excitatory cells present in the wall of the right atrium near the opening of the superior vena cava.
- SA node is also called the pacemaker of the heart because of its involvement in the initiation of heartbeat impulses.
- SA node initiates impulses more rapidly than any other neuromuscular cells of the heart.

2. Atrioventricular node or AV node

- It includes a mass of self-excitatory tissue located in the wall of the atrial septum near the atrioventricular valves.
- Generally, it works on stimulation by the SA node, but it is also capable of initiating its impulses.
- It is also called the pacesetter of the human heart.

3. Bundle of his or atrioventricular bundle

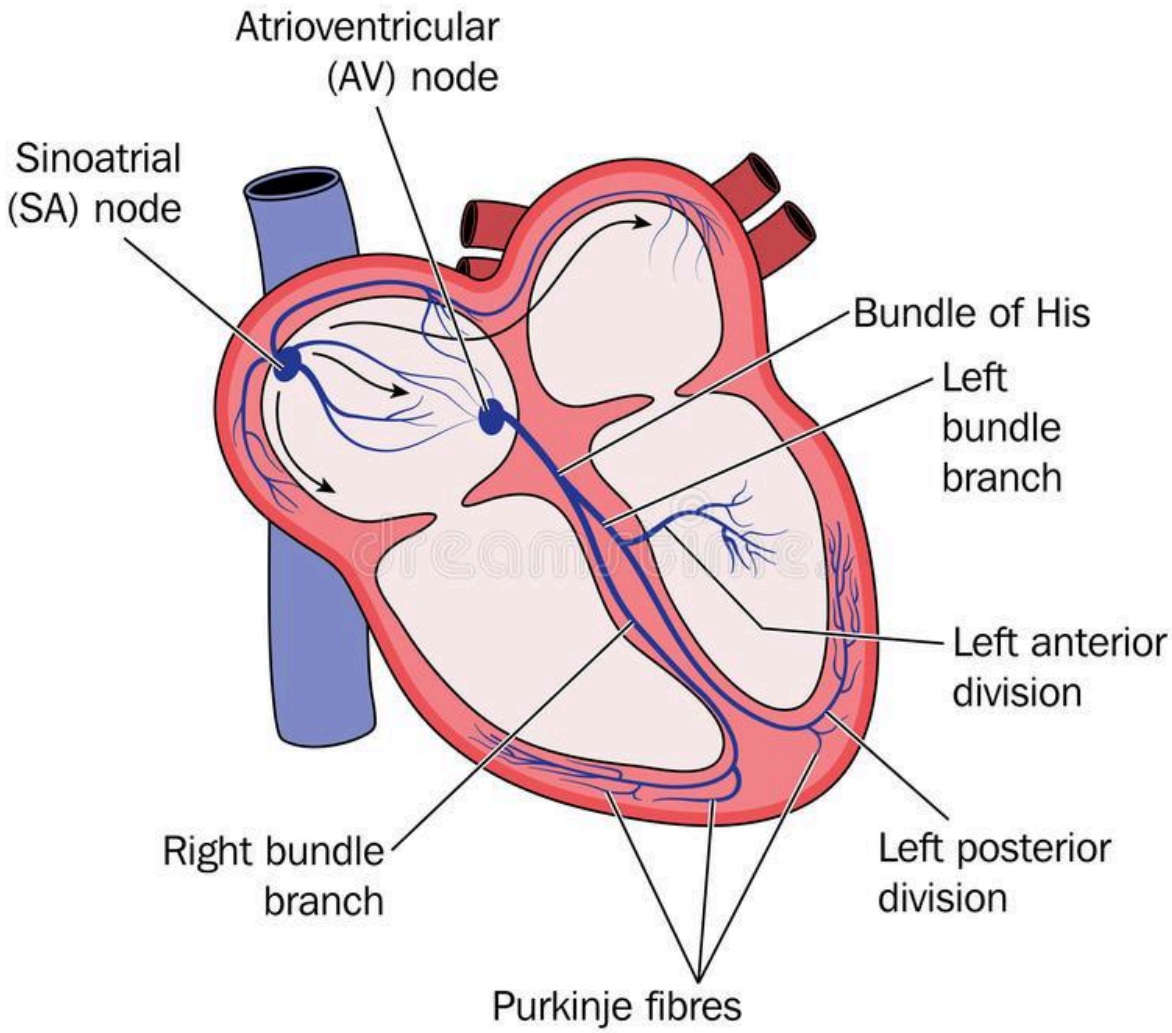
- It is a mass of specialised fibres originating from the AV node.

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- These separate atria from ventricles
- At the upper end of the ventricular septum, it is divided into left and right bundle branches

4. Purkinje fibres

- These are the fine fibres of the AV bundle in the ventricular myocardium
- Both the AV bundle and Purkinje fibres convey impulses of contraction from the AV Node to the apex of the myocardium
- Thus, the AV node and Purkinje fibres bring in ventricular contraction



Solution - E (SA node): The right atrium's sinoatrial (SA) node is where the action potential begins.
C (AV node): After passing through the atrioventricular (AV) node, the impulse experiences a little delay.
The action potential travels from the AV node to the AV bundle, sometimes referred to as the Bundle of His.
D (Bundle branches): The impulse passes through the interventricular septum's right and left bundle branches.
The action potential eventually reaches the Purkinje fibres (B), which disperse the impulse across the ventricles and cause contraction.

Hence, the correct answer is option (1)E-C-A-D-B

Ques:8 Match List I with List II :

	List I		List II
A	**Pons**	**I**	**Provides additional space for Neurons, and regulates posture and balance.**
B	**Hypothalamus**	**II**	**Controls respiration and gastric secretions.**
C	**Medulla**	**III**	**Connects different regions of the brain.**
D	**Cerebellum**	**IV**	**Neuro secretory cells**

Choose the correct answer from the options given below :

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Option 1) A-II, B-III, C-I, D-IV

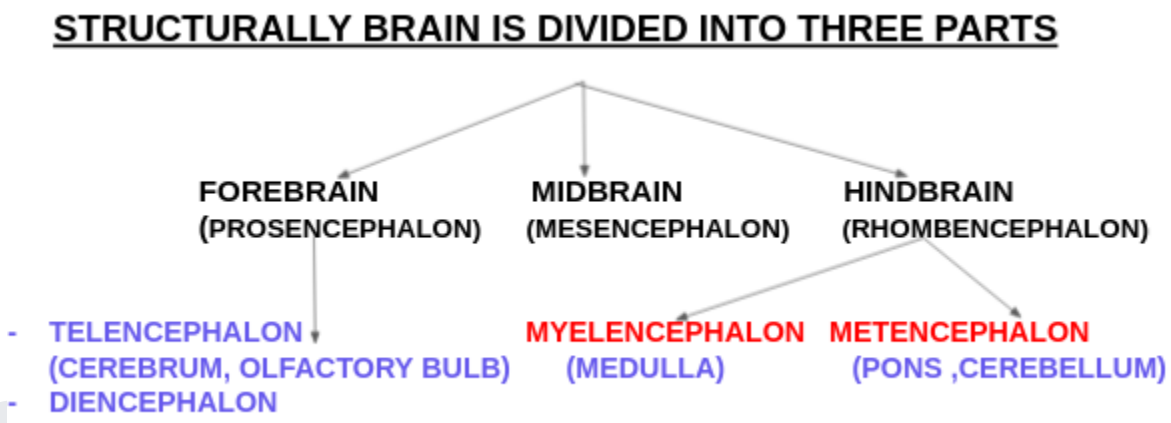
Option 2) A-III, B-IV, C-II, D-I

Option 3) A-I, B-III, C-II, D-IV

Option 4) A-II, B-I, C-III, D-IV

Chapter Name - Neural Control & Coordination
Difficulty - Medium
Concept - The brain is located in the cranial cavity (cranium) of the skull.

- It is enclosed in a bony case called cranium which protects the brain against external injury.
- The brain is surrounded by three membranes called meninges (sing-meninx).
- The brain structure is composed of three main parts: the forebrain, midbrain, and hindbrain, each with multiple parts.



Solution - A. Pons → III. Connects different regions of the brain: The pons facilitate signal transmission and connections between various brain regions.

- B. Hypothalamus → IV. Neurosecretory cells: The hypothalamus contains cells that produce hormones and regulate vital bodily processes like appetite and body temperature.

- C. Medulla → II. Controls respiration and gastric secretions: The medulla aids in digesting and regulates breathing.
- D. Cerebellum → I. Provides additional space for Neurons, and regulates posture and balance: The cerebellum helps with balance and movement, and gives space for brain cells to work.

Hence, the correct answer is option 2)A-III, B-IV, C-II, D-I.

Ques: 9 Which of the following factors are favourable for the formation of oxyhaemoglobin in alveoli?
Option 1) High pO_2 and High pCO_2temperature
Option 2) High pO_2 and Lesser H^+ concentration
Option 3) Low pCO_2 and High H^+ concentration
Option 4) Low pCO_2 and High
Chapter Name - Excretory Products & Their Elimination
Difficulty Level - Medium
Concept - Blood vessels and their system

- The study of vessels is called angiology.
- The blood vessels of a closed circulatory system are of the following three types.

Arteries

- These are elastic vessels that transport blood away from the heart
- The largest artery of the body is the aorta
- The aorta originates from the heart and branches out into smaller arteries.
- The smallest branches are called arterioles, which further branch off into capillaries.
- Arteries carry oxygenated blood except the Pulmonary artery and Umbilical artery.

Veins

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- These are non-elastic blood vessels that transport blood to the heart.
- The smallest veins in the body are called venules.
- These receive blood from the arteries via the capillaries
- Veins carry deoxygenated blood except Pulmonary vein and Umbilical vein

Capillaries

- These are extremely small vessels that are located within the tissues of the body that transport blood from the arteries to the veins.
- Capillaries were discovered by Malpighi in 1661
- The walls of capillaries are thin and consist of only a single layer of endothelium
- The walls of capillaries help to exchange oxygen, carbon dioxide, nutrients and waste

Structure of wall of blood vessels

- The wall of arteries and veins consists of the following three layers

1. Tunica interna- It is the innermost layer consisting of an inner endothelium of long, thin cells of squamous epithelium resting on a basement membrane along with an outer elastic membrane of yellow elastic tissue
2. Tunica media - It is the middle layer which mainly consists of smooth muscle fibres and elastic connective tissue. It is thicker in the artery
3. Tunica externa - It is the outermost layer which is formed of loose connective tissue with longitudinally arranged white and yellow fibres. It is also known as tunica adventitia

Solution - Conditions favourable for the formation of oxyhaemoglobin in alveoli are high pO_2 and less H^+ concentration.
Low pCO_2 and low temperature. A lower concentration of hydrogen ions (higher pH) stabilizes haemoglobin's affinity for oxygen, enhancing oxygen uptake. The low partial pressure of carbon dioxide (pCO_2) reduces competition for haemoglobin binding, further favouring oxygenation. A lower temperature helps maintain haemoglobin's

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oxygen-binding capacity, facilitating efficient oxygen transport in the blood. These conditions collectively ensure optimal oxygen loading onto haemoglobin for distribution to body tissues.
Option (1), (3) and (4) are not correct as they do not favour the formation of oxyhaemoglobin.

Hence, the correct answer is option 2) High pO_2 and Lesser H^+ concentration

Ques: 10 Match List-I with List-II

List - I	List - II
A. GLUT- 4	**I. Hormone**
B. Insulin	**II. Enzyme**
C. Trypsin	**III. Intercellular ground substance**
D. Collagen	**IV. Enables glucose transport into cells**

Choose the correct answer from the options given below.

Option 1) A-IV, B-I, C-II, D-III

Option 2) A-I, B-II, C-III, D-IV

Option 3) A-II, B-III, C-IV, D-I

Option 4) A-III, B-IV, C-I, D-II

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Chapter Name - Chemical Control & Integration

Difficulty Level - Medium

Concept - The pancreas is a soft, oblong, glandular structure that lies transversely in the retroperitoneum behind the stomach.

- Its head is present in the C-shaped curvature of the duodenum.
- The pancreas is a heterocrine gland. It is a mix of exocrine (secreting digestive enzymes) and endocrine (releasing hormones into the blood) functions.
- It is covered by the serosa or serous capsule formed by the external visceral peritoneum and internal dense connective tissue.
- The main pancreatic duct, called the **duct of Wirsung**, is formed from smaller ducts within the pancreas.
- The main pancreatic duct opens into the hepatopancreatic ampulla.
- The exocrine part of the pancreas is made up of rounded lobules called the acini.
- These secrete alkaline pancreatic juice with a pH of 8.4
- The pancreatic juice contains:
- Sodium bicarbonate
- Proenzymes: trypsinogen, chymotrypsinogen and procarboxypeptidase
- Enzymes: pancreatic amylase, DNAse, RNAse, pancreatic lipase

Solution - GLUT-4 is a glucose transporter responsible for enabling glucose to enter cells, particularly in muscle and adipose tissues. Insulin, a hormone, regulates blood glucose levels by facilitating the uptake of glucose into cells. Trypsin, an enzyme, plays a crucial role in the digestion of proteins by breaking down peptide bonds in the small intestine. Collagen, a structural protein, is an important component of the intercellular ground substance, providing strength and support to connective tissues in the body.

Hence, the correct answer is option (1)

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Ques:1 Which of the following statements are correct regarding skeletal muscle?

A. Muscle bundles are held together by a collagenous connective tissue layer called a fascicle.

B. The sarcoplasmic reticulum of muscle fibre is a storehouse of calcium ions.

C. Striated appearance of skeletal muscle fibre is due to the distribution pattern of acting and myosin proteins.

D. M line is considered a functional unit of contraction called sarcomere.

Choose the most appropriate answer from the options given below:

Option 1) A, B and C only

Option 2) B and C only

Option 3) A, C and D only

Option 4) C and D only

Chapter Name - Locomotion & Movement

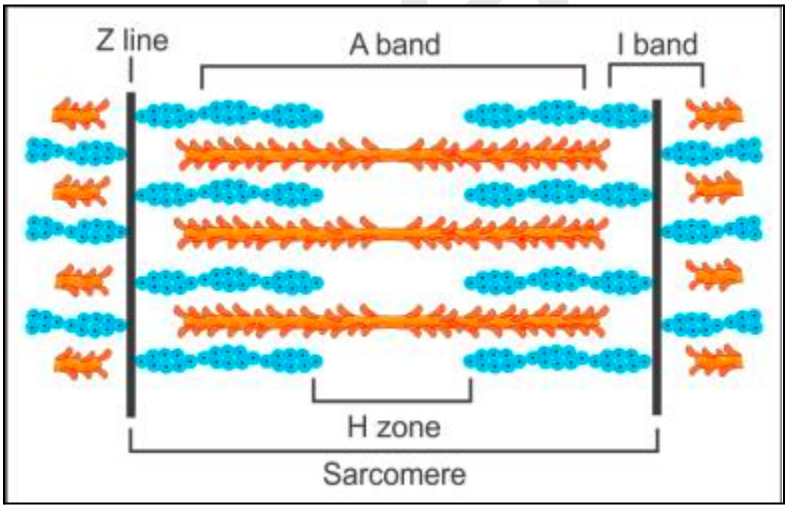
Difficulty Level - Medium

Concept - Sarcomere

- The smallest unit of the skeletal muscle is called the sarcomere.
- It is the structural and functional unit of the myofibril
- The sarcomere consists of the thick and thin filaments.
- The thin filament is made up of the globular protein called actin.
- The thick filament is made up of a protein called myosin.

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- The light bands contain actin and are called the I-band or Isotropic band, whereas the dark band called 'A' or Anisotropic band contains myosin.
- Both proteins are arranged as rod-like structures, parallel to each other and also to the longitudinal axis of the myofibrils.
- In the centre of each 'I' band is an elastic fibre called the 'Z' line which bisects it.
- The thin filaments are firmly attached to the 'Z' line.
- The thick filaments in the 'A' band are also held together in the middle of this band by a thin fibrous membrane called the 'M' line.
- The 'A' and 'I' bands are arranged alternately throughout the length of the myofibrils.
- The portion of the myofibril between two successive 'Z' lines is considered the functional unit of contraction and is called a sarcomere. The length of a sarcomere is about 2 -3 microns.
- In a resting state, the edges of thin filaments on either side of the thick filaments partially overlap the free ends of the thick filaments leaving the central part of the thick filaments.
- This central part of the thick filament, not overlapped by thin filaments is called the 'H' zone.



Solution - Statement B is correct. The sarcoplasmic reticulum of muscle fibres is a specialised type of endoplasmic reticulum that stores and releases calcium ions (Ca^{2+}) during muscle contraction.

Statement C is correct. The striated appearance of skeletal muscle fibres is due to the arrangement and distribution pattern of actin and myosin proteins, which form the contractile units called sarcomeres.

For incorrect options,

Statement A is incorrect. Muscle bundles in skeletal muscle are held together by a collagenous connective tissue layer called perimysium, not fascicle.

Statement D is incorrect. The M line is a structural component of the sarcomere, but the functional unit of contraction in skeletal muscle is the sarcomere itself, not specifically the M line.

Hence, the correct answer is Option 2) B and C only.

Ques: 2 The parts of the human brain that help in the regulation of sexual behaviour, expression of excitement, pleasure, rage, fear, etc are.

Option 1) Limbic system and hypothalamus

Option 2) Corpora quadrigemina and hippocampus

Option 3) Brain stem and epithalamus

Option 4) Corpus callosum and thalamus

Chapter Name - Neural Control & Coordination

Difficulty Level - Medium

Concept - Limbic System

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- The limbic system is the collective name for the parts of the brain that control emotions, motivation, and emotional association with memory.
- Certain components of the cerebrum and diencephalon constitute the limbic system.
- These components are discussed below:

1. Hippocampus:

- It is located inside the temporal lobe.
- It is an essential structure for learning and memory.
- It converts short-term into long-term memory.

2. Amygdala or Amygdaloid nucleus:

- Almond-shaped structure located in the tip of the temporal lobe.
- The amygdala is involved in the experience of emotion and in tying emotional meaning to memories.

3. Septal nuclei:

- These are located within the septal area formed by the regions under the corpus callosum

4. Mammillary bodies:

- These are present behind the infundibulum.

5. Basal ganglia:

- Basal ganglia are the scattered masses of grey matter
- These are submerged in the subcortical substance of cerebral hemispheres.
- It consists of the corpus striatum and lenticular nucleus.
- Parkinson's disease occurs due to damage to the basal ganglia.
- Wilson's disease is due to damage to the lenticular nucleus.
- It is responsible for the following:

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-control of the movements during voluntary motor activity
-control of reflex muscular activity
-control of automatic associated movements

Solution - The limbic system and hypothalamus are the parts of the human brain that play a crucial role in regulating sexual behaviour and expressing emotions such as excitement, pleasure, rage, fear, and more.

The limbic system is a collection of structures located in the cerebral hemispheres of the brain. It includes the hippocampus, amygdala, and parts of the thalamus and hypothalamus. The limbic system is involved in various functions, including emotion, behaviour, motivation, and memory.

The hypothalamus is a small but important region located below the thalamus. It serves as a control centre for many autonomic functions and behaviours, including regulation of body temperature, hunger, thirst, and sexual behaviour. The hypothalamus plays a key role in the release of hormones involved in sexual arousal and reproductive functions.

Hence, the correct answer is option 1) Limbic system and hypothalamus.

Ques: 3 Which of the following statements is correct?

A. An excessive loss of body fluid from the body switches off osmoreceptors.

B. ADH facilitates water reabsorption to prevent diuresis.

C. ANF causes vasodilation.

D. ADH causes an increase in blood pressure.

E. ADH is responsible for the decrease in GFR.

Choose the correct answer from the options given below:

Option 1) A and B only

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Option 2) B, C, and D only

Option 3) A, B and E only

Option 4) C, D and E only

Concept - Regulation of Kidney Function -

- The functioning of the kidneys is efficiently monitored and regulated by hormonal feedback mechanisms involving the hypothalamus, JGA, and to a certain extent, the heart.

Control by ADH:

- Osmoreceptors in the body are activated by changes in blood volume, body fluid volume, and ionic concentration.
- An excessive loss of fluid from the body can activate these receptors which stimulate the hypothalamus to release antidiuretic hormone (ADH) or vasopressin from the neurohypophysis.
- ADH facilitates water reabsorption from the latter parts of the tubule, thereby preventing diuresis.
- An increase in body fluid volume can switch off the osmoreceptors and suppress the ADH release to complete the feedback.
- ADH can also affect kidney function by its constrictor effects on blood vessels. This causes an increase in blood pressure.
- An increase in blood pressure can increase the glomerular blood flow and thereby the GFR.

Control by Juxtaglomerular Apparatus:

- The JGA plays a complex regulatory role.
- A fall in glomerular blood flow/glomerular blood pressure/GFR can activate the JG cells to release renin which converts angiotensinogen in the blood to angiotensin I and further to angiotensin II.

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- Angiotensin II, being a powerful vasoconstrictor, increases glomerular blood pressure and thereby GFR.
- Angiotensin II also activates the adrenal cortex to release Aldosterone.
- Aldosterone causes the reabsorption of Na+ and water from the distal parts of the tubule.
- This also leads to an increase in blood pressure and GFR.
- This complex mechanism is generally known as the Renin-Angiotensin-Aldosterone mechanism.

Control by ANF:

- An increase in blood flow to the atria of the heart can cause the release of Atrial Natriuretic Factor (ANF).
- ANF can cause vasodilation (dilation of blood vessels) and thereby decrease the blood pressure.
- ANF mechanism, therefore, acts as a check on the renin-angiotensin mechanism.

Solution -

B. ADH facilitates water reabsorption to prevent diuresis: This statement is correct. Antidiuretic hormone (ADH), also known as vasopressin, acts on the kidneys to increase water reabsorption. It helps prevent excessive water loss and promotes water retention in the body.

C. ANF causes vasodilation: This statement is correct. Atrial natriuretic peptide (ANF) is a hormone released by the heart in response to increased blood volume and pressure. It promotes vasodilation, which leads to the relaxation of blood vessels and increased excretion of sodium and water.

D. ADH causes an increase in blood pressure: This statement is correct. ADH acts on the blood vessels, causing vasoconstriction. This constriction increases peripheral vascular resistance, which can contribute to an increase in blood pressure.

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For incorrect options,

A. An excessive loss of body fluid from the body switches off osmoreceptors: This statement is incorrect. Osmoreceptors are specialized cells that detect changes in osmotic pressure and regulate the release of antidiuretic hormone (ADH). Excessive loss of body fluid would lead to increased osmotic pressure and activate osmoreceptors, not switch them off.

E. ADH is responsible for a decrease in GFR (glomerular filtration rate): This statement is incorrect. ADH does not directly influence the glomerular filtration rate. It primarily affects water reabsorption in the renal tubules.

Hence, the correct answer is option 2) B, C, and D only.

Ques:4 Which of the following are NOT under the control of thyroid hormone?

A. Maintenance of water and electrolyte balance
B. Regulation of basal metabolic rate
C. Normal rhythm of sleep-wake cycle
D. Development of the immune system
E. Support the process of RBC formation.

Choose the correct answer from the options given below:

Option 1)A and D only

Option 2)B and C only

Option 3)C and D only

Option 4)D and E only

Concept - Thyroid Gland

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- The thyroid gland is a butterfly-shaped organ and is composed of two cone-like lobes called the right lobe and left lobe, connected via the isthmus.
- It develops from the endoderm.
- It is located anterior to the thyroid cartilage of the larynx of the neck.
- It is composed of thyroid follicles made up of cuboidal epithelium and filled with homogeneous material called colloid.
- The stroma of the thyroid gland is made up of loose connective tissue.
- The stroma contains blood capillaries and parafollicular cells or C cells.
- It is the only gland that stores hormones in large quantities.

Solution - Thyroid hormone primarily influences the regulation of metabolism and growth in the body. It affects various physiological processes, but not all processes are directly controlled by thyroid hormone. Let's go through each option:

C. Normal rhythm of sleep-wake cycle: The sleep-wake cycle is primarily regulated by the hormone melatonin, which is produced by the pineal gland. Thyroid hormone does not directly control the sleep-wake cycle.

D. Development of the immune system: The development of the immune system is not directly controlled by thyroid hormone. It is regulated by various other factors and immune cells.

For incorrect options,

A. Maintenance of water and electrolyte balance: This process is not directly regulated by thyroid hormones. It is primarily controlled by hormones such as antidiuretic hormone (ADH) and aldosterone.

B. Regulation of basal metabolic rate: Thyroid hormone plays a crucial role in regulating the basal metabolic rate (BMR). It affects the rate at which the body utilizes energy.

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E. Support the process of RBC formation: The production of red blood cells (RBCs) is primarily controlled by erythropoietin, a hormone produced by the kidneys. Thyroid hormone does not directly support the process of RBC formation.

Hence, the correct answer is option 3) C and D only.

Ques: 5 Match List I with List II.

	List I		**List II**
A.	**P-wave**	**I.**	**Beginning of systole**
B.	**Q -wave**	**II.**	**Repolarisation of ventricles**
C.	**QRS complex**	**III.**	**Depolarisation of atria**
D.	**T-wave**	**IV.**	**Depolarisation of ventricles**

Choose the correct answer from the options given below:

Option 1) A-III, B-I, C-IV, D-II

Option 2) A-IV, B-III, C-II, D-I

Option 3) A-II, B-IV, C-I, D-III

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Option 4) A-I, B-II, C-III, D-IV

Concept - ECG

- The representation of the electric current produced by the excitation of the cardiac muscles in the form of a graph is called an electrocardiogram (ECG)
- The machine by which the electrocardiogram is recorded is called an electrocardiograph.
- The electrocardiograph was invented by Einthoven and he is known as the father of electrocardiography.
- An electrocardiogram shows three consecutive waves namely the P wave, the QRS wave, and the T wave.

1. P wave-

- It is a small upward wave that indicates the depolarisation of the atria leading to contraction of both atria. It is caused by the activation of the SA node.

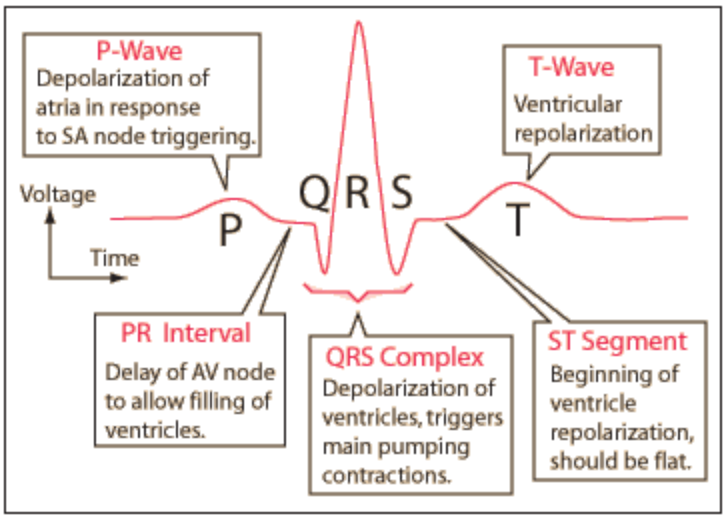
2. QRS wave

- It begins after a fraction of a second of the P wave
- It begins as a small downward deflection (Q) and continues as a large upright (R) and triangular wave, ending as a downward wave (S) at its base
- QRS wave indicates depolarization of ventricles
- It is caused by the impulses of the contraction from the AV node through the Bundle of His and Purkinje fibres and the contraction of ventricular muscles
- In this wave, there is a spread of electrical impulses through ventricles
- The normal duration of the QRS complex is <0.10 seconds

3. T wave

- The T wave is dome-shaped and it indicates ventricular repolarization (ventricular relaxation)
- The end of T waves marks systole

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P-R interval

- It is the time required for an impulse to travel through the atria and AV node to the remaining conductive tissue.
- The normal duration of the P-R interval is <0.12 to 0.2 seconds
- The P-R interval is lengthened during conditions like arteriosclerotic heart disease and rheumatic fever due to the inflammation of atria and AV node

S-T interval

- It is the time between the end of the spread of impulse through ventricles and its repolarization.
- It starts at the end of the S wave and terminates at the beginning of the T wave.
- During acute myocardial infarction, the S-T segment is elevated while it is depressed during the insufficient supply of oxygen to the muscles.

Q-T interval

- The QT interval represents the time of ventricular activity including both depolarization and repolarization.
- It is measured from the beginning of the QRS complex to the end of the T wave.
- The QT interval will vary with the patient's gender, age, and heart rate.

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- The normal duration of the Q-T interval is <0.42 seconds.

Solution - A. P-wave: The P-wave represents the depolarization (contraction) of the atria. It corresponds to the beginning of systole, which is the contraction phase of the cardiac cycle. Therefore, the correct pairing is A-III.

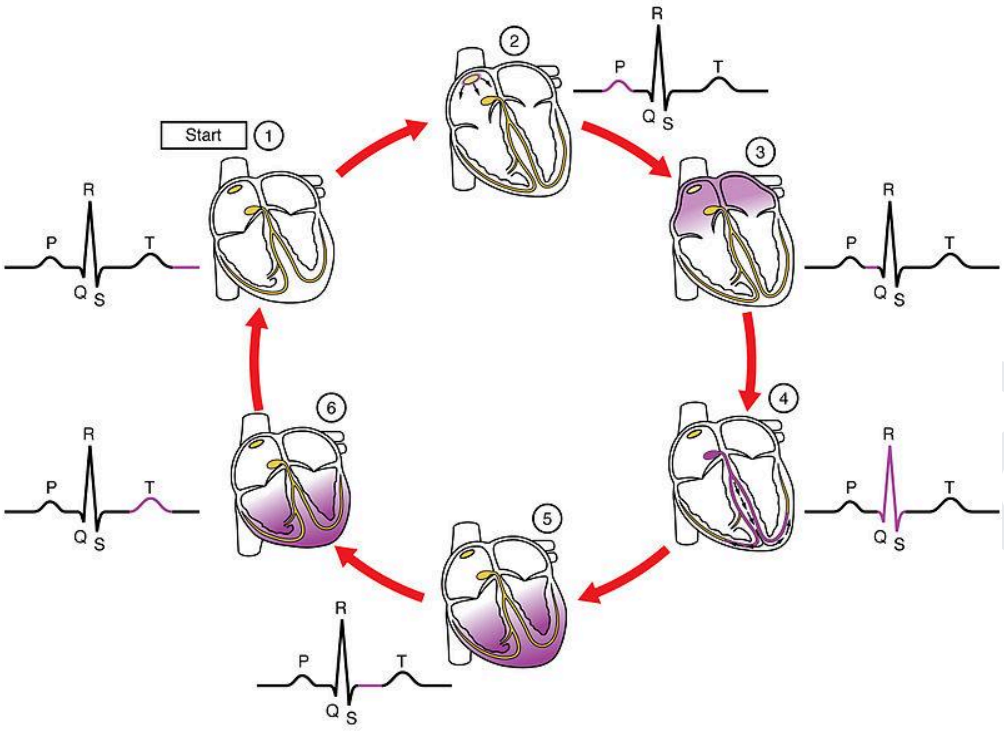
B. Q-wave: The Q-wave is a small downward deflection that appears before the R-wave in the QRS complex. It doesn't have a specific physiological event associated with it. Therefore, the correct pairing is B-I.

C. QRS complex: The QRS complex represents the depolarization (contraction) of the ventricles. It is the largest and most prominent wave complex on the ECG. It occurs during the ventricular systole phase, which is the main pumping phase of the cardiac cycle. Therefore, the correct pairing is C-IV.

D. T-wave: The T-wave represents the repolarization (relaxation) of the ventricles. It occurs during the diastole phase, which is the relaxation phase of the cardiac cycle. Therefore, the correct pairing is D-II.

Hence, the correct answer is option 1)A-III, B-I, C-IV, D-II.

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Solution - A. P-wave: The P-wave represents the depolarization (contraction) of the atria. It corresponds to the beginning of systole, which is the contraction phase of the cardiac cycle. Therefore, the correct pairing is A-III.

B. Q-wave: The Q-wave is a small downward deflection that appears before the R-wave in the QRS complex. It doesn't have a specific physiological event associated with it. Therefore, the correct pairing is B-I.

C. QRS complex: The QRS complex represents the depolarization (contraction) of the ventricles. It is the largest and most prominent wave complex on the ECG. It occurs during the ventricular systole phase, which is the main pumping phase of the cardiac cycle. Therefore, the correct pairing is C-IV.

D. T-wave: The T-wave represents the repolarization (relaxation) of the ventricles. It occurs during the diastole phase, which is the relaxation phase of the cardiac cycle. Therefore, the correct pairing is D-II.

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Hence, the correct answer is option 1)A-III, B-I, C-IV, D-II.

Ques: 6 Match List I with List II.

	List I		List II
A.	**Taenia**	**I.**	**Nephridia**
B.	**Paramoecium**	**II.**	**Contractile vacuole**
C.	**Periplaneta**	**III.**	**Flame cells**
D.	**Pheretima**	**IV.**	**Urecose gland**

Choose the correct answer from the options given below:

Option 1)A-I, B-II, C-III, D-IV

Option 2)A-I, B-II, C-IV, D-III

Option 3)A-III, B-II, C-IV, D-I

Option 4)A-II, B-I, C-IV, D-III

Chapter -Excretory Products & Their Elimination

Difficulty -Medium

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Concept-Excretory Structures in Animals -

- Protonephridia or flame cells are the excretory structures in Platyhelminthes (Flatworms, e.g., Planaria), rotifers, some annelids, and the cephalochordate – Amphioxus.
- Protonephridia are primarily concerned with ionic and fluid volume regulation, i.e., osmoregulation.
- Nephridia are the tubular excretory structures of earthworms and other annelids.
- Nephridia helps to remove nitrogenous wastes and maintain a fluid and ionic balance.
- Malpighian tubules are the excretory structures of most of the insects including cockroaches.
- Malpighian tubules help in the removal of nitrogenous wastes and osmoregulation.
- Coxal glands are found in arachnids for excretion.
- Antennal glands or green glands perform the excretory function in crustaceans like prawns.
- Molluscs show Organs of Bojanus and Keber's organs for excretion.
- Neural glands are present in urochordates such as Herdmania for excretion.

Solution-A. Taenia → III. For excretion and osmoregulation, Taenia, a kind of flatworm, contains flame cells.

B. Paramoecium → II. The contractile vacuole of Paramoecium, a type of protozoan, is used to control water balance and eliminate surplus water.

C. Periplaneta → IV. Urecose gland: Urecose glands are used by cockroaches to excrete waste, such as uric acid.

D. Pheretima → I. Nephridia: The earthworm Pheretima excretes waste from the body by using nephridia.

Hence, the correct answer is option 3)A-III, B-II, C-IV, D-I.

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Ques 7: Once the undigested and unabsorbed substances enter the caecum, their backflow is prevented by

Option 1)Sphincter of Oddi

Option 2)Ileo - caecal valve

Option 3)Gastro-oesophageal sphincter

Option 4)Pyloric sphincter

Chapter-Digestion & Absorption

Difficulty-Medium

Concept-The undigested and unabsorbed substances are passed on to the large intestine.

- No significant digestive activity occurs in the large intestine.
- The functions of the large intestine are:
    - absorption of some water, minerals and certain drugs;
    - secretion of mucus which helps in adhering the waste (undigested) particles together and lubricating it for an easy passage
- The undigested, unabsorbed substances called faeces enter into the caecum of the large intestine through the ileocaecal valve, which prevents the backflow of the faecal matter.
- It is temporarily stored in the rectum until defecation.

Solution - The ileo-caecal valve is a one-way valve located between the ileum (the last part of the small intestine) and the caecum (the first part of the large intestine). It prevents the backflow of undigested and unabsorbed substances from the caecum back into the ileum, ensuring that the materials move in the correct direction through the digestive system.

Hence the correct answer is option 2) ileocaecal valve

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Ques 8: Assertion (A): Nephrons are of two types: Cortical & Juxta medullary, based on their relative position in the cortex and medulla.

Reason (R): Juxta medullary nephrons have a short loop of Henle whereas, cortical nephrons have a longer loop of Henle.

In the light of the above statements, choose the correct answer from the options given below:

Option 1)Both A and R are true and R is the correct explanation of A.

Option 2)Both A and R are true but R is NOT the correct explanation of A.

Option 3)A is true but R is false.

Option 4)A is false but R is true.

Chapter- Excretory Products & Their Elimination

Difficulty- Difficult

Concept-Types of Nephrons -

Juxtamedullary Nephrons:

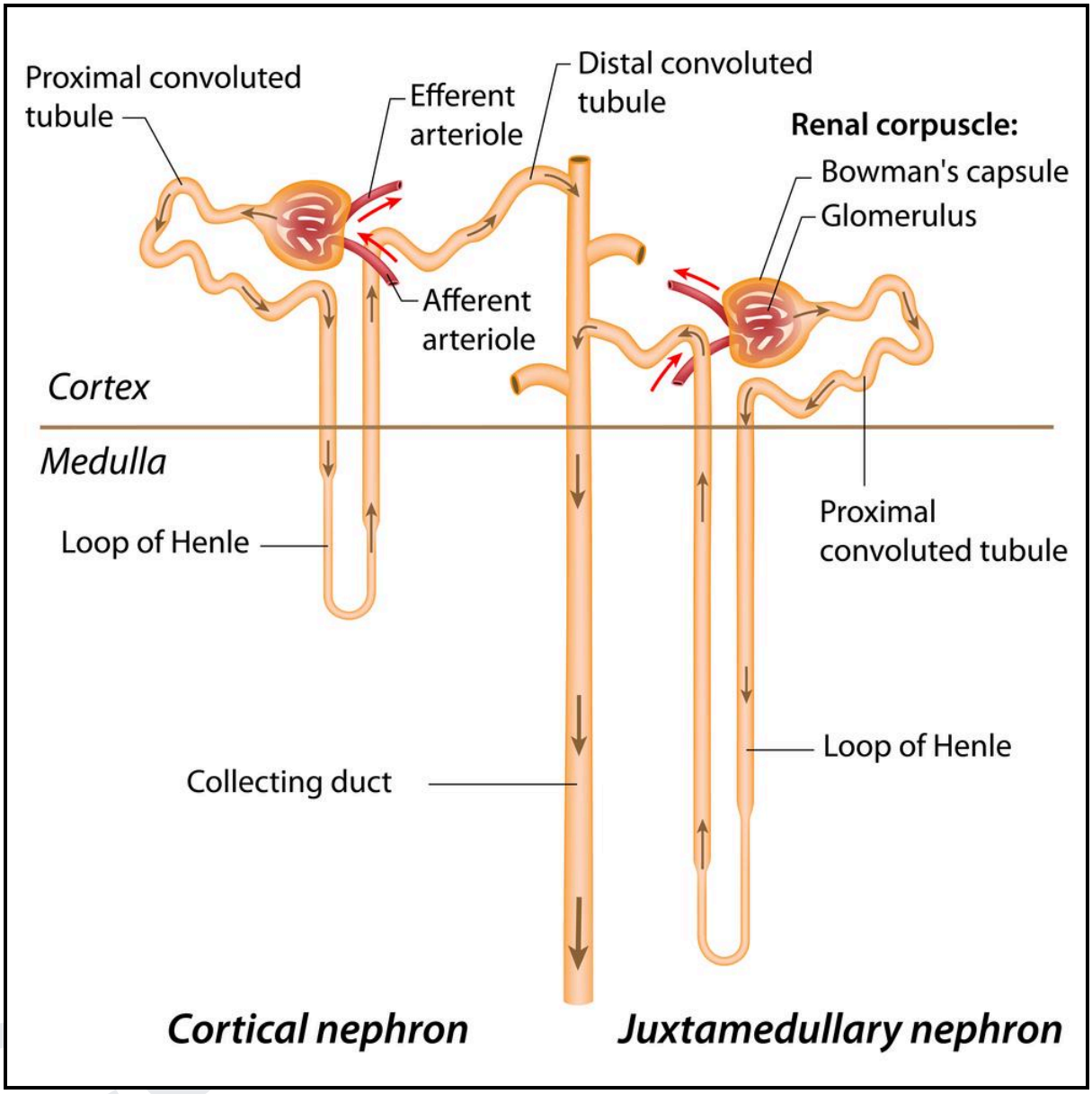
- They form about 15% of the total nephrons.
- Their glomeruli are found at the junction of the cortex and medulla.
- These are larger in size and the loop of Henle is present deeper into the medulla.
- The loop of Henle is associated with the vasa recta.
- They are responsible for controlling the volume of plasma when the water supply is short.

Cortical Nephrons:

- They form about 85% of the total nephrons.
- They lie in the renal cortex. The glomeruli are present in the outer cortex.

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- The loop of Henle is short and extends a short distance into the medulla.
- **The vasa recta are absent.**



Solution-Based on their relative locations in the cortex and medulla, nephrons can be classified as either cortical or juxtamedullary, according to Assertion A.

But Reason R is untrue. In contrast to cortical nephrons, which have a shorter loop of Henle that does not extend as far into the medulla, juxtamedullary nephrons have a larger loop of Henle that extends further into the medulla. The loop of Henle in cortical

and juxtamedullary nephrons is arranged in the exact opposite way as stated in Reason R.

Hence, the correct answer is option 3) which is A is true but R is false.

Ques 9: Match List I with List II.

	List I (Type of Joint)		**List II (Found between)**
A.	**Cartilaginous Joint**	**I.**	**Between flat skull bones**
B.	**Ball and Socket Joint**	**II.**	**Between adjacent vertebrae in the vertebral column**
C.	**Fibrous Joint**	**III.**	**Between the carpal and metacarpal of the thumb**
D.	**Saddle Joint**	**IV.**	**Between Humerus and Pectoral girdle**

Choose the correct answer from the options given below:

Option 1)A-III, B-I, C-II, D-IV

Option 2)A-II, B-IV, C-I, D-III

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Option 3)A-I, B-IV, C-III, D-II

Option 4)A-II, B-IV, C-III, D-I

Chapter- Locomotion & Movement

Difficulty- Medium

Concept-Gliding Joint: Flat or slightly curved surfaces of bones articulate. Sliding or twisting in various planes is possible. Examples: the joints between the bones of the wrist and between the bones of the ankle.

- Hinge Joint: The convex surface of one bone articulates with the concave surface of another. Up-and-down motion in one plane is possible. Examples: the elbow and knee joints.
- Ball-and-socket joint: The ball-shaped head of one bone fits into the cup-shaped socket of another. Movement in all planes, as well as rotation, is possible. Examples: the shoulder and hip joints.
- Saddle joint: Each bone is saddle-shaped and fits into the complementary regions of the other. A variety of movements are possible. Example: the joint between the carpal and metacarpal bones of the thumb.
- Pivot joint: A small, cylindrical projection of one bone pivots within the ring formed of bone and ligament of another bone. Only rotation is possible. Examples: the joint between the proximal ends of the radius and ulna, and the joint between the atlas and axis.
- Condyloid joint: The oval-shaped condyle of one bone fits into the elliptical cavity of another. Movement in different planes is possible, but rotation is not. Examples: the joints between the metacarpals and phalanges.

Solution -

A. Cartilaginous Joint is found between flat skull bones.

B. Ball and Socket Joint is found between the Humerus and Pectoral girdle.

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C. Fibrous Joint is found between the carpal and metacarpal of the thumb.
D. Saddle Joint is found between adjacent vertebrae in the vertebral column.

Therefore, the correct match is:
A-II, B-IV, C-I, D-III.

Hence, the correct answer is Option 2) A-II, B-IV, C-I, D-III.

Ques 10: Match List I with List II with respect to the human eye.

	List I		List II
A.	**Fovea**	**I.**	**The visible coloured portion of the eye regulates the diameter of the pupil.**
B.	**Iris**	**II.**	**The external layer of the eye is formed of dense connective tissue.**
C.	**Blind spot**	**III.**	**Point of greatest visual acuity or resolution.**
D.	**Sclera**	**IV.**	**leaves the eyeball and photoreceptor cells are absent.**

Choose the correct answer from the options given below :

Option 1)A–III, B–I, C–IV, D–II

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Option 2)A-IV, B–III, C–II, D–I

Option 3)A–I, B–IV, C–III, D–II

Option 4)A-II, B–I, C–III, D–IV

Chapter- Neural Control & Coordination

Difficulty- Difficult

Concept- The inner layer of the eyeball is called the retina.

- From outside to inside, it is made up of pigmented epithelium, photoreceptor cells, bipolar cells and ganglion cells.
- The photoreceptor cells are Rods and Cones.
- The photosensitive pigment of rods is called rhodopsin (a derivative of vitamin A)
- Rods enable night vision. Cones are responsible for colour vision and are of three types depending on the photopigments:
    - The erythrosine in the red-sensitive cones.
    - The chloropicrin in the green-sensitive cones.
    - The cyanopsin in the blue-sensitive cones.
- Cone cells contain the pigment iodopsin.
- The optic nerve leaves the eye from a point called a blind spot or optic disc.
- The blood vessels also leave and enter via the blind spot.
- There is no photoreception at the blind spot so it is not involved in vision.
- At the posterior pole of the eye, a bit inferior-lateral to the blind spot, there is a yellowish spot called macula lutea with a central depression called fovea centralis.
- The fovea is rich in cone cells and is the maximum visual resolution point.
- The following diagram depicts the arrangement of various layers of cells in the retina.

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Solution - A. Fovea - III. Point of greatest visual acuity or resolution
The fovea is a small depression in the eye's retina that contains a high concentration of cone cells, which are responsible for detailed and colour vision. It is the area of the retina with the highest visual acuity.

B. Iris - I. The visible coloured portion of the eye that regulates the diameter of the pupil
The iris is the coloured part of the eye surrounding the pupil. It controls the size of the pupil and regulates the amount of light entering the eye.

C. Blind spot - IV. Leaves the eyeball, and photoreceptor cells are absent
The blind spot, also known as the optic disc, is the area on the retina where the optic nerve leaves the eye. It does not contain photoreceptor cells, so there is no vision in that area.

D. Sclera - II. The external layer of the eye is formed of dense connective tissue
The sclera is the tough, white, outer layer of the eye that forms the protective covering of the eyeball. It is composed of dense connective tissue.

Hence, the correct matching is Option 1. A–III, B–I, C–IV, D–II.

Ques 11: Match List I with List II.

	List I (cells)		**List II (Secretion)**
A.	Peptic cells	I.	Mucus
B.	Goblet cells	II.	Bile juice

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C.	Oxyntic cells	III.	Proenzyme pepsinogen
D.	Hepatic cells	IV.	HCl and intrinsic factor for absorption of vitamin B12

Choose the **correct** answer from the options given below:

Option 1)A-IV, B–III, C–II, D–I

Option 2)A-II, B–I, C–III, D–IV

Option 3)A–III, B–I, C–IV, D–II

Option 4)A-II, B–IV, C–I, D–III

Chapter -Human Physiology

Difficulty -Medium

Concept -The mucosa consists of an epithelium, the lamina propria underneath, and a thin bit of smooth muscle called the muscularis mucosae.

- The submucosa lies under this and consists of fibrous connective tissue.
- The epithelium of the stomach forms deep pits called fundic oxyntic or gastric glands.
- The cells of the gastric glands are as follows:
    - The cells at the base of these pits are **chief cells zymogenic cells or peptic cells** that are responsible for the production of pepsinogen, an inactive precursor of pepsin, which degrades proteins.

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    - Further up the pits, **parietal cells or oxyntic cells** produce gastric acid and a vital substance, an intrinsic factor. The gastric acid kills most of the bacteria in food, stimulates hunger, and activates pepsinogen into pepsin. It denatures the complex protein molecule as a precursor to protein digestion through enzyme action in the stomach and small intestines. The intrinsic factor helps in the absorption of Vit B12.
    - Near the top of the pits, closest to the contents of the stomach, there are mucus-producing cells called **goblet cells** that help protect the stomach from self-digestion.
    - The **G-cells or enteroendocrine cells** secrete gastrin hormone.
    - The **argentaffin cells** secrete serotonin, somatostatin, and histamine.
- The muscularis externa is made up of three layers of smooth muscle as follows:
    - The innermost layer is obliquely oriented: this is not seen in other parts of the digestive system. This layer is responsible for creating the motion that churns and physically breaks down the food.
    - The next layer is made up of circular muscles.
    - It is followed by longitudinal muscles.

Solution - A. Peptic cells - III. Proenzyme pepsinogen
Peptic cells secrete the proenzyme pepsinogen, which is later activated by the enzyme pepsin and helps in the digestion of proteins.

B. Goblet cells - I. Mucus
Goblet cells secrete mucus, which helps in lubricating and protecting the epithelial lining of various organs.

C. Oxyntic cells - IV. HCl and intrinsic factor for absorption of vitamin B12
Oxyntic cells, also known as parietal cells, secrete hydrochloric acid (HCl) to aid in the digestion of food and also produce intrinsic factors, which are essential for the absorption of vitamin B12 in the small intestine.

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D. Hepatic cells - II. Bile juice

Hepatic cells, found in the liver, secrete bile juice, which is stored in the gallbladder and aids in the digestion and absorption of fats.

Hence, the correct answer is option 3)A–III, B–I, C–IV, D–II.

Ques 12: Match **List I** with **List II**.

	List I		**List II**
A.	CCK	I.	Kidney
B.	GIP	II.	Heart
C.	ANF	III.	Gastric gland
D.	ADH	IV.	Pancreas

Choose the **correct** answer from the options given below:

Option 1)A-IV, B-III, C-II, D-I

Option 2)A-III, B-II, C-IV, D-I

Option 3)A-II, B-IV, C-I, D-III

Option 4)A-IV, B-II, C-III, D-I

Chapter - Excretory Products & Their Elimination

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Difficulty -Easy

Concept -**Hormones of Adenohypophysis:**

1. Growth Hormone (GH)

2. Adrenocorticotropic Hormone (ACTH)

3. Thyroid Stimulating Hormone (TSH)

4. Follicle Stimulating Hormone (FSH)

5. Luteinizing Hormone (LH)

6. Prolactin (PRL) or Mammotropin Hormone or Luteotrophic Hormone.

Solution - A. CCK - IV. Pancreas
CCK (cholecystokinin) is a hormone released by the small intestine and stimulates the pancreas to release digestive enzymes.

B. GIP - III. Gastric gland
GIP (gastric inhibitory peptide) is a hormone released by the small intestine and inhibits gastric acid secretion.

C. ANF - II. Heart
ANF (atrial natriuretic peptide) is a hormone produced by the heart and is involved in regulating blood pressure and fluid balance.

D. ADH - I. Kidney
ADH (antidiuretic hormone), also known as vasopressin, is produced by the hypothalamus and released by the pituitary gland. It acts on the kidneys to regulate water balance and urine concentration.

Hence, the correct answer is option 1)A-IV, B-III, C-II, D-I.

Ques 13: The vital capacity of the lung is_____.

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Option 1)ERV + ERV

Option 2)IRV + ERV + TV + RV

Option 3)ERV + ERV + TV + RV

Option 4)IRV + ERV + TV

Chapter - Breathing & Exchange of Gases

Difficulty - Easy

Concept - By adding up a few respiratory volumes described in the previous concept, one can derive various pulmonary capacities, which can be used in clinical diagnosis.

Vital Capacity (VC): The maximum volume of air a person can breathe in after forced expiration. This includes ERV, TV and IRV or the maximum volume of air a person can breathe out after a forced inspiration.

Inspiratory Capacity (IC): The total volume of air a person can inspire after a normal expiration. This includes tidal volume and inspiratory reserve volume (TV+IRV).

Expiratory Capacity (EC): The total volume of air a person can expire after a normal inspiration. This includes tidal volume and expiratory reserve volume (TV+ERV).

Functional Residual Capacity (FRC): Volume of air that will remain in the lungs after a normal expiration. This includes ERV+RV.

Total Lung Capacity (TLC): Total volume of air accommodated in the lungs at the end of forced inspiration. This includes RV, ERV, TV and IRV or vital capacity + residual volume.

Solution - The vital capacity of the lungs is the maximum amount of air that can be exhaled after a maximum inhalation. It is calculated by adding the following lung volumes:

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IRV (Inspiratory Reserve Volume): the additional air that can be forcefully inhaled after a normal inhalation.
ERV (Expiratory Reserve Volume): the additional air that can be forcefully exhaled after a normal exhalation.
TV (Tidal Volume): the amount of air inhaled or exhaled during normal breathing.
The residual volume (RV), which is the amount of air that remains in the lungs after maximum exhalation, is not included in the vital capacity calculation.

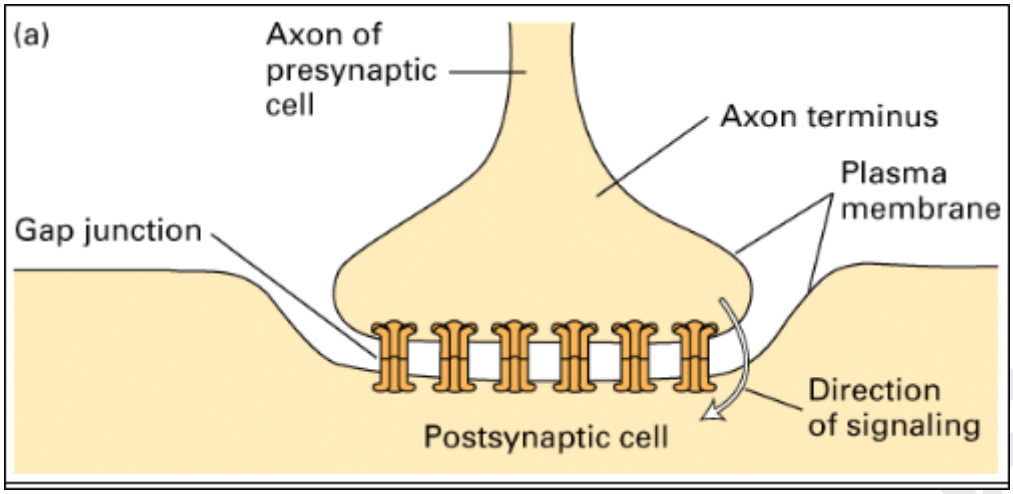
Hence, the correct answer is option 4)IRV + ERV + TV.

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Ques 1: Select the incorrect statement regarding synapses:
Option 1) Electrical current can flow directly from one neuron into the other across the electrical synapse.
Option 2) Chemical synapses use neurotransmitters.
Option 3) Impulse transmission across a chemical synapse is always faster than an electrical synapse.
Option 4) The membranes of presynaptic and postsynaptic neurons are nearby in an electrical synapse.
Chapter - Neural Control & Coordination
Difficulty - Medium
Concept -Transmission of Impulses: At Electrical Synapse

- At electrical synapses, there is continuity between the presynaptic and postsynaptic membranes of the neurons.
- This continuity is due to the gap junctions.
- Hence, there is a minimal synaptic delay due to the direct flow of ions from one neuron to another through gap junctions.
- These are relatively rare and are found in the cardiac muscles, smooth muscles of the intestine, and epithelial cells of the lens.

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Solution -The electrical synapse is faster than the chemical synapse. An electrical synapse is faster because it relies on gap junctions, which are specialized protein channels that directly connect the cytoplasm of two adjacent neurons. These gap junctions allow ions and small molecules to pass directly from one neuron to another, enabling an electrical signal to travel almost instantaneously.

Hence, the correct answer is option 3) Impulse transmission across a chemical synapse is always faster than an electrical synapse.

Ques 2: Which of the following are not the effects of Parathyroid hormone?

(a) Stimulates the process of bone resorption

(b) Decrease the level of blood

(c) Reabsorption of Ca^{2+} by renal tubules

(d) Decrease the absorption of digested food

(e) Increase metabolism of carbohydrates

Choose the most appropriate answer from the options given below :

Option 1)(b), (d), and (e) only

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Option 2)(a) and (e) only

Option 3)(b) and (c) only

Option 4)(a) and (c) only

Chapter - Chemical Control & Integration

Difficulty - Medium

Concept - Parathyroid Gland & Its Disorders -

In humans, four parathyroid glands are present on the backside of the thyroid gland, one pair each in the two lobes of the thyroid gland.
The cells of the parathyroid glands are arranged in a compact mass and are of two types: small chief cells or principal cells and large oxyphil cells (or eosinophil cells).
The chief cells of the parathyroid secrete a hormone called parathyroid hormone (PTH) or parathormone also called Collip's hormone.
This hormone regulates the calcium and phosphate balance between the blood and other tissues.
It mobilises the release of calcium into the blood from the bones.
PTH increases calcium absorption from the intestines.
It increases calcium resorption from the nephrons (and inhibits phosphate reabsorption) of the kidneys.
Parathyroids are under the feeding back control of blood calcium level.
A fall in blood calcium stimulates them to secrete parathormone, and a rise in blood calcium inhibits parathormone secretion from them.
Thus PTH has an effect that opposes the effect of calcitonin**.**

Solution - Parathyroid hormone (PTH) or Collop's hormone increases Ca+2 levels in the blood plasma. This is achieved by

1. Increasing reabsorption by the bones
2. Increasing reabsorption by Kidney

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3. Increasing absorption by intestines

Hence, the correct answer is option 1) (b), (d), and (e) only.

(b) Decrease the level of blood.

(d) Decrease the absorption of digested food.

(e) Increase metabolism of carbohydrates.

Ques 3: Which of the following statements is correct?

Option 1)The tricuspid and the bicuspid valves open due to the pressure exerted by the simultaneous contraction of the atria.

Option 2)Blood moves freely from the atrium to the ventricle during joint diastole.

Option 3) Increased ventricular pressure causes the closing of the semilunar valves.

Option 4) The atrioventricular node (AVN) generates an action potential to stimulate aerial contraction

Chapter - Body fluids & Circulation

Difficulty - Medium

Concept -

Solution - 1) Tricuspid & Bicuspid open due to blood & atrial systole

2) True

3) Increased ventricular pressure causes the opening of semilunar valves

4) Action potentials are generated by the SA node as it is the pacemaker

Hence, Option (2) is correct.

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Ques 4: Match List I with List II

	List I (Biological Molecules)		List II (Biological functions)
(a)	Glycogen	(i)	Hormone
(b)	Globulin	(ii)	Biocatalyst
(c)	Steroids	(iii)	Antibody
(d)	Thrombin	(iv)	Storage product

Choose the correct answer from the options given below:

Option 1) a-iv, b-ii, c-i, d-iii

Option 2) a-iiii, b-iv, c-iii, d-i

Option 3) a-iv, b-iii, c-i, d-ii

Option 4) a-iii, b-ii, c-iv, d-i

Chapter - Chemical Control & Integration

Difficulty - Medium

Concept - The liver makes 800 ml to 1200 ml of bile per day.

- Its pH is 8.6 in the liver and 7.6 in the gallbladder.

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- It is golden yellow or green.
- It is a mixture of water, bile salts, bile pigments, phospholipids (such as lecithin), electrolytes, cholesterol, and triglycerides.
- It is released into the duodenum when stimulated by cholecystokinin (CCK).
- The bile salts are as follows:
    - Sodium glycocholate
    - Sodium taurocholate
    - Potassium glycolate
    - Potassium taurocholate
- Bile salts act as emulsifying agents, so they are also important for the absorption of digested lipids.
- The bile pigments are formed by the degradation of the haemoglobin.
- Bilirubin and biliverdin are the main bile pigments.

Solution - Glycogen serves as a storage product, storing energy in the form of glucose for later use; globulin functions as an antibody (AB), playing a crucial role in immune defence; steroid acts as a hormone, regulating various physiological processes such as metabolism and immune response; and thrombin acts as a biocatalyst, facilitating the conversion of fibrinogen into fibrin during blood clotting.

Hence, the correct answer is Option (3) a-iv, b-iii, c-i, d-ii

Ques 5: Nitrogenous waste is excreted in the form of pellet or paste by :

Option 1)Salamandra

Option 2)Hippocampus

Option 3)Pavo

Option 4)Ornithorhyuchus

Chapter - Excretory Products & Their Elimination

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Difficulty - Medium

Concept - Types of Animals Based on Excretory Wastes -

Ammonotelism:

- The process of excreting ammonia is Ammonotelism.
- Many bony fishes, aquatic amphibians, and aquatic insects are ammonotelic in nature.
- Ammonia, as it is readily soluble, is generally excreted by diffusion across body surfaces or through gill surfaces (in fish) as ammonium ions.
- Kidneys do not play any significant role in its removal.

Ureotelism:

- Terrestrial adaptation necessitated the production of lesser toxic nitrogenous wastes like urea and uric acid for the conservation of water.
- Mammals, many terrestrial amphibians, and marine fishes mainly excrete urea and are called ureotelic animals.
- Ammonia produced by metabolism is converted into urea in the liver of these animals and released into the blood which is filtered and excreted by the kidneys.
- Some amount of urea may be retained in the kidney matrix of some of these animals to maintain the desired osmolarity.

Uricotelism:

- Reptiles, birds, land snails, and insects excrete nitrogenous wastes as uric acid in the form of a pellet or paste with a minimum loss of water and are called uricotelic animals.

Solution - Pavo (the peafowl or peacock) excretes nitrogenous waste in a pellet or paste.

To save water, the nitrogenous waste—primarily uric acid—is concentrated and expelled

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as a paste or solid in this kind of excretion, which is typical in birds. This aids birds in staying hydrated, particularly in areas with limited water supplies.

Hence, the correct answer is option 3)Pavo.

Ques 6: Which of the following is not the function of conducting part of the respiratory system?

Option 1)Inhaled air is humidified

Option 2)Temperature of inhaled air is brought to body temperature

Option 3)Provides surface for the diffusion of O2 and CO_2

Option 4)It clears inhaled air from foreign particles

Chapter - Breathing & Exchange of Gases

Difficulty - Medium

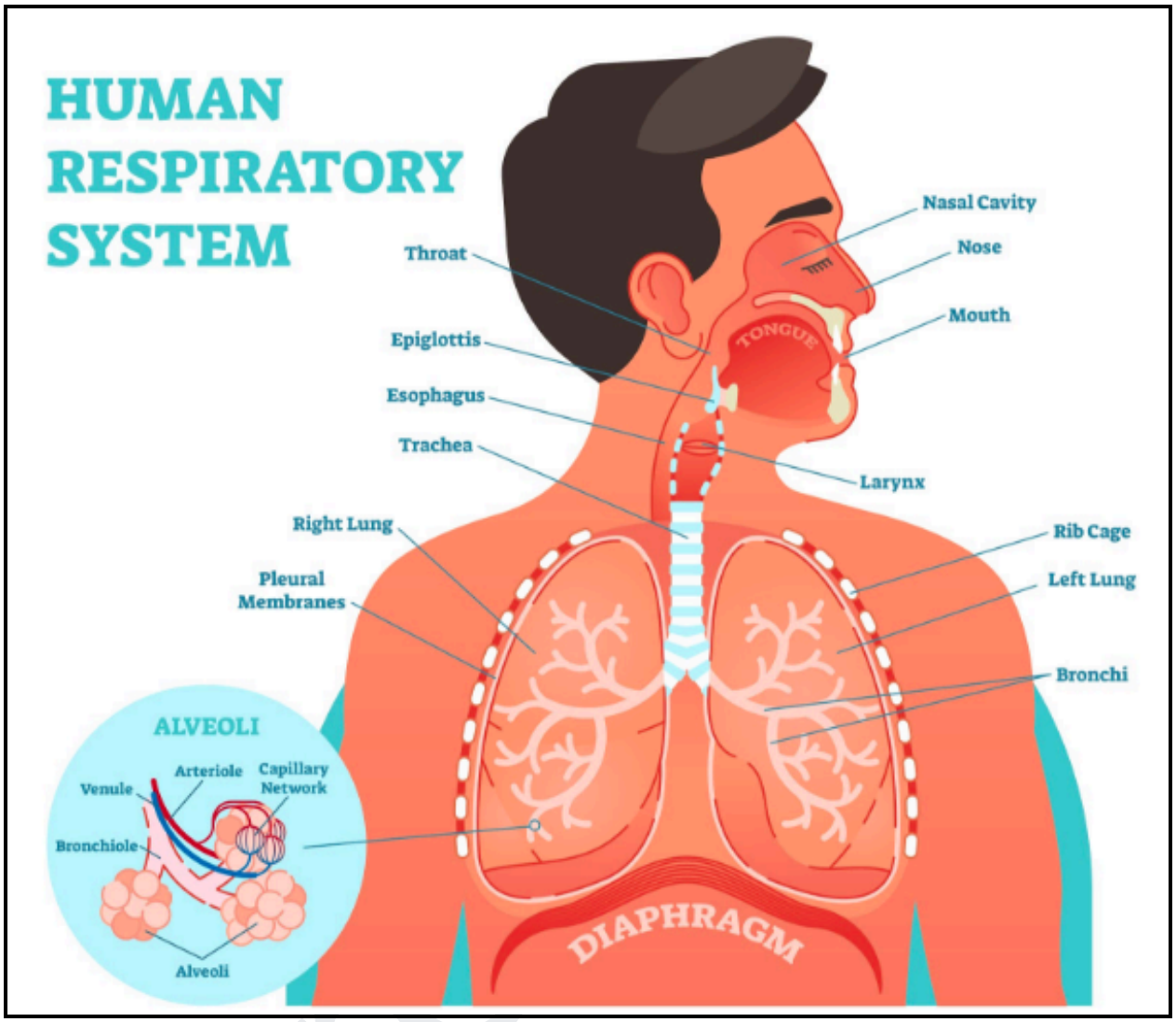
Concept - Respiratory System in Humans -

- The human respiratory system is divisible into two portions:
    - Conducting Portion:
        - Nose
        - Pharynx
        - Larynx
        - Trachea
        - Bronchi
        - Lobular and terminal bronchioles
    - Respiratory Portion:
        - Respiratory bronchioles
        - Alveolar ducts
        - Alveolar sacs and alveoli

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- There is a pair of external nostrils opening out above the upper lips. It leads to a nasal chamber through the nasal passage.
- The nasal chamber opens into the pharynx.
- The pharynx opens through the larynx region into the trachea.
- The larynx is a cartilaginous box that helps in sound production and is hence called the soundbox. The larynx is made up of 9 cartilages out of which three are paired and three are unpaired.
- During swallowing, the glottis can be covered by a thin elastic cartilaginous flap called the epiglottis to prevent the entry of food into the larynx.
- The trachea is a straight tube extending up to the mid-thoracic cavity, which divides at the level of the 5th thoracic vertebra into a right and left primary bronchi.
- Each bronchus undergoes repeated divisions to form the secondary and tertiary bronchi and bronchioles ending up in very thin terminal bronchioles.
- The tracheae, primary, secondary, and tertiary bronchi, and initial bronchioles are supported by incomplete cartilaginous rings.
- Each terminal bronchiole gives rise to several very thin, irregular-walled, and vascularised bag-like structures called alveoli.
- The branching network of bronchi, bronchioles, and alveoli comprise the lungs.

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Solution - The respiratory system is divided into the conducting part and the Respiratory Part. The conducting Part is for transporting the gases. Respiratory is for exchange. The alveoli are tiny air sacs where oxygen is transferred into the blood and carbon dioxide is expelled through the process of diffusion.

Hence, the correct answer is option 3) which Provides a surface for the diffusion of O_2 and CO_2.

Ques 7: Which of the following is the correct match for the disease and its symptoms?

Option 1) Tetany - high Ca^{2+} level causing rapid spasms

Option 2) Myasthenia gravis - Genetic disorder resulting in weakening and paralysis of skeletal muscle

Option 3) Muscular dystrophy - An autoimmune disorder causing progressive degeneration of skeletal muscle.

Option 4) Arthritis - Inflamed joints

Chapter - Chemical Control & Integration

Difficulty - Medium

Concept - 1. Arthritis:

- It is caused by the inflammation of the joints. Some types of arthritis are described here:
    - Rheumatoid Arthritis (RA): It is diagnosed by the presence of rheumatoid factor (a type of immunoglobulin IgM). Rheumatoid arthritis is an inflammation of the synovial membrane in synovial joints.
    - Osteoarthritis: It is also called degenerative joint disease. It is the most common type of joint disease. It is characterized by the progressive erosion of articular cartilage in synovial joints.
    - Infectious Arthritis: These types of arthritis mostly occur due to bacterial and viral infection and are called bacterial and viral arthritis respectively.
    - Gout and Gouty Arthritis: This disease is due to a defect in purine metabolism that causes an excess of uric acid and its salts (urates). The level of uric acid is raised in the blood and crystals of its salts (e.g., sodium urate) accumulate in the joints resulting in gouty arthritis.

2. Osteoporosis:

- Osteoporosis is a disease in which bone loses minerals and fibres from its matrix.

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- There are more chances of fractures.
- It is caused by imbalances of hormones like calcitonin of the thyroid, parathormone of parathyroids, sex hormones and deficiencies of calcium and vitamin D.

3. Osteomalacia or Rickets:

- Osteomalacia, called rickets when it occurs in childhood, is a disease in which the bones contain insufficient amounts of calcium and phosphorus.
- It can be caused due to—kidney disease, vitamin D deficiency, and an inherited defect.

4. Myasthenia gravis: Autoimmune disorder affecting neuromuscular junction leading to fatigue, weakening and paralysis of skeletal muscle.

5. Muscular dystrophy: Progressive degeneration of skeletal muscle mostly due to a genetic disorder.

6. Tetany: Rapid spasms (wild contractions) in muscle due to low Ca++ in body fluid.

Solution - (1) Rapid spasms $\rightarrow$ Tetany is due to low Ca^{+2} levels.

(2) Myasthenia Gravis is an Autoimmune disorder where Ab destroys the body cells.

(3) Muscular dystrophy - It is a Genetic disorder.

(4) Arthritis $\rightarrow$ Inflamed joints.

Hence, the correct answer is option 4)Arthritis - Inflamed joints.

Ques 8: Given below are two statements:

Statement I: The coagulum is formed of a network of threads called thrombins.

Statement II: The spleen is the graveyard of erythrocytes.

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In light of the statements, choose the most appropriate answer from the options given below :

Option 1)Both the Statement I and Statement II are incorrect

Option 2)Statement I is correct but Statement II is incorrect

Option 3)Statement I is incorrect but Statement II is correct

Option 4)Both the Statement I and Statement II are correct

Chapter - Body Fluids & Circulation

Difficulty - Medium

Concept - Blood coagulation

- Whenever some injury occurs to any part of the body, it starts bleeding
- Blood coagulation or blood clotting refers to the process of formation of a blood clot to prevent bleeding
- The process of blood coagulation involves a cascade of reactions and these are carried out by clotting factors
- Thirteen clotting factors have been identified and these are as follows :
- The cascade reactions can be divided into three major steps :

Step I Formation of prothrombinase

- At the site of injury, the blood platelets disintegrate and release a phospholipid, called platelets factor -3 (platelet thromboplastin)
- Injured tissues also release a lipoprotein called thromboplastin.
- These two factors combine with Ca ions and certain proteins of the blood plasma to form an enzyme called prothrombinase.

Step II Catalyzation of prothrombin into thrombin

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- The prothrombinase inactivates heparin (anti-coagulant) in the presence of calcium.m
- Enzyme prothrombinase catalyzes the breakdown n prothrombin (inactive plasma protein) into an active protein called thrombin.
- Prothrombin is factor II. It is glycoproteinaceous and synthesised in the liver by vitamin K

Step III Conversion of fibrinogen into fibrin and formation of a clot

- Thrombin acts as an enzyme and brings about the depolymerization of fibrinogen (a soluble plasma protein) into its monomers
- Later thrombin stimulates the polymerization of these monomers into long insoluble fibre-like polymers called fibrin
- Fibrins form a dense network upon the wound and trap blood cells resulting in a clot.
- The blood clot seals the wound and thus the bleeding stops.
- Vitamin K is essential for clotting as it is required for the synthesis of prothrombin in the liver.

Solution - Statement 1: "Coagulum is a blood clot. It is composed of a network of strands called fibrils."
This statement is true. Coagulum is another term for a blood clot, which is a network of fibrin threads known as fibrils, that stabilise the clot. Coagulation may be defined as the change of fibrinogen into fibrin, which then serves to form this network that facilitates wound sealing and thereby prevents blood loss.
Statement 2: "RBCs live for 120 days and when they die it is in the spleen. Spleen thus is the cemetery."
This is also true. In general, red blood cells live for about 120 days. By this time, they are usually removed from circulation and broken down in the spleen, a location commonly referred to as the "graveyard" for RBCs because it is a site for filtering out and recycling old red blood cells.

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Hence the correct answer is Option 3. Statement I is incorrect but Statement II is correct.

Ques 9: Which of the following functions is not performed by secretions from salivary glands?

Option 1)Digestion of complex carbohydrates

Option 2)Lubrication of the oral cavity

Option 3)Digestion of disaccharides

Option 4)Control bacterial population in the mouth

Chapter- Digestion & Absorption

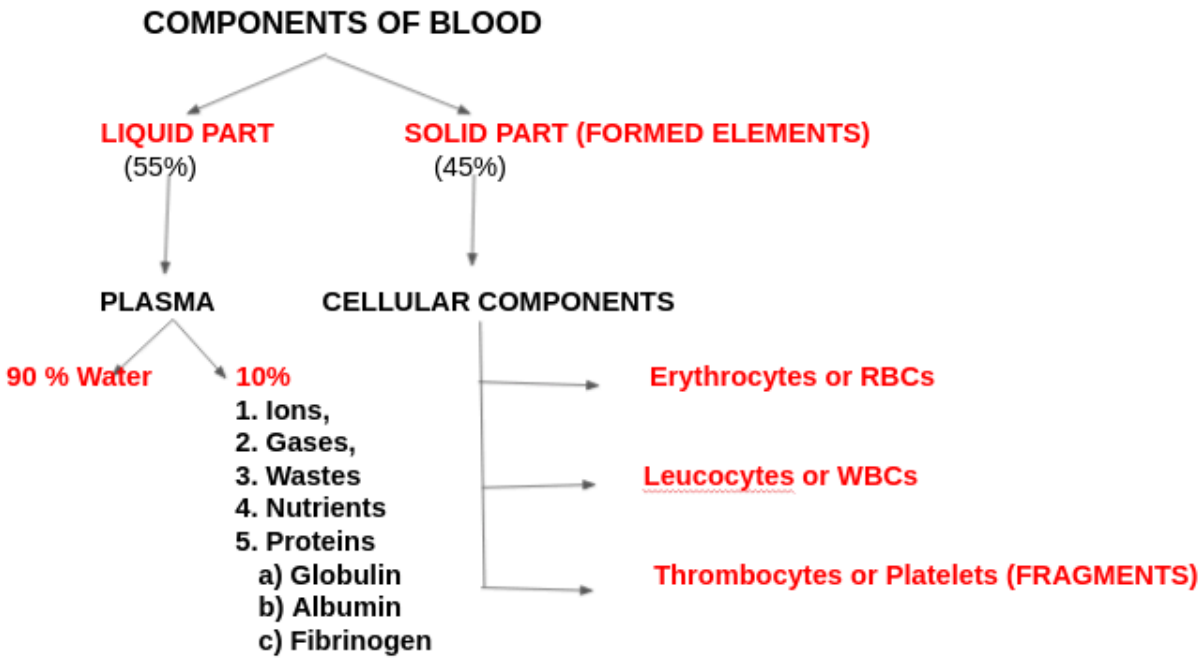
Difficulty -Easy

Concept -Blood

Blood

- Blood refers to a mobile connective tissue which mainly is a composition of a fluid called plasma and floating cells called blood corpuscles or formed elements.
- Blood is a slightly alkaline fluid which forms about 30-35 % of the extracellular fluid.

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Blood plasma

- Blood plasma is a slightly alkaline nonliving intercellular substance. It is a pale yellow but transparent and clear fluid. It constitutes 55% of the blood.
- Plasma contains minerals (Na, Ca, Mg, etc), nutrients (glucose, amino acids, lipids etc.), proteins (albumin, globulin, fibrinogens etc), defence compounds (immunoglobulins, lysozyme, properdin), excretory substances, dissolved gases, anticoagulants (heparin), hormones, vitamins, blood clotting factors etc.
- Plasma without blood clotting factors is called serum.
- The proteins present in blood plasma perform the following functions:

1. Prevention of blood loss (Fibrinogen and prothrombin play a role in blood clotting)
2. Retention of fluid in the blood (albumin and globulins retain water in blood plasma)
3. Body immunity (Some immunoglobulins act as antibodies)
4. Maintenance of pH
5. Transport of certain materials

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6. Distribution of heat
7. As enzymes

Formed elements (blood corpuscles)

- Formed elements or blood cells constitute 45% of the blood.
- Blood corpuscles are of three types i.e. leukocytes (white blood cells), erythrocytes (red blood cells) and thrombocytes (blood platelets).
- The functions of RBC include the transport of O2 and CO2, maintenance of pH etc.
- The function of leukocytes in the defence against microbes and foreign particles
- The function of thrombocytes is in blood clotting.

Haemoglobin:
It is a conjugated protein which is made up of a protein called globin and a non-protein group heme. Heme is an iron porphyrin complex. The mammalian haemoglobin molecule includes 4 heme molecules which are joined

NOTE: Platelets are formed from Megakaryocytes by a process called Fragmentation and hence Platelets are known as Fragments

Let's understand the components

A) ERYTHROCYTES

1. Most abundant
2. 5 million to 5.5 million of RBCs per cu.mm
3. Biconcave and without nuclei
4. Has carbonic anhydrase
5. 12-16 gms of Hb in every 100mL
6. The formation is Erythropoiesis Occurs in red bone marrow in adults

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7. Fetal erythropoiesis takes place first in the mesenchymal tissues and subsequently in the liver and spleen. The production of red blood cells in the bone marrow gradually starts during the second trimester. This is the main site of production at the time of birth even in preterm newborns.
8. Spleen is the graveyard. The life span is 120 days.

B) LEUCOCYTES

1. Colourless, have nuclei and other organelles
2. Shows diapedesis and phagocytosis
3. 2 types: Granular and Agranular
4. Agranulocytes are Lymphocytes and Monocytes
5. Lymphocytes: 20-25 %. Numerous. Small cells with round nuclei. 2 forms B and T
6. Monocytes: Largest. Have kidney or horseshoe-shaped nuclei. Some monocytes enter the tissues and these are Macrophages.
7. Granulocytes are Eosinophils, Basophils and Neutrophils
8. Eosinophils: 2-3%. Nucleus with two lobes. Coarse granules which have hydrolytic enzymes and peroxidases. Eosinophils resist infections and are associated with allergies. Similar to lysosomes
9. Basophils: least abundant. 0.5-1%. The nucleus is three-lobed. Cells secrete serotonin, heparin and histamine
10. Neutrophils: Most abundant Phagocytic cells
11. Formation in bone marrow

C) PLATELETS

1. Thrombocytes
2. Smallest
3. Life span is about a week
4. These are fragments
5. Important for blood clotting

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6. Reduction is Thrombocytopenia

Solution - Salivary amylase brings about the digestion of starch (complex Carbohydrates into disaccharides (Maltose). It cannot digest Maltose further. Once the starch is broken down into maltose, the digestion of maltose is carried out by a different enzyme, maltase, which is found in the small intestine.

Hence, the correct answer is option 3) Digestion of disaccharides.

Ques 10: Under normal physiological conditions in human beings, every 100 of oxygenated blood can deliver _____ ml of O2 to the tissues.

Option 1) 5mL

Option 2) 4mL

Option 3) 10mL

Option 4) 2mL

Chapter - Breathing & Exchange of Gases

Difficulty - Medium

Concept - Oxygen Transport and Oxygen Dissociation Curve -

- Blood is the medium of transport for O_2 and CO_2.
- About 97 per cent of O_2 is transported by RBCs in the blood.
- The remaining 3 per cent O_2 is carried in a dissolved state through the plasma.
- Haemoglobin is a red-coloured iron-containing pigment present in the RBCs.
- O_2 Can reversibly bind with haemoglobin to form oxyhaemoglobin. This is called the O2-Hb association.
- Each haemoglobin molecule can carry a maximum of four molecules O_2.
- The binding of oxygen with haemoglobin is primarily related to the partial pressure O_2.

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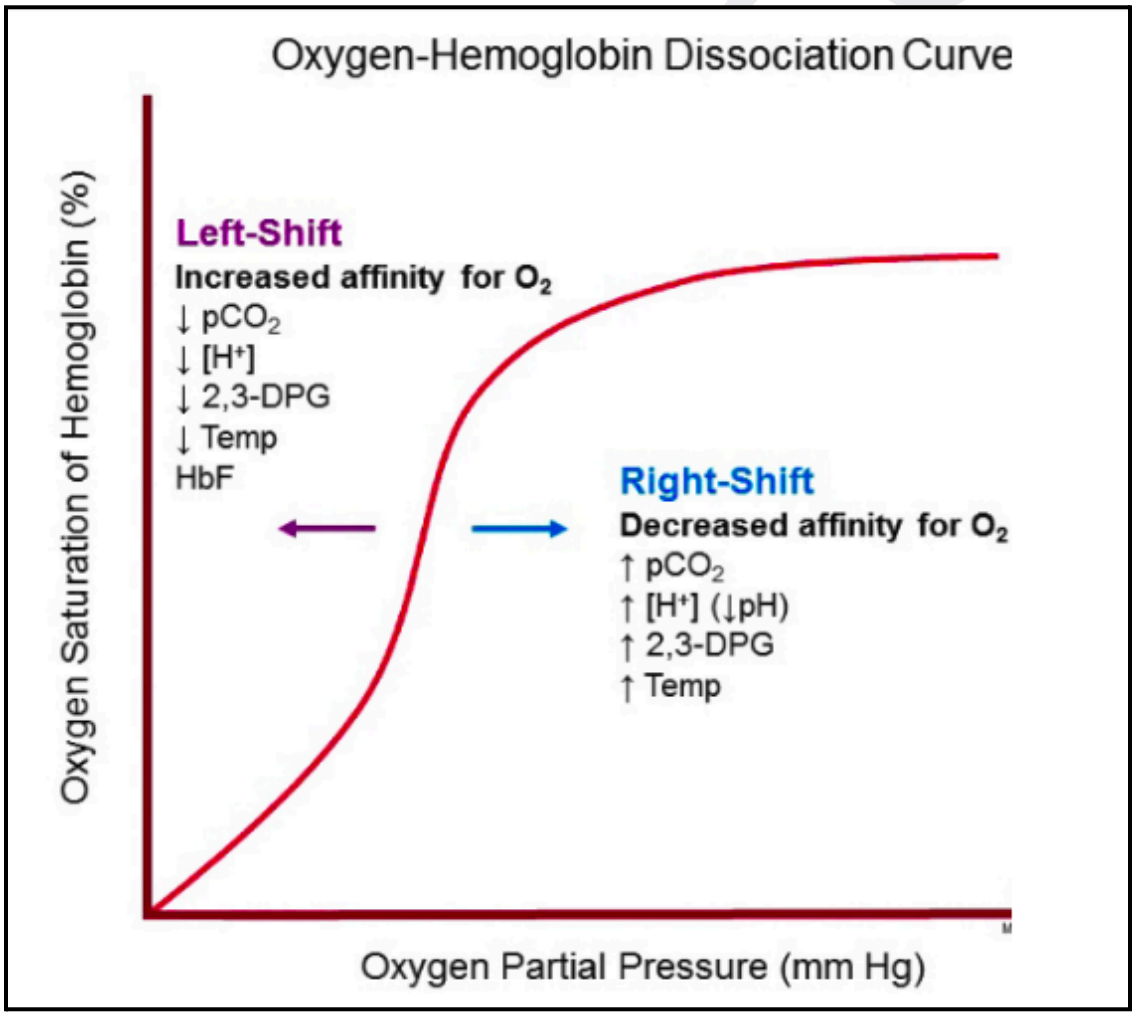
- pO_2 Is high in alveoli so association occurs while at the tissues, the pO_2 decreases which causes dissociation of oxygen.
- The partial pressure of CO_2, hydrogen ion concentration and temperature are the other factors which can interfere with this binding.
- A sigmoid curve is obtained when the percentage saturation of haemoglobin with O_2 is plotted against the pO_2.
- This curve is called the Oxygen dissociation curve.
- This curve is highly useful in studying the effect of factors like pCO_2 H+ concentration, etc, on the binding of O_2 haemoglobin.
- In the alveoli, there is
    - high pO_2
    - low pCO_2
    - lesser H+ concentration
    - high pH
    - lower temperature
    - low 2,3-BPG
- These factors are all favourable for the formation of oxyhaemoglobin.
- In the tissues, there is
    - low pO_2
    - high pCO_2
    - high H+ concentration
    - low pH
    - higher temperature
    - high 2,3-BPG
- 2,3-BPG is Bisphophoglyceric acid. It lowers the affinity of oxygen to Hb and thus it is more near the tissues. This compound thus helps in the dissociation of oxyhemoglobin.
- These conditions are favourable for the dissociation of oxygen from oxyhaemoglobin.

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- This indicates that O_2 gets bound to haemoglobin in the lung surface gets haemoglobin in the lung surface and gets dissociated in the tissues.
- Every 100 ml of oxygenated blood can deliver around 5 ml O_2 to the tissues under normal physiological conditions.

Solution - Under normal physiological conditions, every 100 ml of oxygenated blood can deliver approximately 5 ml of oxygen to the tissues. Oxygen is primarily transported in the blood bound to haemoglobin in red blood cells. Each haemoglobin molecule can carry up to four oxygen molecules.

Hence, the correct answer is option 1) 5ml



Solution -Under normal physiological conditions, every 100 ml of oxygenated blood can deliver approximately 5 ml of oxygen to the tissues. Oxygen is primarily transported in the blood bound to haemoglobin in red blood cells. Each haemoglobin molecule can carry up to four oxygen molecules.

Hence, the correct answer is option 1) 5ml

Ques 11: Given below are two statements

Statement I: Fatty acids and glycerols cannot be observed in the blood.

Statement II: Specialized lymphatic capillaries called lacteals carry chylomicrons into lymphatic vessels and ultimately into the blood.

In light of the above statements, choose the most appropriate answer from the options given below:

Option 1)Both Statement I and Statement II are incorrect

Option 2) Statement I is correct but Statement II is incorrect

Option 3) Statement I is incorrect but Statement II is correct

Option 4)Both Statement I and Statement II are correct

Chapter - Digestion & Absorption

Difficulty - Medium

Concept -Absorption is the process by which the end products of digestion pass through the intestinal mucosa into the blood or lymph.

- **It is carried out by passive, active, or facilitated transport mechanisms.**
- **Small amounts of monosaccharides like glucose, and amino acids, and some electrolytes like chloride ions are generally absorbed by simple diffusion.**

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- **The passage of these substances into the blood depends upon the concentration gradients.**
- **However, some substances like glucose and amino acids are absorbed with the help of carrier proteins.**
- **This mechanism is called facilitated transport.**
- **The transport of water depends upon the osmotic gradient.**
- **Active transport occurs against the concentration gradient and hence requires energy.**
- **Various nutrients like amino acids, monosaccharides like glucose, and electrolytes like Na+ are absorbed into the blood by this mechanism.**
- **Fatty acids and glycerol being insoluble, cannot be absorbed into the blood.**
- **They are first incorporated into small droplets called micelles which move into the intestinal mucosa.**
- **They are re-formed into very small protein-coated fat globules called chylomicrons which are transported into the lymph vessels (lacteals) in the villi.**
- **These lymph vessels ultimately release the absorbed substances into the bloodstream.**
- **Absorption of substances takes place in different parts of the alimentary canal, like the mouth, stomach, small intestine, and large intestine.**
- **However, maximum absorption occurs in the small intestine.**

Solution - Statement 1 is correct because Fatty acids and Glycerol are insoluble in water. Hence these cannot be absorbed directly into the blood. Hence these cannot be absorbed directly into the blood.

Statement 2 is correct because Fatty acid and Glycerol are first converted into micelles. These move from the intestinal lumen into mucosa These are then transformed into very small protein-coated fat globules called chylomicrons. These enter lacteals

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(lymphatic capillaries) because chylomicrons are bigger and cannot enter blood capillaries. Inside the lacteals, they break down and finally enter blood:

Hence, option 4 is the correct answer.

NEET 2021

Ques 1: **Assertion (A) :**

A person goes to high altitude and experiences 'altitude sickness' with symptoms like breathing difficulty and heart palpitations.

Reason (R) :

Due to low atmospheric pressure at high altitudes, the body does not get sufficient oxygen.

In light of the above statements, choose the **correct** answer from the options given below.

Option 1)Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**
Option 2) Both **(A)** and **(R)** are true but **(R)** is not the correct explanation of **(A)**
Option 3) **(A)** is true but **(R)** is false
Option 4) **(A)** is false but **(R)** is true
Chapter - Breathing & Exchange of Gases
Difficulty - Medium
Concept - Disorders of the Respiratory System -

- **Asthma** is a difficulty in breathing causing wheezing due to inflammation of the bronchi and bronchioles. Allergens (which cause allergy) stimulate the release of histamine from mast cells. Histamine has several effects, one of which is to cause bronchiolar smooth muscle to contract.
- **Emphysema**: The word “emphysema” means “inflation” or “full of air”. Emphysema is inflation or abnormal distension of the bronchioles or alveolar sacs

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of the lungs. Major causes are cigarette smoking and the inhalation of smoke or toxic substances over some time. As the alveolar septa collapse, the surface area for gas exchange is greatly reduced. There is a loss of elasticity in the walls of bronchioles or alveolar sacs. As a result, the alveolar sacs remain filled with air even after expiration. The exhalation becomes more difficult. The lungs remain inflated.

- **Occupational Respiratory Disorders:** As the name indicates, these disorders are due to the occupation of the individual. These are caused by harmful substances, such as gas fumes or specks of dust, present in the environment where a person works. Silicosis and asbestosis are common examples, which occur due to chronic exposure of silica and asbestos dust in the mining industry. It is characterised by the proliferation of fibrous connective tissue (fibrosis) of the upper part of the lungs, causing inflammation.
- **Pneumonia**: It is an acute infection or inflammation of the alveoli of the lungs. Pneumonia is caused mainly by the bacteria *Streptococcus pneumoniae*. Sometimes, other bacteria or fungi, protozoans, viruses and mycoplasma may cause pneumonia.
- **Diphtheria**: It is caused by *Corynebacterium diphtheriae* resulting in enlarged mucous membranes of the oropharynx, nasopharynx and larynx. It may obstruct airways and cause death from asphyxiation.
- **Coryza** (= Common Cold): Hundreds of viruses can cause coryza or the common cold. However, a group of viruses called rhinovirus is responsible for about 40% of all colds in adults. Its symptoms include sneezing, excessive nasal secretion, dry cough and congestion.
- **Influenza (Flu)**: It is caused by the Influenza virus. Its symptoms include chills, fever, headache and muscle pain. Cold-like symptoms appear as the fever subsides.
- **SARS — Severe Acute Respiratory Syndrome:** It is caused by Human coronavirus (HCV). It is a new member of the influenza virus family which is considered a mutant form of influenza virus. Human coronavirus spreads

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through contact, respiratory secretions and cockroaches. Symptoms include cold dry cough, headache, loss of appetite, high temperature, fever and hypoxia at a later stage followed by muscular stiffness, and malaise (feeling of bodily discomfort, but without clear signs of a particular illness).

Solution - At high altitudes, many individuals experience altitude sickness, which includes symptoms such as difficulty in breathing and heart palpitations.

- At high altitudes, the atmospheric pressure is lower, which results in a lower partial pressure of oxygen. This leads to insufficient oxygen availability for the body, causing symptoms of altitude sickness.

Hence the correct option is 1)Both (A) and (R) are true and (R) is the correct explanation of (A).

Ques 2: Match List - I with List - II.

List - I		**List - II**	
(a)	**Scapula**	**(i)**	**Cartilaginous joints**
(b)	**Cranium**	**(ii)**	**Flat bone**
(c)	**Sternum**	**(iii)**	**Fibrous joints**

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(d)	Vertebral Column	(iv)	Triangular flat bone

Choose the correct answer from the options given below.

Option 1)(a)-(i), (b)-(iii), (c)-(ii), (d)-(iv)

Option 2)(a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)

Option 3)(a)-(iv), (b)-(ii), (c)- (iii), (d)- (i)

Option 4)(a)-(iv), (b)- (iii), (c)-(ii), (d)-(i)

Chapter - Skeletal system and its functions; Joints, Locomotion & Movement

Difficulty - Difficult

Concept -The skull forms the skeleton of the head.

- **It consists of a total of 29 bones.**
- **It includes Cranial Bones, Facial bones, Ear ossicles and Hyoid bone.**
- **The skull region articulates with the superior region of the vertebral column with the help of two occipital condyles (dicondylic skull).**

A) Cranial Bones:

- **The cranium is formed by 8 bones.**
- **The bones which form the cranium are:**

PAIRED BONES

- **PARIETAL**
- **TEMPORAL**

UNPAIRED BONES

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- **FRONTAL**
- **OCCIPITAL**
- **SPHENOID**
- **ETHMOID**
- **The cranial cavity, in which the brain is lodged, posteriorly opens by foramen magnum.**
- **The brain is connected to the spinal cord at this foramen.**

Solution - The scapula is a triangular flat bone that forms the posterior part of the shoulder girdle, providing attachment for various muscles. The cranium consists of multiple bones connected by fibrous joints, which are immovable and protect the brain. The sternum, a flat bone in the centre of the chest, plays a crucial role in protecting the heart and lungs. The vertebral column is composed of vertebrae connected by cartilaginous joints, allowing flexibility and support. Together, these skeletal components contribute to body structure, movement, and protection of vital organs.

List - I	List - II
(a) Scapula	(iv) Triangular flat bone
(b) Cranium	(iii) Fibrous joints
(c) Sternum	(ii) Flat bone
(d) Vertebral Column	(i) Cartilaginous joints

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Hence the correct option is 4)(a)-(iv), (b)- (iii), (c)-(ii), (d)-(i).

Ques 3: During muscular contraction which of the following events occur?

(a) The 'H' zone disappears

(b) 'A' band widens

(c) The 'I' band reduces in width

(d) Myosine hydrolyzes ATP, releasing the ADP_____and Pi

(e) Z-lines attached to actins are pulled in

Choose the correct answer from the option given below.

Option 1)(a),(c),(d),(e) only

Option 2)(a),(b),(c),(d) only

Option 3)(b),(c),(d),(e) only

Option 4)(b),(d),(e),(a) only

Chapter - Locomotion & Movement

Difficulty - Medium

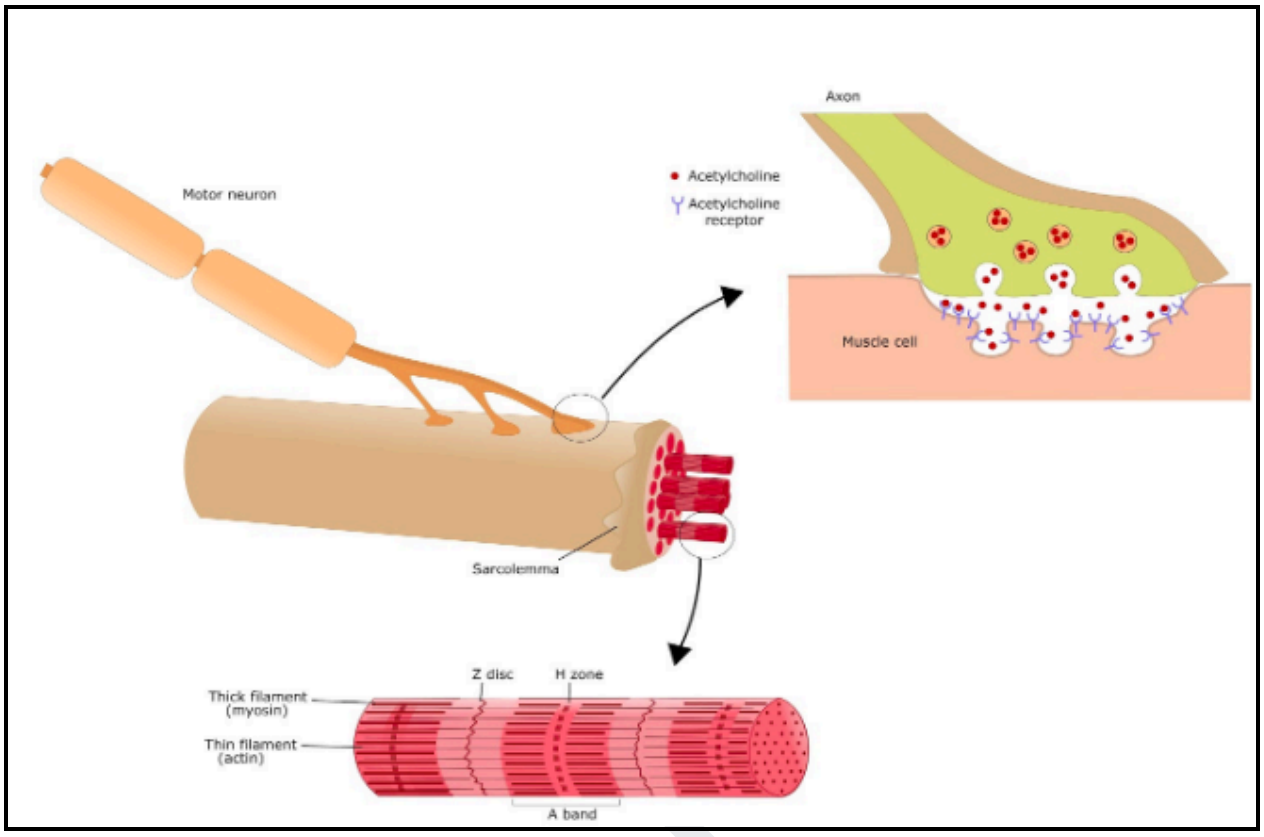
Concept - Mechanism of Muscle Contraction -

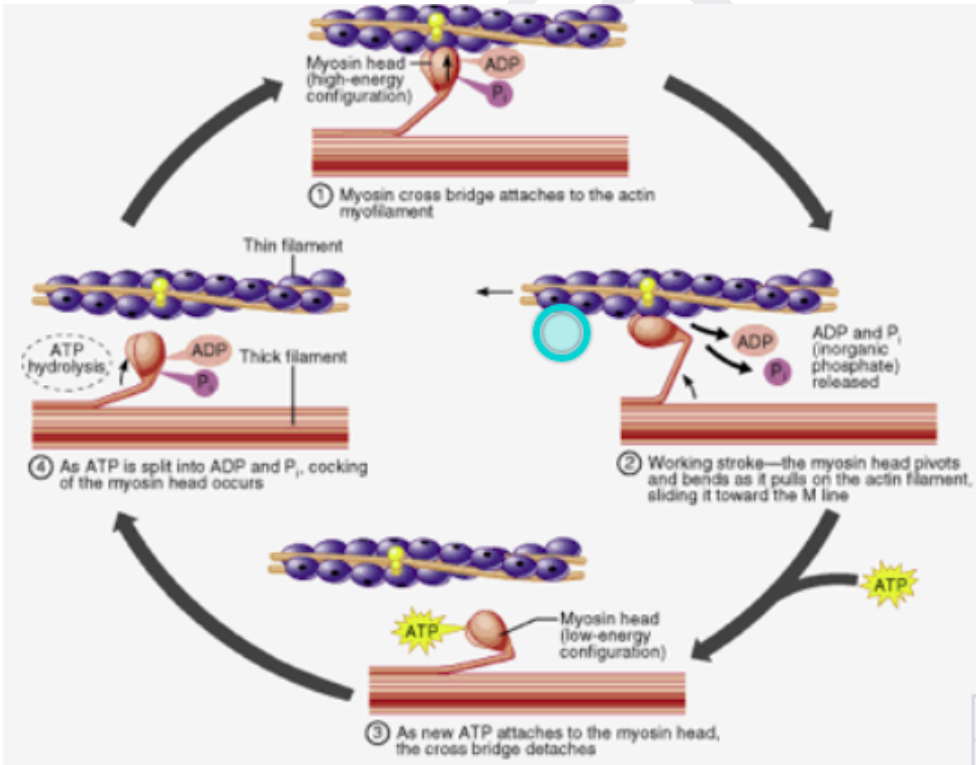
- **The mechanism of muscle contraction is best explained by the sliding filament theory which states that the contraction of a muscle fiber takes place by the sliding of the thin filaments over the thick filaments.**
- **Muscle contraction is initiated by a signal sent by the central nervous system (CNS) via a motor neuron.**
- **A motor neuron along with the muscle fibers connected to it constitute a motor unit.**

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- **The junction between a motor neuron and the sarcolemma of the muscle fibre is called the neuromuscular junction or motor-end plate.**
- **A neural signal reaching this junction releases a neurotransmitter (Acetylcholine) which generates an action potential in the sarcolemma.**
- **This spreads through the muscle fibre and causes the release of calcium ions into the sarcoplasm.**
- **An increase in Ca++ level leads to the binding of calcium with a subunit of troponin (TpC) on actin filaments and thereby removes the masking of active sites for myosin.**
- **Utilizing the energy from ATP hydrolysis, the myosin head now binds to the exposed active sites on actin to form a cross bridge.**
- **This pulls the attached actin filaments towards the centre of the 'A' band.**
- **The 'Z' line attached to the actins is also pulled inwards thereby causing a shortening of the sarcomere, i.e., contraction.**
- **It is clear from the above steps, that during shortening of the muscle, i.e., contraction, the 'I' bands get reduced, whereas the 'A' bands retain the length.**
- **The myosin releases the ADP and Pi goes back to its relaxed state.**
- **A new ATP binds and the cross-bridge is broken.**
- **The ATP is hydrolyzed by the myosin head and the cycle of cross-bridge formation and breakage is repeated causing further sliding.**
- **The process continues till the Ca++ ions are pumped back to the sarcoplasmic cisternae resulting in the masking of actin filaments.**
- **This causes the return of 'Z' lines back to their original position, i.e., relaxation.**
- **Since ATP is essential for muscular contraction, ADP produced needs to be converted back to ATP immediately. This is achieved in the muscles by the presence of another high-energy compound called creatine phosphate.**
- **ADP combines with creatine phosphate to produce ATP and creatinine.**

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Axon
Acetylcholine
Acetylcholine receptor
Motor neuron
Muscle cell
Sarcolemma
Z disc
H zone
Thick filament (myosin)
Thin filament (actin)
A band


Myosin head (high-energy configuration)
ADP
Pi
1 Myosin cross bridge attaches to the actin myofilament
Thin filament
ATP hydrolysis
ADP
Pi
Thick filament
4 As ATP is split into ADP and Pi, cocking of the myosin head occurs
ADP
Pi
ADP and Pi (inorganic phosphate) released
2 Working stroke—the myosin head pivots and bends as it pulls on the actin filament, sliding it toward the M line
ATP
ATP
Myosin head (low-energy configuration)
3 As new ATP attaches to the myosin head, the cross bridge detaches

Solution -During muscular contraction:

(a) 'H' zone disappears: The H zone disappears as actin filaments slide over myosin filaments during contraction.

(b) 'A' band widens: The A band remains the same width throughout the contraction process.

(c) 'I' band reduces in width: The I band decreases in width as actin filaments are pulled toward the centre of the sarcomere.

(d) Myosin hydrolyzes ATP, releasing the ADP and Pi: Myosin heads hydrolyze ATP into ADP and inorganic phosphate, enabling the power stroke during contraction.

(e) Z-lines attached to actins are pulled in: The Z-lines are pulled closer together, shortening the sarcomere and resulting in muscle contraction.

Hence the correct option is 1)(a),(c),(d),(e) only.

Ques 4: Sphincter of oddi is present at :

Option 1) Ileo-Caecal junction

Option 2) Junction of hepato-pancreatic duct and duodenum

Option 3) Gastro-oesophageal junction

Option 4) Junction of jejunum and duodenum

Chapter - Digestion & Absorption

Difficulty - Medium

Concept - The liver is the largest gland in the body, weighing about three pounds in an adult.

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- It is present in the right upper quadrant of the abdomen, just below the diaphragm.
- The liver is heavier in males than in females.
- The liver is connected to the abdominal wall and diaphragm by five peritoneal folds referred to as ligaments.
- The lesser omentum tethers the liver to the lesser curvature of the stomach.

Histology of Liver:

- The liver is covered by two sheaths or layers.
- The outer sheath is the serous capsule. It is made up of the visceral peritoneum.
- The inner sheath is called the Glisson's capsule. It is made up of dense connective tissue.
- The liver is divided into two primary lobes: a large right lobe and a smaller left lobe.
- Hepatic lobules are the functional units of the liver. These are present in each lobe of the liver.
- 50,000 to 1,00,000 hepatic lobules are present in the liver.
- The hepatic lobule is made up of hepatocytes.
- A hepatocyte is the liver's main cell type, accounting for around 80 per cent of the liver's volume.
- These cells play a role in a wide variety of secretory, metabolic, and endocrine functions.
- The hepatocytes are arranged in the form of hepatic plates.
- Each plate is made up of two columns of hepatocytes.
- In between the two columns, there are narrow ducts called bile canaliculi.
- These small ducts accumulate bile produced by hepatocytes.
- From here, bile flows first into bile ductules and then into bile ducts.
- The bile ducts unite to form the larger right and left hepatic ducts, which themselves merge and exit the liver as the common hepatic duct.

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- In between the neighbouring hepatic plates, a blood space called a sinusoid is present.
- The sinusoid is made up of loose connective tissue and is composed of Kupffer cells, arterioles, and venules.
- The hepatic sinusoids combine and send blood to a central vein.
- Blood then flows through a hepatic vein into the inferior vena cava.
- The portal triad is a distinctive arrangement around the perimeter of hepatic lobules, consisting of three basic structures: a bile duct, a hepatic artery branch, and a hepatic portal vein branch.

Solution - The Sphincter of Oddi is a muscular valve located at the junction where the hepatopancreatic duct opens into the duodenum, the first part of the small intestine. This sphincter regulates the flow of bile and pancreatic juices into the duodenum, playing a vital role in digestion.

Hence, the correct answer is option 2) the Junction of the hepato-pancreatic duct and duodenum.

Ques 5: Select the favourable conditions required for the formation of oxyhaemoglobin at the alveoli.

Option 1) High pO_2, low pCO_2, less H^+ lower temperature

Option 2) Low pO_2, high pCO_2, more H^+ higher temperature

Option 3) High pO_2, high pCO_2, less H^+ higher temperature

Option 4) Low pO_2, low pCO_2, more H^+ higher temperature

Chapter - Breathing & Exchange of Gases

Difficulty - Difficult

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Concept - Oxygen Transport and Oxygen Dissociation Curve -

- Blood is the medium of transport for O_2 and CO_2.
- About 97 percent of O_2 is transported by RBCs in the blood.
- The remaining 3 percent O_2 is carried in a dissolved state through the plasma.
- Haemoglobin is a red-coloured iron-containing pigment present in the RBCs.
- O_2 can reversibly bind with haemoglobin to form oxyhaemoglobin. This is called the O2-Hb association.
- Each haemoglobin molecule can carry a maximum of four molecules O_2.
- The binding of oxygen with haemoglobin is primarily related to the partial pressure O_2.
- pO_2 is high in alveoli so association occurs while at the tissues, the pO_2 decreases which cause dissociation of oxygen.
- The partial pressure of CO_2, hydrogen ion concentration and temperature are the other factors that can interfere with this binding.
- A sigmoid curve is obtained when the percentage saturation of haemoglobin with O_2 is plotted against the pO_2.
- This curve is called the Oxygen dissociation curve.
- This curve is highly useful in studying the effect of factors like pCO_2 H+ concentration, etc, on the binding of O_2 haemoglobin.
- In the alveoli, there is
    - high pO_2
    - low pCO_2
    - lesser H+ concentration
    - high pH
    - lower temperature
    - low 2,3-BPG
- These factors are all favourable for the formation of oxyhaemoglobin.
- In the tissues, there is
    - low pO_2
    - high pCO_2

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    - high H+ concentration
    - low pH
    - higher temperature
    - high 2,3-BPG
- 2,3-BPG is Bisphophoglyceric acid. It lowers the affinity of oxygen to Hb and thus it is more near the tissues. This compound thus helps in the dissociation of oxyhemoglobin.
- These conditions are favourable for the dissociation of oxygen from oxyhaemoglobin.
- This indicates that O_2 gets bound to haemoglobin in the lung surface gets haemoglobin in the lung surface and gets dissociated at the tissues.
- Every 100 ml of oxygenated blood can deliver around 5 ml O_2 to the tissues under normal physiological conditions.

Ques 6: *Succus entericus* is referred to as :

Option 1)Pancreatic juice

Option 2)Intestinal juice

Option 3)Gastric juice

Option 4)Chyme

Chapter - Digestion & Absorption

Difficulty - Medium

Concept -Various types of movements are generated by the muscular layer of the small intestine.

- These movements help in a thorough mixing of the food with various secretions in the intestine and thereby facilitate digestion.

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- The bile, pancreatic juice and intestinal juice are the secretions released into the small intestine.
- Pancreatic juice and bile are released through the hepato-pancreatic duct.
- The pancreatic juice contains inactive enzymes – trypsinogen, chymotrypsinogen, procarboxypeptidase, amylases, lipases and nucleases.
- Trypsinogen is activated by an enzyme, enterokinase, secreted by the intestinal mucosa into active trypsin, which in turn activates the other enzymes in the pancreatic juice.
- The bile released into the duodenum contains bile pigments (bilirubin and biliverdin), bile salts, cholesterol and phospholipids but no enzymes.
- Bile helps in the emulsification of fats, i.e., breaking down the fats into very small micelles.
- Bile also activates lipases.
- The intestinal mucosal epithelium has goblet cells which secrete mucus.
- The secretions of the brush border cells of the mucosa along with the secretions of the goblet cells constitute the intestinal juice or succus entericus.
- This juice contains a variety of enzymes like disaccharidases (e.g., maltase), dipeptidases, lipases, nucleosidases, etc.
- The mucus along with the bicarbonates from the pancreas protects the intestinal mucosa from acid as well as provides an alkaline medium (pH 7.8) for enzymatic activities.
- Submucosal glands (Brunner's glands) also help in this.

Solution - Succus entericus is the intestinal juice of the glands present in the wall of the small intestine. The fluid mainly includes water, mucus, and several digestive enzymes such as disaccharidases, dipeptidases, lipases, and nucleosidases. Succus entericus plays an important role in digestion because it breaks down carbohydrates, proteins, and fats that are absorbed as nutrients in the small intestine; it also has the function of neutralizing acidic chyme that arrives from the stomach.

Hence, the correct answer is Option (2) Intestinal juice.

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Ques 7: Which enzyme is responsible for the conversion of inactive fibrinogens to fibrins?

Option 1) Thrombin

Option 2) Renin

Option 3)Epinephrine

Option 4)Thrombokinase

Chapter - Body Fluids & Circulation

Difficulty - Easy

Concept - **Blood coagulation**

- Whenever some injury occurs to any part of the body, it starts bleeding
- Blood coagulation or blood clotting refers to the process of formation of a blood clot to prevent bleeding
- The process of blood coagulation involves a cascade of reactions and these are carried out by clotting factors
- The cascade reactions can be divided into three major steps :

Step I Formation of prothrombinase

- At the site of injury, the blood platelets disintegrate and release a phospholipid, called platelets factor -3 (platelet thromboplastin)
- Injured tissues also release a lipoprotein called thromboplastin.
- These two factors combine with Ca ions and certain proteins of the blood plasma to form an enzyme called prothrombinase.

Step II Catalyzation of prothrombin into thrombin

- The prothrombinase inactivates heparin (anti-coagulant) in the presence of calcium.m

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- Enzyme prothrombinase catalyzes the breakdown of prothrombin (inactive plasma protein) into an active protein called Thrombin.
- Prothrombin is factor II. It is glycoproteinaceous and synthesised in the liver by vitamin K

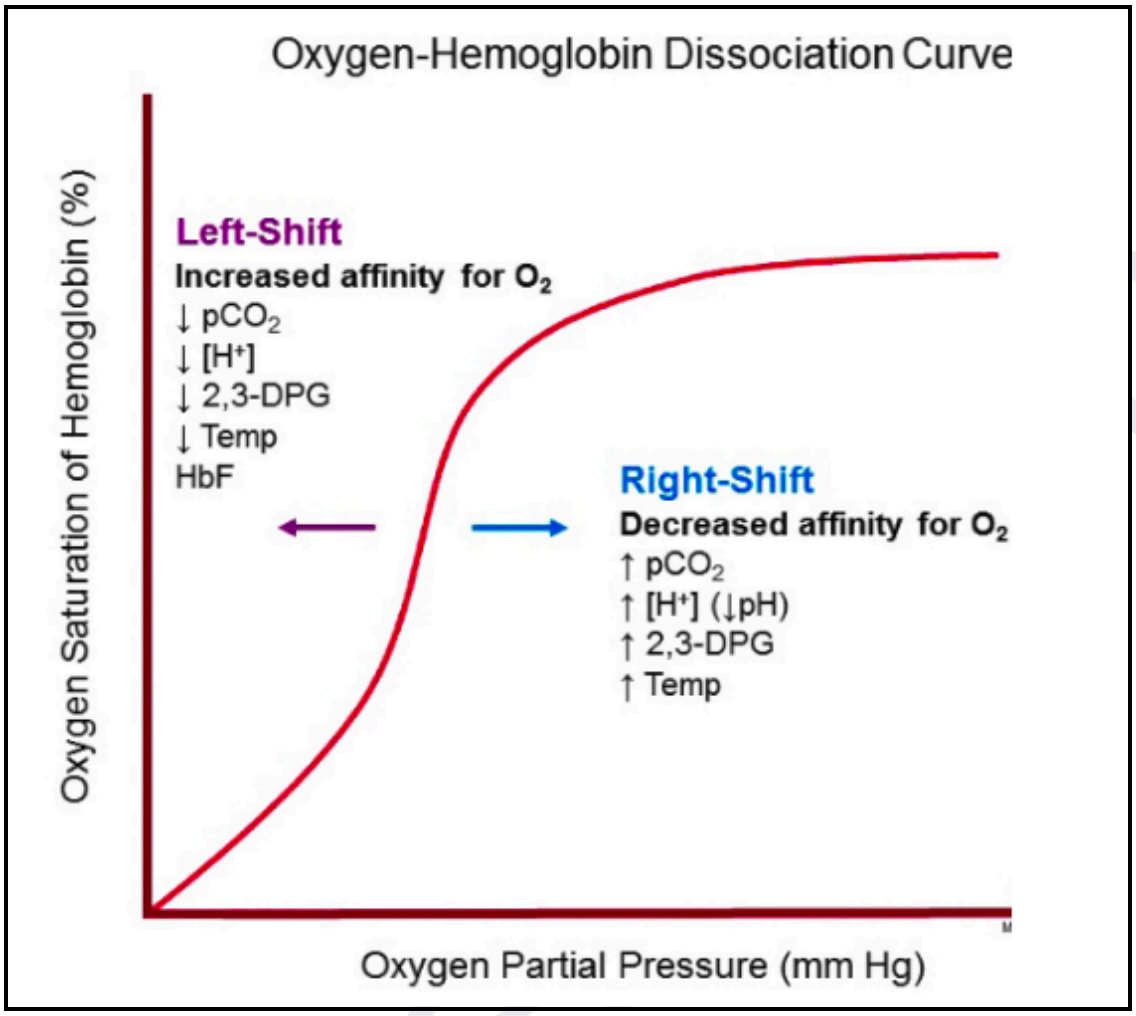
Step III Conversion of fibrinogen into fibrin and formation of a clot

- Thrombin acts as an enzyme and brings about the depolymerization of fibrinogen (a soluble plasma protein) into its monomers
- Later thrombin stimulates the repolymerization of these monomers into long insoluble fibre-like polymers called fibrin
- Fibrins form a dense network upon the wound and trap blood cells resulting in a clot.
- The blood clot seals the wound and thus the bleeding stops.
- Vitamin K is essential for clotting as it is required for the synthesis of prothrombin in the liver.

Solution - Thrombin is an enzyme in the blood that plays a crucial role in the coagulation (clotting) process. It is responsible for converting inactive fibrinogen (a soluble plasma protein) into fibrin (an insoluble protein) during the coagulation cascade. This fibrin network traps blood cells and helps form a stable blood clot at the site of injury. The process prevents excessive blood loss and allows wound healing to begin.

Hence, the correct answer is option 1)Thrombin.

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Solution - The formation of oxyhemoglobin at the alveoli is favoured by high partial pressure of oxygen (pO_2), low partial pressure of carbon dioxide (pCO_2), lower hydrogen ion concentration (less H^+, indicating higher pH), and lower temperature. These conditions increase haemoglobin's affinity for oxygen, facilitating effective oxygen loading in the lungs.

Hence, the correct answer is option 1)High pO_2, low pCO_2, less H^+ lower temperature.

Ques 8: A chronic autoimmune disorder affecting neuromuscular junction leading to fatigue, weakening, and paralysis of skeletal muscle is called:

Option 1) Arthritis

Option 2) Muscular dystrophy

Option 3) Myasthenia gravis

Option 4) Gout

Chapter - Disorders of muscular and skeletal system-Myasthenia gravis, Tetany, Muscular dystrophy, Arthritis, Osteoporosis, Gout

Difficulty - Medium

Concept - Disorders of Muscular and Skeletal System -

1. Arthritis:

- It is caused by the inflammation of the joints. Some types of arthritis are described here:
    - Rheumatoid Arthritis (RA): It is diagnosed by the presence of rheumatoid factor (a type of immunoglobulin IgM). Rheumatoid arthritis is an inflammation of the synovial membrane in synovial joints.
    - Osteoarthritis: It is also called degenerative joint disease. It is the most common type of joint disease. It is characterized by the progressive erosion of articular cartilage in synovial joints.
    - Infectious Arthritis: These types of arthritis mostly occur due to bacterial and viral infection and are called bacterial and viral arthritis respectively.
    - Gout and Gouty Arthritis: This disease is due to a defect in purine metabolism that causes an excess of uric acid and its salts (urates). The level of uric acid is raised in the blood and crystals of its salts (e.g., sodium urate) accumulate in the joints resulting in gouty arthritis.

2. Osteoporosis:

- Osteoporosis is a disease in which bone loses minerals and fibres from its matrix.
- There are more chances of fractures.

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- It is caused by imbalances of hormones like calcitonin of the thyroid, parathormone of parathyroids, and sex hormones, and deficiencies of calcium and vitamin D.

3. Osteomalacia or Rickets:

- Osteomalacia, called rickets when it occurs in childhood, is a disease in which the bones contain insufficient amounts of calcium and phosphorus.
- It can be caused due to—kidney disease, vitamin D deficiency, and an inherited defect.

4. Myasthenia gravis: Autoimmune disorder affecting neuromuscular junction leading to fatigue, weakening, and paralysis of skeletal muscle.

5. Muscular dystrophy: Progressive degeneration of skeletal muscle mostly due to a genetic disorder.

6. Tetany: Rapid spasms (wild contractions) in muscle due to low Ca++ in body fluid.

Solution - An autoimmune chronic condition is myasthenia gravis. It has an impact on the neuromuscular junction, which is where muscles and nerves connect. The immune system unintentionally targets acetylcholine receptors, a neurotransmitter essential for muscular contraction, in this disorder.

The muscles weaken as a result, making it more difficult for the body to communicate with them. Fatigue, weakened muscles, and in certain situations, paralysis, result from this. The most frequently impacted muscles are those that govern respiration, swallowing, facial expressions, and eye movement.

Typically, symptoms become better with rest and get worse with activity. It may be difficult for people with this illness to do daily duties. Medication to increase muscular strength and, in certain situations, surgery to remove the thymus gland are part of the treatment. With the right attention, many people can handle the condition effectively.

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Hence, the correct option is 3)Myasthenia gravis.

Ques 9: The partial pressures (in mm Hg) of oxygen O_2 and carbon dioxide CO_2 at alveoli (the site of diffusion) are:

Option 1) pO_2 = 104 and pCO_2 = 40

Option 2) pO_2 = 40 and pCO_2 = 45

Option 3) pO_2 = 95 and pCO_2 = 40

Option 4) pO_2 = 160 and pCO_2

Chapter - Breathing & Exchange of Gases

Difficulty - Medium

Concept - <u>Exchange of Gases -</u>

- Alveoli are the primary sites of exchange of gases. The exchange of gases also occurs between blood and tissues.
- There are approx. 30 crore alveoli in the human lungs.
- O2 and CO2 are exchanged in these sites by simple diffusion mainly based on pressure/concentration gradient.
- The solubility of the gases, as well as the thickness of the membranes involved in diffusion, are also some important factors that can affect the rate of diffusion.
- **Dalton's Law**: Pressure contributed by an individual gas in a mixture of gases is called partial pressure and is represented as pO2 for oxygen and pCO2 for carbon dioxide. It is equal to the total pressure of the mixture divided by the percentage of that gas in the mixture. For instance, the pressure of atmospheric gas at sea level is 760 mmHg, and oxygen forms 21% of the air, so pO2 will be 21% of 760 or 159 mmHg.

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Solution - The partial pressures of oxygen (O_2) and carbon dioxide (CO_2) in the alveoli, which are the minute gas exchange units in the lungs, are crucial for understanding respiratory physiology. Here, the values are provided:

- Partial pressure of O_2 in alveoli: 104 mm Hg
- Partial pressure of CO_2 in alveoli: 40 mm Hg

To clarify, alveoli are the locations where gases like oxygen and carbon dioxide are transferred between the inhaled air and the circulating blood. The O_2 partial pressure is greater in alveoli compared to deoxygenated blood, which is approximately 40 mm Hg, enabling efficient diffusion into the bloodstream. Conversely, the CO_2 partial pressure is lesser in alveoli relative to the blood's content, typically around 45 mm Hg, thus aiding in the diffusion of CO_2 from the blood to the alveoli for exhalation. This gradient-driven process is fundamental to maintaining homeostasis in respiratory gas exchange.

Hence, the correct answer is option 1) $pO_2 = 104$ and $pCO_2 = 40$.

Ques 10: Erythropoietin hormone which stimulates R.B.C. formation is produced by :

Option 1) Alpha cells of the pancreas

Option 2) The cells of the rostral adenohypophysis

Option 3) The cells of bone marrow

Option 4) Cells of the kidney

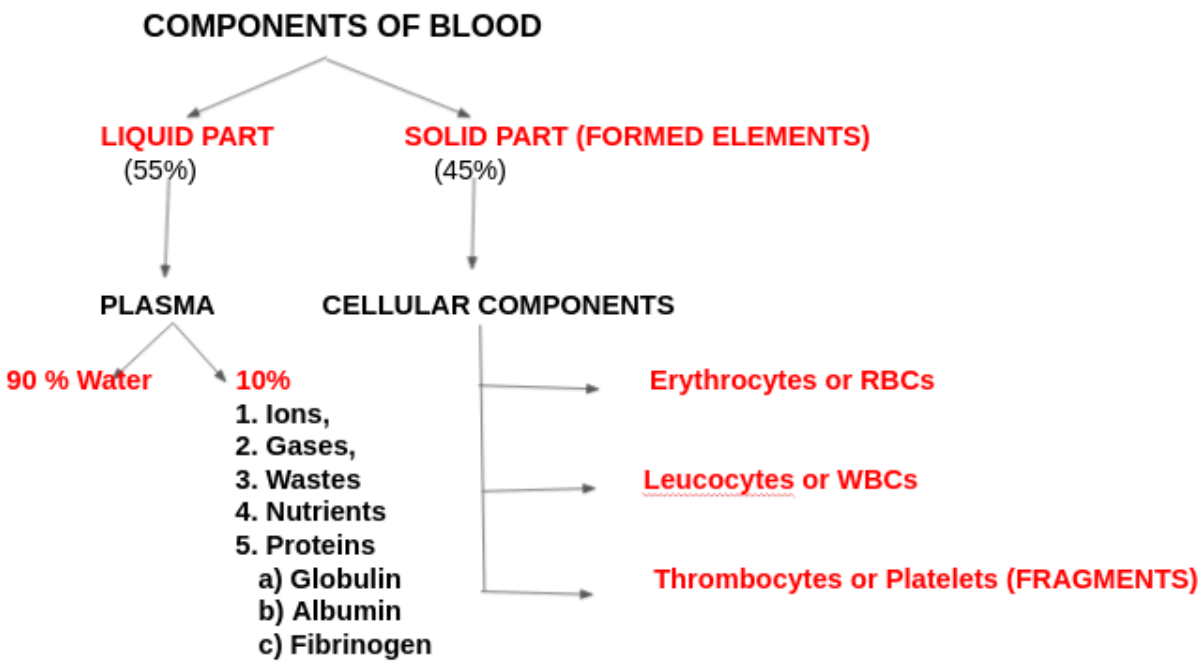
Chapter - Body Fluids & Circulation

Difficulty - Medium

Concept - **Blood**

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- Blood refers to a mobile connective tissue which mainly is a composition of a fluid called plasma and floating cells called blood corpuscles or formed elements.
- Blood is a slightly alkaline fluid which forms about 30-35 % of the extracellular fluid.



Blood plasma

- Blood plasma is a slightly alkaline nonliving intercellular substance. It is a pale yellow but transparent and clear fluid. It constitutes 55% of the blood.
- Plasma contains minerals (Na, Ca, Mg, etc), nutrients (glucose, amino acids, lipids etc.), proteins (albumin, globulin, fibrinogens etc), defence compounds (immunoglobulins, lysozyme, properdin), excretory substances, dissolved gases, anticoagulants (heparin), hormones, vitamins, blood clotting factors etc.
- Plasma without blood clotting factors is called serum.
- The proteins present in blood plasma perform the following functions:

1. Prevention of blood loss (Fibrinogen and prothrombin play a role in blood clotting)

2. Retention of fluid in the blood (albumin and globulins retain water in

blood plasma)

3. Body immunity (Some immunoglobulins act as antibodies)
4. Maintenance of pH
5. Transport of certain materials
6. Distribution of heat
7. As enzymes

Formed elements (blood corpuscles)

- Formed elements or blood cells constitute 45% of the blood.
- Blood corpuscles are of three types i.e. leukocytes (white blood cells), erythrocytes (red blood cells) and thrombocytes (blood platelets).
- The functions of RBC include the transport of O2 and CO2, maintenance of pH etc.
- The function of leukocytes in the defence against microbes and foreign particles
- The function of thrombocytes is in blood clotting.

Haemoglobin:
It is a conjugated protein which is made up of a protein called globin and a non-protein group heme. Heme is an iron porphyrin complex. The mammalian haemoglobin molecule includes 4 heme molecules which are joined

NOTE: Platelets are formed from Megakaryocytes by a process called Fragmentation and hence Platelets are known as Fragments

Let's understand the components

A) ERYTHROCYTES

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1. Most abundant
2. 5 million to 5.5 million of RBCs per cu.mm
3. Biconcave and without nuclei
4. Has carbonic anhydrase
5. 12-16 gms of Hb in every 100mL
6. The formation is Erythropoiesis Occurs in red bone marrow in adults
7. Fetal erythropoiesis takes place first in the mesenchymal tissues and subsequently in the liver and spleen. The production of red blood cells in the bone marrow gradually starts during the second trimester. This is the main site of production at the time of birth even in preterm newborns.
8. Spleen is the graveyard. The life span is 120 days.

B) LEUCOCYTES

1. Colourless, have nuclei and other organelles
2. Shows diapedesis and phagocytosis
3. 2 types: Granular and Agranular
4. Agranulocytes are Lymphocytes and Monocytes
5. Lymphocytes: 20-25 %. Numerous. Small cells with round nuclei. 2 forms B and T
6. Monocytes: Largest. Have kidney or horseshoe-shaped nuclei. Some monocytes enter the tissues and these are Macrophages.
7. Granulocytes are Eosinophils, Basophils and Neutrophils
8. Eosinophils: 2-3%. Nucleus with two lobes. Coarse granules which have hydrolytic enzymes and peroxidases. Eosinophils resist infections and are associated with allergies. Similar to lysosomes
9. Basophils: least abundant. 0.5-1%. The nucleus is three-lobed. Cells secrete serotonin, heparin and histamine
10. Neutrophils: Most abundant Phagocytic cells
11. Formation in bone marrow

C) PLATELETS

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1. Thrombocytes
2. Smallest
3. Life span is about a week
4. These are fragments
5. Important for blood clotting
6. Reduction is Thrombocytopenia

Solution - The hormone erythropoietin (EPO), which stimulates the formation of red blood cells (RBCs), is produced by the kidneys.

- Erythropoietin is primarily secreted by the peritubular cells in the renal cortex of the kidneys in response to low oxygen levels in the blood (hypoxia).
- It stimulates the bone marrow to increase the production of red blood cells, helping to enhance the oxygen-carrying capacity of the blood.

Hence, the correct answer is option 4) Cells of the kidney.

Ques 11: Persons with the 'AB' blood group are called "Universal recipients". This is due to :

Option 1) Absence of antigens A and B on the surface of RBCs

Option 2) Absence of antigens A and B in plasma

Option 3) Presence of antibodies, anti-A and anti- B on RBCs

Option 4) Absence of antibodies, anti-A and anti-B, in plasma

Chapter - Body Fluids & Circulation

Difficulty - Medium

Concept - Blood groups: ABO Blood Groups

- There are more than 30 antigens present on the surfaces of blood cells. These antigens give rise to blood groups.

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- Antigens can induce an immune response by secreting a class of proteins called antibodies that bind in a specific fashion with antigens.
- Depending on the nature of antigens present on the membrane of RBCs various types of blood grouping have been done. Two such blood group systems are ABO (antigen A, B) and Rh (antigen Rh) blood grouping.

ABO blood group:

- ABO blood groups in human beings were reported for the first time by Karl Landsteiner. According to this system, humans can have 4 types of blood groups, A, B, AB and O.
- A, B and O blood groups were discovered by Landsteiner while the AB blood group was discovered by de Castello and Steini.
- ABO blood groups are determined by the gene Ig which has three alleles Ia, Ib, and Io of this gene
- Allele Ia and Ib produce antigen A and antigen B respectively.
- RBCs of each blood group contain some antigens and they possess antibodies against other antigens in their plasma.
- During blood transfusion, it is very important to know the blood groups to avoid clumping reactions.
- Clumping is also known as Agglutination. It occurs when Antibodies stick to the wrong Antigens and this can result in clumps which can be fatal. Hence blood group recognition is very important before transfusion (Antigens are Agglutinogen and Antibodies are Agglutinins)
- Since individuals with blood group AB do not have antibodies against any of the antigens, they can receive blood from individuals with A, B and O blood groups. This is why blood group AB is called the universal recipient.
- Individuals with blood groups do not have any kind of Antigen and thus can donate blood to individuals with A, B and AB blood groups. Hence, blood group O is called the universal donor.

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Solution - The ABO blood group system is based on the presence of specific antigens that can be found on the outer surface of red blood cells. There are four main blood groups: A, B, AB, and O.
Blood Group A has A antigens and anti-B antibodies.
Blood Group B has B antigens and anti-A antibodies.
Blood Group AB carries both A and B antigens, but shows no antibodies against A or B antigens, and thus AB blood group recipients are universal recipients.
A +B antigens of Blood Group O are absent and both anti-A and anti-B antibodies are present.
AB blood group is formed by both A antigens and is not exposed to corresponding antibodies; thus, the blood recipients who possess this group accept all other types of blood from them without agglutination risk.

Hence, the correct answer is Option (4) Absence of antibodies, anti-A and anti-B, in plasma.

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Ques 1: The enzyme enterokinase helps in the conversion of:
Option 1) protein into polypeptide
Option 2) trypsinogen into trypsin
Option 3) caseinogen into casein
Option 4) pepsinogen into pepsin
Chapter - Role of digestive enzymes and gastrointestinal hormones
Difficulty - Easy
Concept - Various types of movements are generated by the muscularis layer of the small intestine.

- These movements help in a thorough mixing of the food with various secretions in the intestine and thereby facilitate digestion.
- The bile, pancreatic juice and intestinal juice are the secretions released into the small intestine.

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- Pancreatic juice and bile are released through the hepato-pancreatic duct.
- The pancreatic juice contains inactive enzymes – trypsinogen, chymotrypsinogen, procarboxypeptidase, amylases, lipases and nucleases.
- Trypsinogen is activated by an enzyme, enterokinase, secreted by the intestinal mucosa into active trypsin, which in turn activates the other enzymes in the pancreatic juice.
- The bile released into the duodenum contains bile pigments (bilirubin and biliverdin), bile salts, cholesterol and phospholipids but no enzymes.
- Bile helps in the emulsification of fats, i.e., breaking down the fats into very small micelles.
- Bile also activates lipases.
- The intestinal mucosal epithelium has goblet cells which secrete mucus.
- The secretions of the brush border cells of the mucosa along with the secretions of the goblet cells constitute the intestinal juice or succus entericus.
- This juice contains a variety of enzymes like disaccharidases (e.g., maltase), dipeptidases, lipases, nucleosidases, etc.
- The mucus along with the bicarbonates from the pancreas protects the intestinal mucosa from acid as well as provides an alkaline medium (pH 7.8) for enzymatic activities.
- Submucosal glands (Brunner's glands) also help in this.
- Proteins, proteases and peptones (partially hydrolysed proteins) in the chyme reaching the intestine are acted upon by the proteolytic enzymes of pancreatic juice as given below:
- Carbohydrates in the chyme are hydrolysed by pancreatic amylase into disaccharides.
- Fats are broken down by lipases with the help of bile into monoglycerides.
- Nucleases in the pancreatic juice act on nucleic acids to form nucleotides and nucleosides.
- The enzymes in the succus entericus act on the end products of the above reactions to form the respective simple absorbable forms.

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- These final steps in digestion occur near the mucosal epithelial cells of the intestine.

Solution - Trypsinogen is converted to trypsin with the help of the enzyme enterokinase.

The duodenum (small intestine) cells release enterokinase, which activates trypsinogen, an inactive precursor (zymogen) of the trypsin enzyme. Trypsin is an essential digestive enzyme that breaks down proteins into smaller peptides once it is activated.

Hence, the correct answer is option 2)trypsinogen into trypsin.

Ques 2: Goblet cells of the alimentary canal are modified from:
Option 1)Squamous epithelial cells
Option 2)Columnar epithelial cells
Option 3)Chondrocytes
Option 4)Compound epithelial cells
Chapter - Alimentary canal and digestive glands
Difficulty - Easy
Concept -The wall of the large intestine has a few enzyme-secreting cells.

- There are no circular folds or villi.
- Other than in the anal canal, the mucosa of the colon is simple columnar epithelium made mostly of enterocytes (absorptive cells) and goblet cells.
- The wall of the large intestine has far more intestinal glands.
- These goblet cells secrete mucus that eases the movement of faeces and protects the intestine from the effects of the acids and gases produced by enteric bacteria.
- The enterocytes absorb water and salts as well as vitamins produced by intestinal bacteria.

Solution - Goblet cells of the alimentary canal are modified from columnar epithelial cells. Goblet cells are specialized epithelial cells found throughout the mucous

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membranes of the alimentary canal, especially in the intestines. Their primary function is to secrete mucus, which lubricates and protects the lining of the digestive tract.

Hence, the correct answer is option 2)Columnar epithelial cells.

Ques 3: Select the correct statement.
Option 1)Glucocorticoids stimulate gluconeogenesis
Option 2)Glucagon is associated with hypoglycemia
Option 3)Insulin acts on pancreatic cells and adipocytes
Option 4)Insulin is associated with hyperglycemia
Chapter - Human endocrine system-Hypothalamus, Pituitary, Pineal, Thyroid, Parathyroid, Adrenal, Pancreas, Gonads
Difficulty - Medium
Concept - Adrenal Glands or Suprarenal Glands: Adrenal Cortex -

- Adrenal Glands are paired structures located on the top of the kidneys.
- Each adrenal gland has two parts:
    - external adrenal cortex
    - internal adrenal medulla
- The cortex is surrounded by a fibrous capsule.

Adrenal Cortex:

- The adrenal cortex is derived from the mesoderm of the embryo.
- It is subdivided into three zones:
    - Zona glomerulosa: This is the outer zone that lies just below the capsule. It constitutes about 15% of the gland. Mineralocorticoids are secreted from this region.
    - Zona fasciculata: This is the middle zone which is the widest of the three zones. It constitutes about 50% of the gland. Glucocorticoids are secreted from this region.
    - Zona reticularis: This is the inner zone that constitutes about 7% of the gland. Gonadocorticoids are secreted from this region.

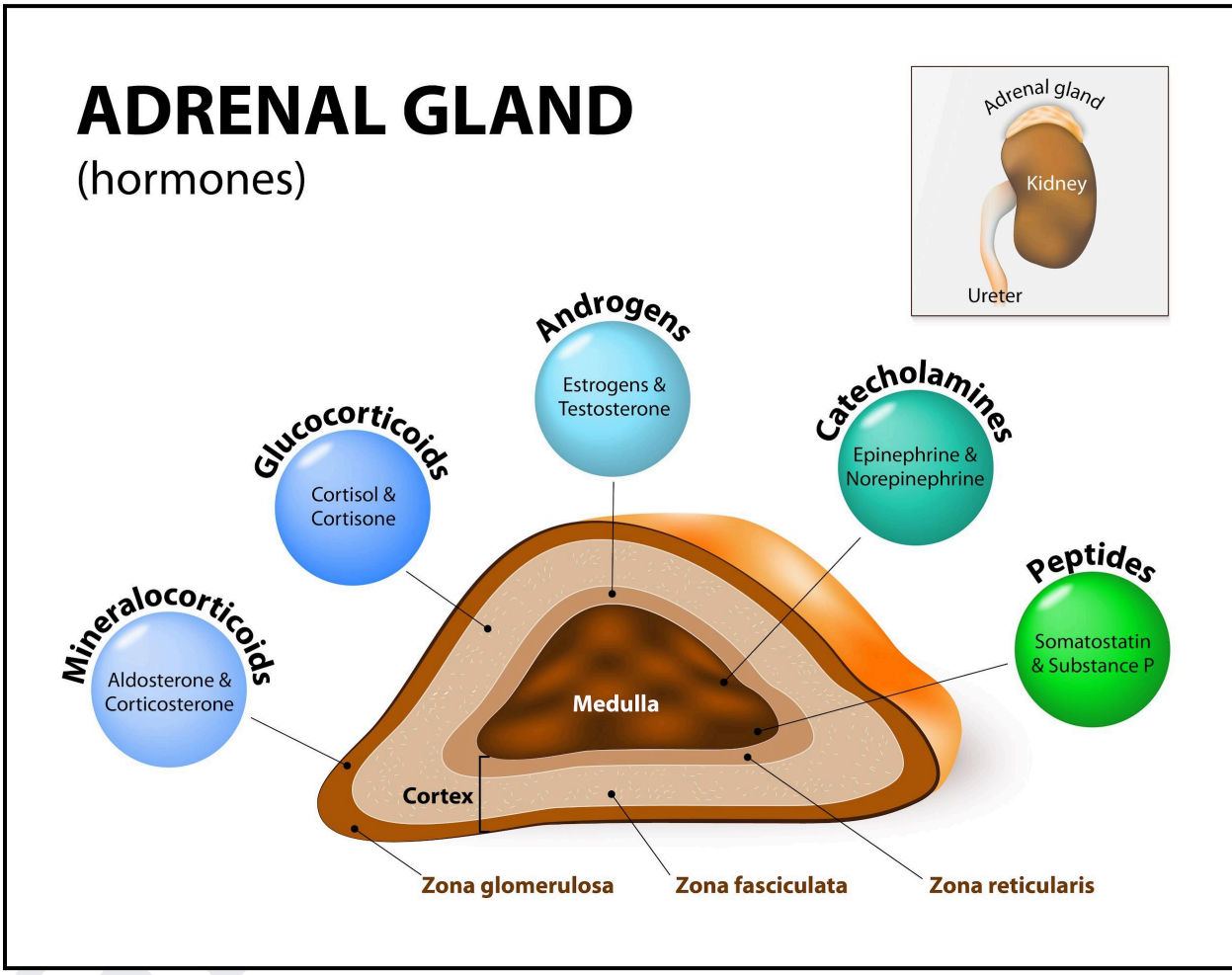
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Hormones of the Adrenal Cortex:

- **Mineralocorticoids**:
    - They are responsible for the regulation of mineral metabolism.
    - **Aldosterone** (salt-retaining hormone) is the principal mineralocorticoid (90 to 95%) in humans.
    - It is secreted when the sodium level is low.
    - It acts on the kidneys to cause more sodium to be returned to the blood and more potassium to be excreted.
    - As the sodium concentration in the blood increases, water follows it by osmosis, so the blood volume also increases.
    - Thus the effect of aldosterone is to increase both sodium and water in the blood.
- **Glucocorticoids:**
    - They affect carbohydrate metabolism, however, they also affect the metabolism of proteins and fats.
    - Glucocorticoids include three main hormones: cortisol (= hydrocortisone), corticosterone, and cortisone.
    - **Cortisol** is the most abundant (about 95%).
    - It stimulates the liver to synthesize carbohydrates from non-carbohydrates such as amino acids and glycerol.
    - The third effect of cortisol is to stimulate the breakdown of fats in adipose tissue and release fatty acids into the blood.
    - It retards the phagocytic activities of WBCs and thus suppresses 'inflammation reaction'.
    - Cortisol is also "immunosuppressive".
    - It suppresses the synthesis of antibodies by inhibiting the production of lymphocytes in lymphoid tissues.
    - Cortisol can cope with stress.
- **Gonadocorticoids:**
    - They are also called sex hormones of adrenal glands.

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- Large quantities of male-than-female sex corticoids (sex hormones) are produced.
- These male sex hormones are called androgens which are important in the development of a male foetus.



Solution - Glucocorticoids, like cortisol, stimulate gluconeogenesis (the production of glucose from non-carbohydrate sources) in the liver.

- Glucagon is associated with increasing blood glucose levels, not hypoglycemia.
- Insulin acts on liver, muscle, and adipose (fat) cells to promote glucose uptake but does not act directly on pancreatic cells.
- Insulin lowers blood glucose levels, counteracting hyperglycemia (high blood glucose) by promoting glucose uptake into cells.

Hence the correct option is 1)Glucocorticoids stimulate gluconeogenesis.

Ques 4: Which of the following would help in the prevention of diuresis?

Option 1)More water reabsorption due to underreaction of ADH.

Option 2) Reabsorption of Na and water from renal tubules to aldosterone.

Option 3) The atrial natriuretic factor causes vasoconstriction.

Option 4) Decreases in the secretion of renin by JG cells.

Chapter - Regulation of kidney function-Renin-angiotensin, Atrial Natriuretic Factor, ADH and Diabetes insipidus

Difficulty level - Medium

Concept - The functioning of the kidneys is efficiently monitored and regulated by hormonal feedback mechanisms involving the hypothalamus, JGA and to a certain extent, the heart.

Control by ADH:

- Osmoreceptors in the body are activated by changes in blood volume, body fluid volume and ionic concentration.
- An excessive loss of fluid from the body can activate these receptors which stimulate the hypothalamus to release antidiuretic hormone (ADH) or vasopressin from the neurohypophysis.
- ADH facilitates water reabsorption from the latter parts of the tubule, thereby preventing diuresis.
- An increase in body fluid volume can switch off the osmoreceptors and suppress the ADH release to complete the feedback.
- ADH can also affect the kidney function by its constrictor effects on blood vessels. This causes an increase in blood pressure.

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- An increase in blood pressure can increase the glomerular blood flow and thereby the GFR.

Control by Juxtaglomerular Apparatus:

- The JGA plays a complex regulatory role.
- A fall in glomerular blood flow/glomerular blood pressure/GFR can activate the JG cells to release renin which converts angiotensinogen in the blood to angiotensin I and further to angiotensin II.
- Angiotensin II, being a powerful vasoconstrictor, increases glomerular blood pressure and thereby GFR.
- Angiotensin II also activates the adrenal cortex to release Aldosterone.
- Aldosterone causes the reabsorption of Na+ and water from the distal parts of the tubule.
- This also leads to an increase in blood pressure and GFR.
- This complex mechanism is generally known as the Renin-Angiotensin-Aldosterone mechanism.

Control by ANF:

- An increase in blood flow to the atria of the heart can cause the release of Atrial Natriuretic Factor (ANF).
- ANF can cause vasodilation (dilation of blood vessels) and thereby decrease the blood pressure.
- ANF mechanism, therefore, acts as a check on the renin-angiotensin mechanism.

Solution - Aldosterone promotes the reabsorption of sodium (Na^+) and water from the renal tubules back into the bloodstream which helps in the prevention of diuresis (excessive urine production).

The adrenal glands generate the hormone aldosterone which primarily increases the kidneys' ability to reabsorb sodium. Due to osmotic pressures, water is also reabsorbed along with salt increasing the body's water retention. Maintaining blood pressure and

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volume lowers the amount of urine generated.

Aldosterone, therefore ensures that both water and salt are maintained when it is active preventing excessive fluid loss (diuresis).

Hence, the correct answer is option 2)Reabsorption of Na and water from renal tubules to aldosterone.

Ques 5: The QRS complex in a standard ECG represents

Option 1) Repolarisation of auricles

Option 2) Depolarisation of auricles

Option 3) Depolarisation of ventricles

Option 4) Repolarisation of ventricles

Chapter - Cardiac cycle, cardiac output, ECG, Double circulation

Difficulty - Biology

Concept - ECG

- The representation of the electric current produced by the excitation of the cardiac muscles in the form of a graph is called an electrocardiogram (ECG)
- The machine by which the electrocardiogram is recorded is called an electrocardiograph.
- The electrocardiograph was invented by Einthoven and he is known as the father of electrocardiography.
- An electrocardiogram shows three consecutive waves namely the P wave, the QRS wave, and the T wave.

1. P wave-

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- It is a small upward wave that indicates the depolarisation of the atria leading to contraction of both atria. It is caused by the activation of the SA node.

2. QRS wave

- It begins after a fraction of a second of the P wave
- It begins as a small downward deflection (Q) and continues as a large upright (R) and triangular wave, ending as a downward wave (S) at its base
- QRS wave indicates depolarization of ventricles
- It is caused by the impulses of the contraction from the AV node through the Bundle of His and Purkinje fibres and the contraction of ventricular muscles
- In this wave, there is a spread of electrical impulses through ventricles
- The normal duration of the QRS complex is <0.10 seconds

3. T wave

- The T wave is dome-shaped and it indicates ventricular repolarization (ventricular relaxation)
- The end of T waves marks systole

P-R interval

- It is the time required for an impulse to travel through the atria and AV node to the remaining conductive tissue.
- The normal duration of the P-R interval is <0.12 to 0.2 seconds
- The P-R interval is lengthened during conditions like arteriosclerotic heart disease and rheumatic fever due to the inflammation of atria and AV node

S-T interval

- It is the time between the end of the spread of impulse through ventricles and its repolarization.
- It starts at the end of the S wave and terminates at the beginning of the T wave.

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- During acute myocardial infarction, the S-T segment is elevated while it is depressed during the insufficient supply of oxygen to the muscles.

Q-T interval

- The QT interval represents the time of ventricular activity including both depolarization and repolarization.
- It is measured from the beginning of the QRS complex to the end of the T wave.
- The QT interval will vary with the patient's gender, age, and heart rate.
- The normal duration of the Q-T interval is <0.42 seconds.

Solution - The QRS complex in a standard ECG represents the depolarization of the ventricles, a critical phase during which electrical impulses spread through the ventricular walls. This depolarization triggers the contraction of the ventricles, enabling them to pump blood—the right ventricle to the lungs for oxygenation and the left ventricle to the rest of the body. This phase is essential for maintaining effective circulation.

Hence, the correct answer is option 3)Depolarisation of ventricles.

Ques 6: Match the following column and select the correct option.

	Column-1		**Column -2**
(a)	**Floating Ribe**	**(i)**	**Located between Second and seventh Ribs**
(b)	**Acromion**	**(ii)**	**Head of the humerus**
(c)	**Scapuls**	**(iii)**	**Clavicle**

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(d)	**Glenoid cavity**	**(iv)**	**Do not connect with the sternum**

Option 1)

(a)	**(b)**	**(c)**	**(d)**
(ii)	**(iv)**	**(i)**	**(iii)**

Option 2)

(a)	**(b)**	**(c)**	**(d)**
(i)	**(iii)**	**(ii)**	**(iv)**

Option 3)

(a)	**(b)**	**(c)**	**(d)**
(iii)	**(ii)**	**(iv)**	**(i)**

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Option 4)

(a)	(b)	(c)	(d)
(iv)	(iii)	(i)	(ii)

Chapter - Skeletal system and its functions; Joints

Difficulty - Medium

Concept - Ribs and Rib Cage -

- **12 pairs of ribs form the bony lateral walls of the thoracic cage.**
- **Each rib is a thin flat bone connected dorsally to the vertebral column and ventral attachments differ.**
- **All the ribs have two articulation surfaces on their dorsal end and is hence are called bicephalic Ribs.**
- **Depending on the ventral attachment of the ribs, there are three types of Ribs.**

1. **The first seven pairs are called true ribs because their anterior ends are attached directly to the sternum using small pieces of hyaline cartilage, the costal cartilage. These are Vertebrosternal Ribs**
2. **The eighth, ninth, and tenth pairs of ribs are called false ribs or vertebral ribs. They articulate by cartilage with costal cartilage of the seventh rib and thus are attached indirectly to the sternum.**
3. **The last pairs (11th and 12th) of ribs are not connected ventrally and are, therefore, called floating ribs. These are also known as Vertebral Ribs as only dorsal attachment is present.**

- **Thoracic vertebrae, ribs, and sternum together form the rib cage.**

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Solution - (a) Floating Rib – (iv) Do not connect with the sternum

- **(b) Acromion – (iii) Clavicle**
- **(c) Scapula – (ii) Head of the humerus**
- **(d) Glenoid cavity – (i) Located between Second and seventh Ribs**

Floating Rib: They are called floating ribs because they do not attach to the sternum.

Acromion: The acromion is part of the scapula and articulates with the clavicle.

Scapula: The scapula is the shoulder blade, and the head of the humerus fits into the glenoid cavity.

Glenoid cavity: The glenoid cavity is a shallow socket located in the scapula between the ribs.

Hence, the correct answer is option (4)

(a)	**(b)**	**(c)**	**(d)**
(iv)	**(iii)**	**(i)**	**(ii)**

Ques 7: Identify the wrong statement concerning the transport of oxygen.

Option 1)The binding of oxygen with haemoglobin is mainly related to the partial pressure of O_2.

Option 2)The partial pressure of CO_2 can interfere with O_2 binding with haemoglobin.

Option 3)Higher H^+ conc. In alveoli favors the formation of oxyhemoglobin.

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Option 4)Low pCO_2 in alveoli favours the formation of oxyhemoglobin.

Chapter - Body Fluids & Circulation

Difficulty - Medium

Concept - Blood is the medium of transport for and.

- About 97 per cent of is transported by RBCs in the blood.
- The remaining 3 per cent is carried in a dissolved state through the plasma.
- Haemoglobin is a red-coloured iron-containing pigment present in the RBCs.
- Can reversibly bind with haemoglobin to form oxyhaemoglobin. This is called the O2-Hb association.
- Each haemoglobin molecule can carry a maximum of four molecules.
- The binding of oxygen with haemoglobin is primarily related to the partial pressure.
- Is high in alveoli so association occurs while at the tissues, the decreases which cause dissociation of oxygen.
- The partial pressure of, hydrogen ion concentration and temperature are the other factors which can interfere with this binding.
- A sigmoid curve is obtained when the percentage saturation of haemoglobin is plotted against the.
- This curve is called the Oxygen dissociation curve.
- This curve is highly useful in studying the effect of factors like concentration, etc, on the binding of haemoglobin.
- In the alveoli, there is
    - high
    - low
    - lesser H+ concentration
    - high pH
    - lower temperature
    - low 2,3-BPG

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- These factors are all favourable for the formation of oxyhaemoglobin.
- In the tissues, there is
    - low
    - high
    - high H+ concentration
    - low pH
    - higher temperature
    - high 2,3-BPG
- 2,3-BPG is Bisphosphoglyceric acid. It lowers the affinity of oxygen to Hb and thus it is more near the tissues. This compound thus helps in the dissociation of oxyhemoglobin
- These conditions are favourable for the dissociation of oxygen from oxyhaemoglobin.
- This indicates that gets bound to haemoglobin in the lung surface gets haemoglobin in the lung surface and gets dissociated at the tissues.
- Every 100 ml of oxygenated blood can deliver around 5 ml to the tissues under normal physiological conditions.

Solution - Several key factors favour the formation of oxyhemoglobin in the alveoli:
High pO2: The partial pressure of oxygen is high in the alveoli, which facilitates oxygen binding to haemoglobin.
Low pCO2: A lower partial pressure of carbon dioxide increases the affinity of haemoglobin for oxygen.
Less H+ Concentration: A lower concentration of hydrogen ions, that is a higher pH, stabilizes haemoglobin and increases its oxygen-binding capacity.
Lower Temperature: The binding of oxygen haemoglobin-bin also favours cooler temperatures.

Hence, the correct answer is option (3). Higher H^+ conc. In alveoli favours the formation of oxyhemoglobin

Ques 8: Select the correct events that occur during the inspiration

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A Contraction of the diaphragm
B Contraction of external intercostal muscles
C Pulmonary volume decreases.
D Intrapulmonary pressure increases

Option 1) (a) and (b)

Option 2) (c) and (d)

Option 3) (a), (b) and (d)

Option 4) only (d)

Chapter - Breathing & Exchange of Gases

Difficulty - Medium

Concept - Mechanism of Breathing: Inspiration -

- Breathing involves two stages:
- inspiration during which atmospheric air is drawn in
- expiration by which the alveolar air is released out
- Two main mechanisms take place to carry out Breathing.
  1) Pressure gradient
- The movement of air into and out of the lungs is carried out by creating a pressure gradient (pressure difference) between the lungs and the atmosphere.
- Human lungs are negative-pressure lungs.
- Inspiration can occur if the pressure within the lungs (intrapulmonary pressure) is less than the atmospheric pressure, i.e., there is a negative pressure in the lungs with respect to atmospheric pressure.
- Expiration occurs when pressure within the lungs is more than the atmospheric pressure.
- 2) Boyle's Law
- Breathing is due to Boyle's Law

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- This law states that when the temperature is constant, Pressure and Volume are inversely proportional to each other and this results in the movement of air.
- INSPIRATION
  Muscles of Inspiration:
- Core muscles:
    - External intercostal muscles: These contract to elevate the ribs
    - Diaphragm: It contracts to expand the thoracic cavity
- Accessory muscles:
    - Sternocleidomastoid: It contracts to elevate the sternum
    - Pectoralis minor: It contracts to pull ribs outwards
- Mechanism of Inspiration :
- The diaphragm and a specialized set of muscles – external and internal intercostals between the ribs, help in the generation of such gradients.
- Inspiration is initiated by the contraction of the diaphragm which increases the volume of the thoracic chamber in the anteroposterior axis.
- The contraction of external intercostal muscles lifts the ribs and the sternum causing an increase in the volume of the thoracic chamber in the dorsoventral axis.
- The overall increase in the thoracic volume causes a similar increase in pulmonary volume.
- An increase in pulmonary volume decreases the intrapulmonary pressure to less than the atmospheric pressure which forces the air from outside to move into the lungs, i.e., inspiration.

Solution - During inspiration, the diaphragm contracts and moves downward, flattening its dome shape, while the external intercostal muscles contract, pulling the ribcage outward and upward. These coordinated muscle actions increase the thoracic cavity's volume, leading to an expansion of the lungs due to their attachment to the thoracic walls via the pleural membranes. The increase in pulmonary volume reduces the intrapulmonary pressure below atmospheric pressure, creating a pressure gradient. This pressure difference drives air into the lungs, allowing oxygen-rich air to flow

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through the airways into the alveoli for gas exchange. This process is energy-dependent, as the contraction of muscles requires ATP. The entire mechanism ensures efficient ventilation and oxygen delivery to the body.

Hence, the correct answer is option 1 (a) and (b)

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